

Comprehensive Report: Building a Virtualized Data Center and Enterprise Domain Environment

Author: Muhammad Bilal

Undergraduate of Bachelors in Computer Science

bilalmubeen99@gmail.com

Abstract:

This document presents a detailed report on designing and implementing a virtualized data center and an enterprise domain environment using VMware Workstation. The project aimed to simulate a virtual infrastructure, set up an Active Directory domain, enforce security policies, and gain hands-on experience with virtualization technologies. Key steps included configuring VMs for Windows and ESXi Server, creating a domain controller, implementing GPOs, and integrating client machines into the domain. Challenges encountered, solutions implemented, and lessons learned are discussed in detail.

Keywords:

Virtualization, VMware Workstation, Active Directory, Group Policy Objects, Enterprise Network Management

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1. Project Summary

The project involved designing and implementing a virtualized data center and an enterprise domain environment using VMware Workstation. The main objectives were to simulate the setup of a virtual infrastructure, create user accounts, apply security policies, and gain hands-on experience with virtualization technologies and domain administration. This project aligns with the goal of developing practical skills in virtualization, operating systems, hypervisors, and Active Directory configurations.

2. Project Scope

- Install VMware Workstation on a physical machine with minimum requirements of 16GB RAM and 200GB HDD.
- Create and configure virtual machines (VMs) for Windows 7, Windows 10, ESXi Server, and Windows Server 2016.
- Set up an Active Directory domain, create user accounts, and implement group policies (GPOs).
- Customize user environments with security and usability enhancements, including password policies and desktop wallpapers.

3. Key Steps and Implementation

3.1. Virtual Machine Setup

- Created VMs for Windows 7, Windows 10, and ESXi server, ensuring proper disk space allocation and hardware requirements.
- Installed operating systems using ISO images and configured the initial network and user settings.

3.2. ESXi Server Configuration

- Installed ESXi server on a dedicated VM.
- Accessed the ESXi server through a web browser to explore datastore, network, and monitoring functionalities.

3.3. Active Directory Setup

- Installed Windows Server 2016 on a VM and configured it as a domain controller.
- Set up a domain (e.g., abc.com) and created organizational units (OUs) for different campus locations.

3.4. Group Policy Implementation

Configured GPOs to enforce password policies, restrict user profiles, and standardize desktop environments:

- User1: Password expires after 30 days.

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- User2: Password never expires.
- Remove recycle bin icon from user desktops.
- Apply a company's customized wallpaper across all desktops.

3.5. Domain Integration

Added Windows 7 and Windows 10 VMs to the domain and validated login with created user accounts.

4. Challenges Faced

4.1. Resource Constraints

Insufficient RAM or disk space led to delays and required adjustments to virtual machine configurations.

4.2. Compatibility Issues

Some older operating systems exhibited challenges in driver installation and domain integration.

4.3. Network Configuration

Misconfigured network adapters led to connectivity issues between VMs and the ESXi server.

4.4. GPO Conflicts

Overlapping policies created conflicts, especially with password rules for different user accounts.

5. Solutions Implemented

1. Upgraded the host machine's hardware to meet the project's demands.
2. Utilized compatibility tools and updated drivers for seamless integration.
3. Standardized network settings and tested connectivity after each configuration change.
4. Organized OUs to manage policy inheritance effectively, resolving GPO conflicts.

6. Lessons Learned

6.1. Resource Planning

Proper planning and assessment of hardware requirements are critical to project success.

6.2. Systematic Troubleshooting

Step-by-step debugging helped identify and resolve network and configuration issues efficiently.

6.3. Policy Design

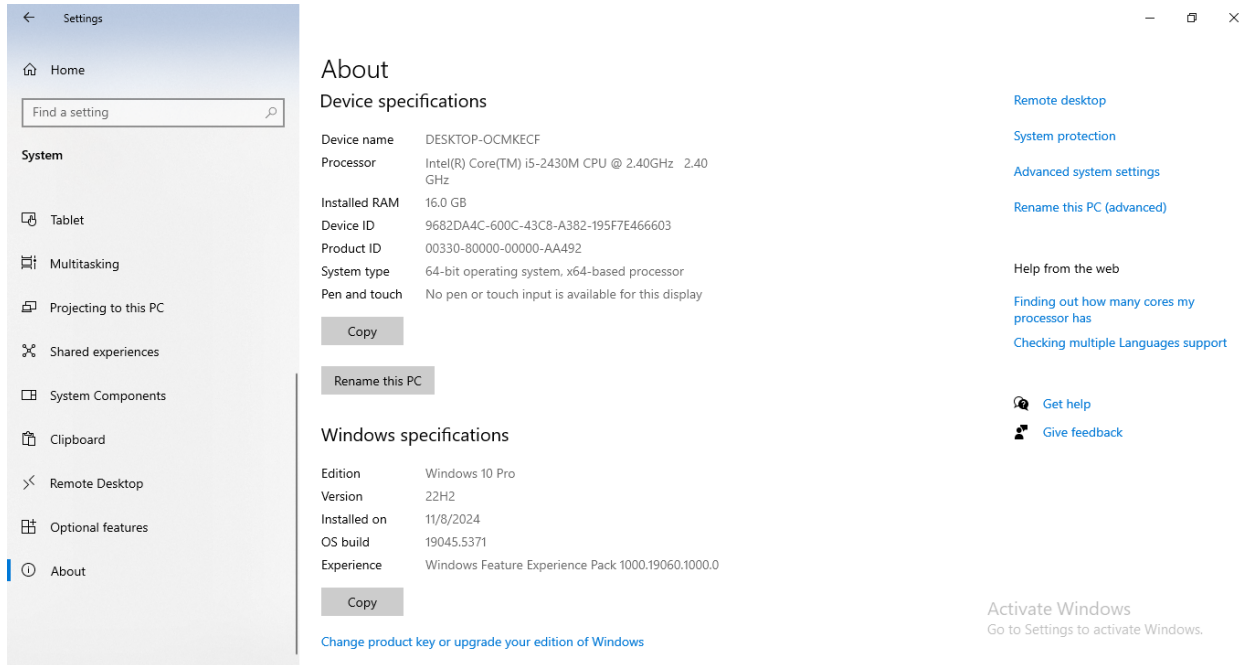
Clear understanding and documentation of GPO hierarchy are essential to avoid conflicts and ensure proper implementation.

VIRTUALIZED DATA CENTER & DOMAIN SETUP

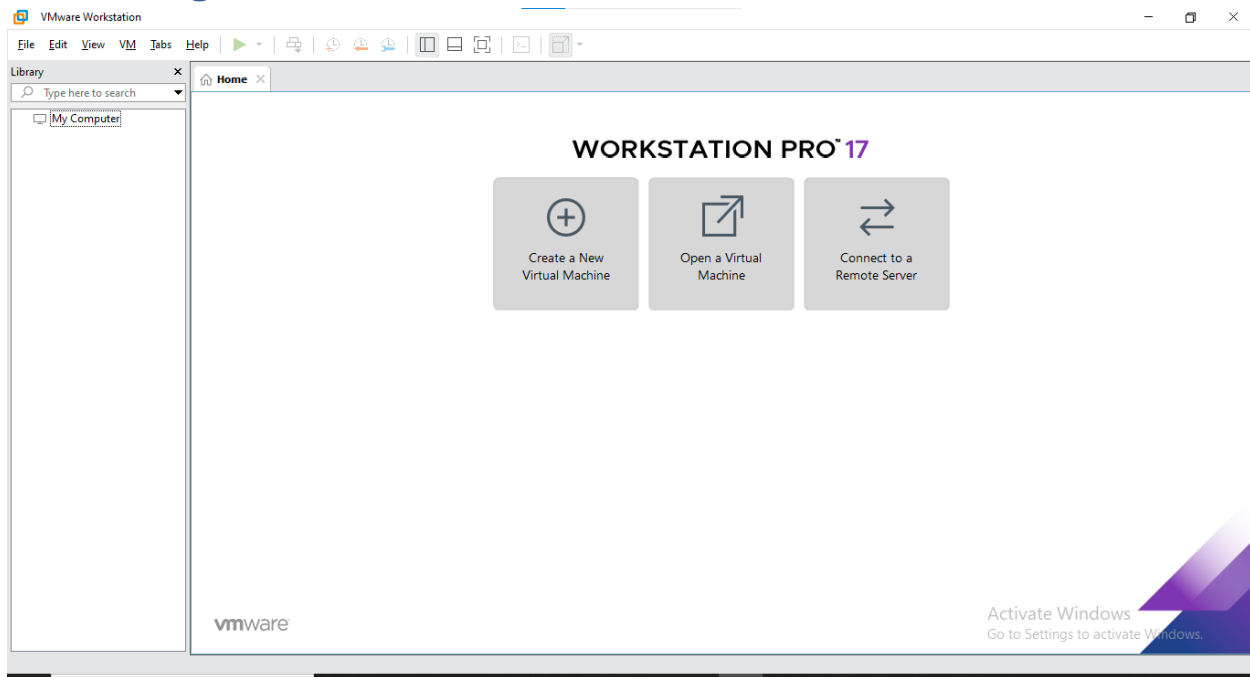
6.4. Hands-on Practice

Practical exposure to virtualization tools like VMware and Active Directory is invaluable for skill development.

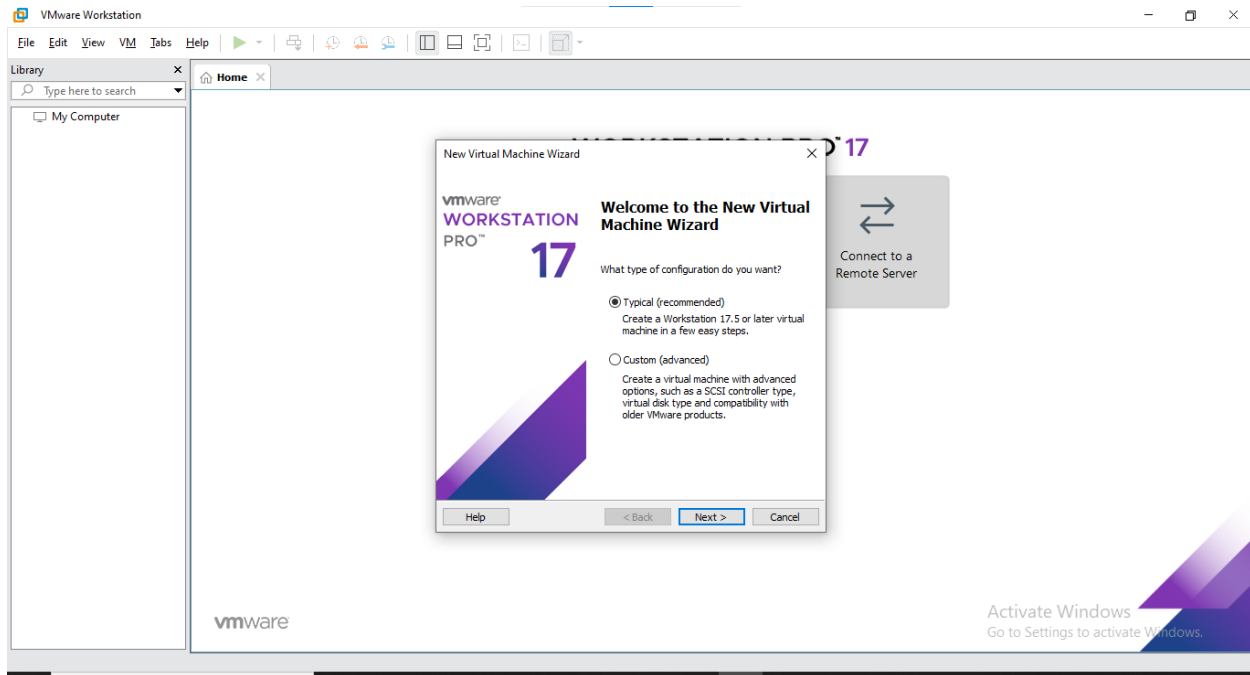
7. Preparing a Windows Machine



8. Creating a New Virtual Machine

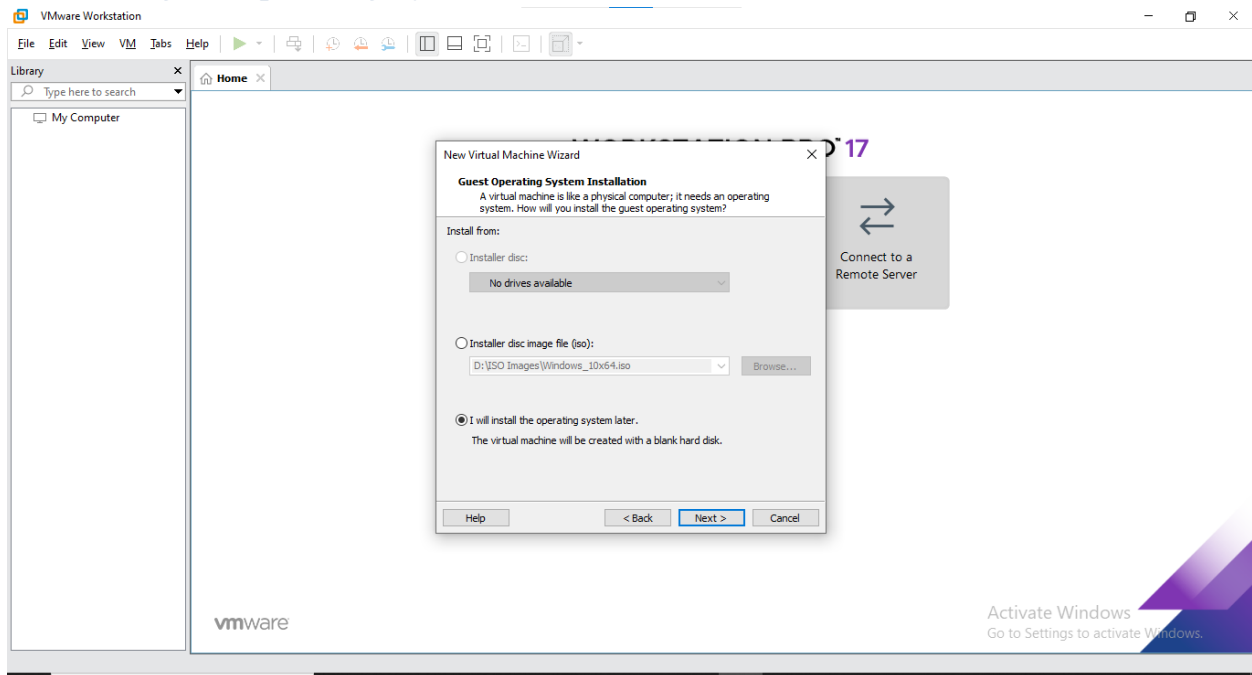


8.1. Creating a Virtual Machine and Installing Windows 7 Pro

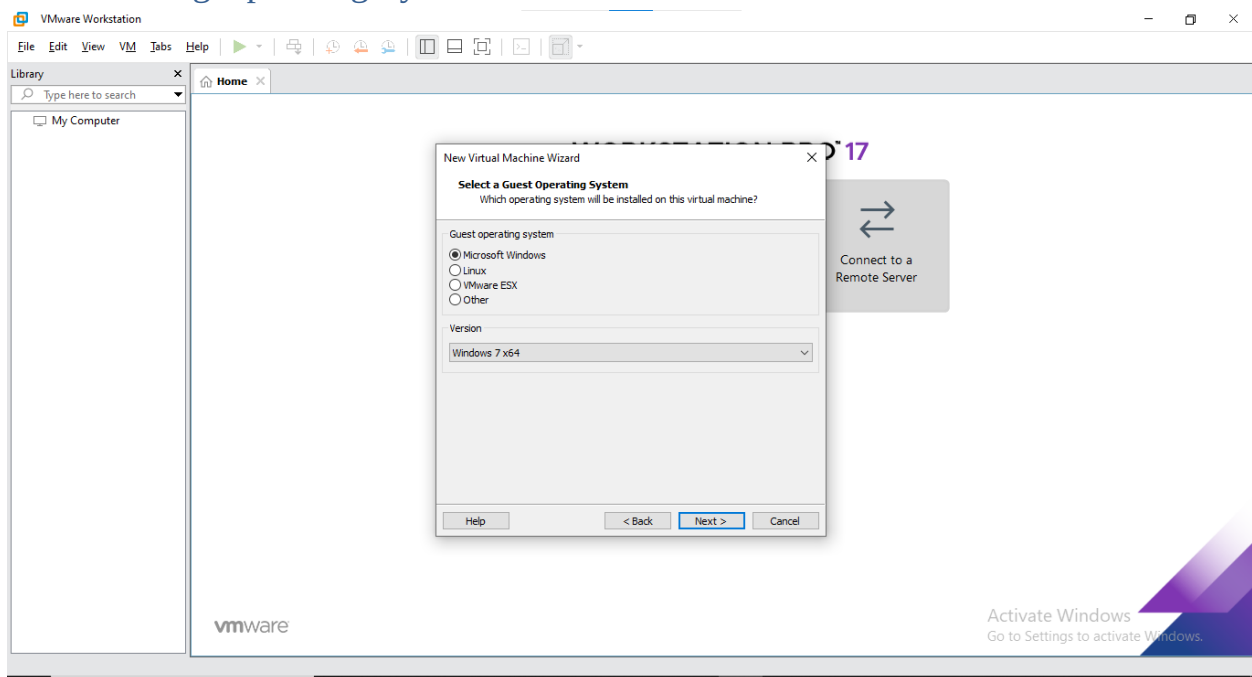


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8.2. Configure Operating System Information

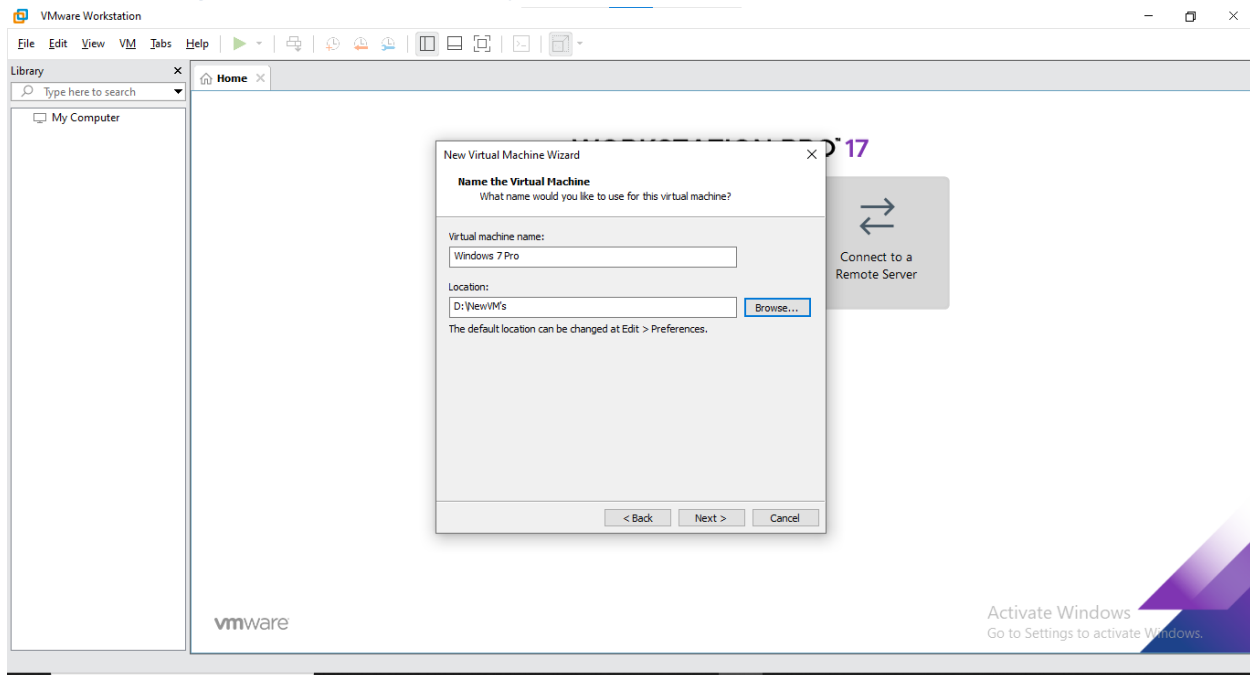


8.3. Selecting Operating System

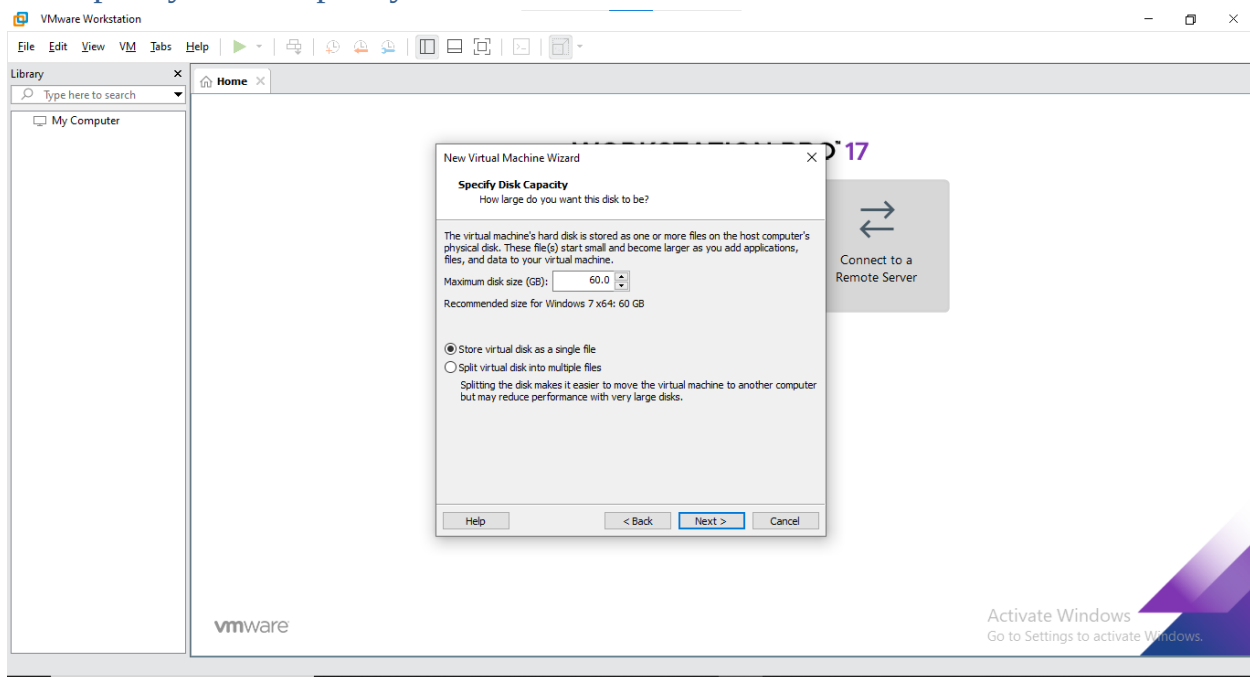


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8.4. Selecting Name and Directory to Store VM

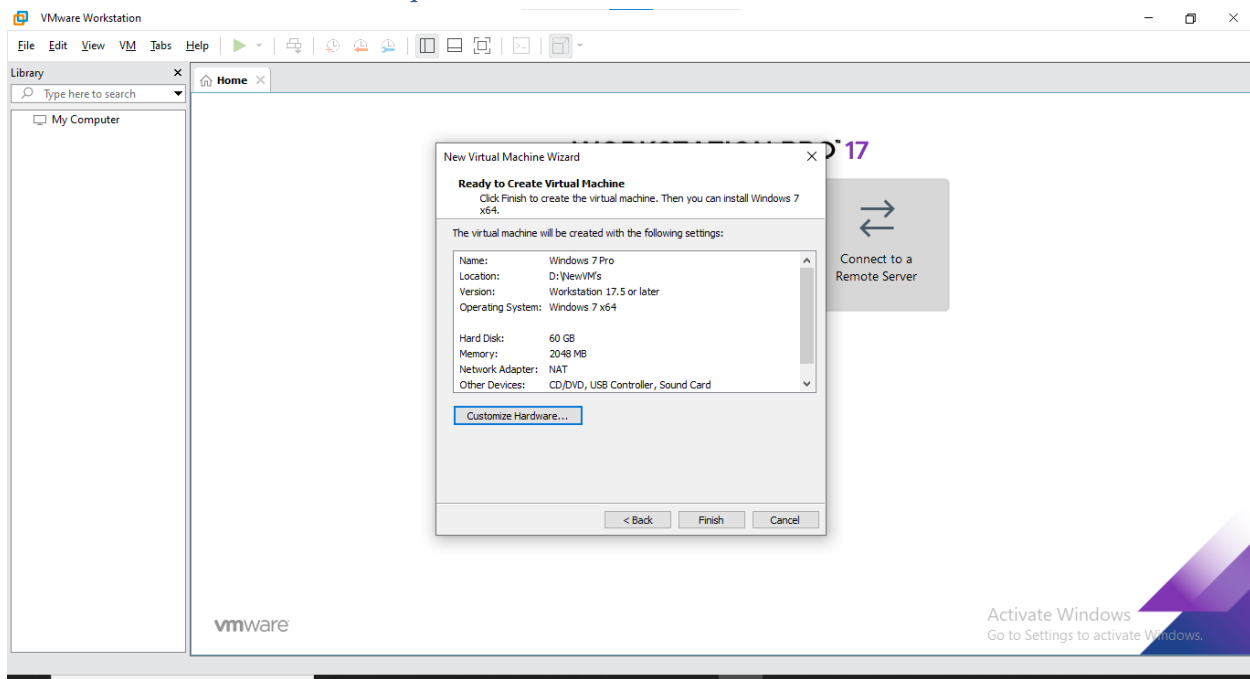


8.5. Specify Disk Capacity

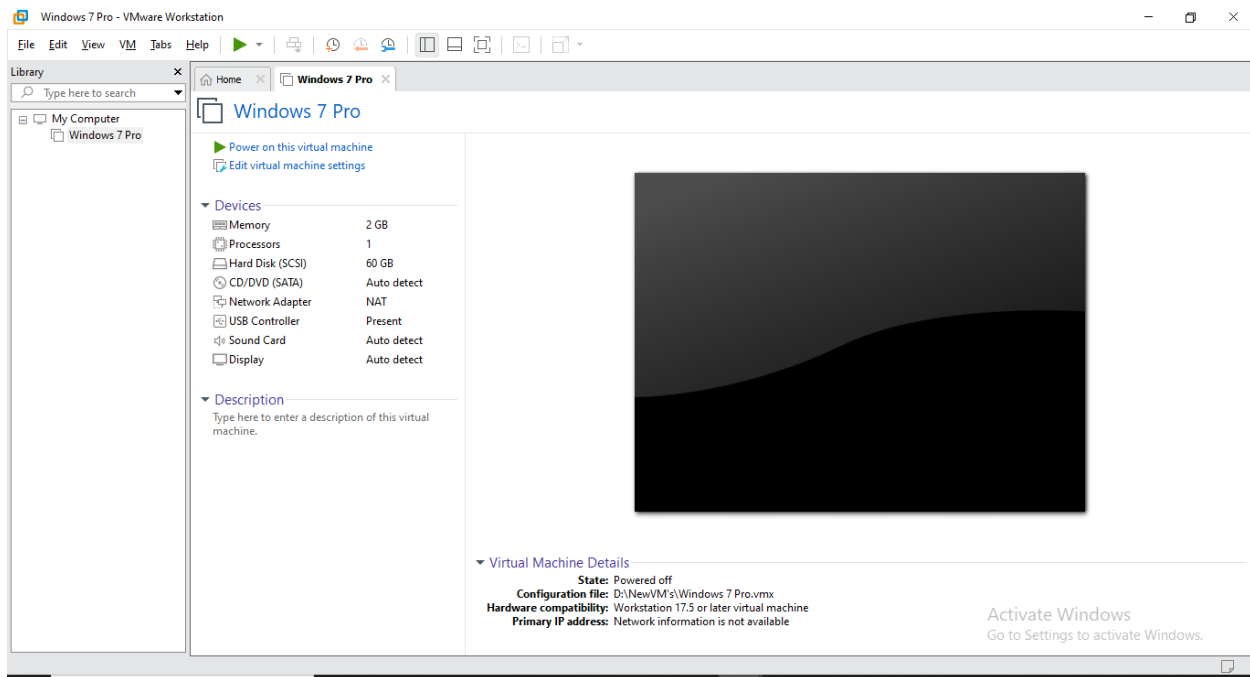


VIRTUALIZED DATA CENTER & DOMAIN SETUP

8.6. Finalize Hardware Requirement

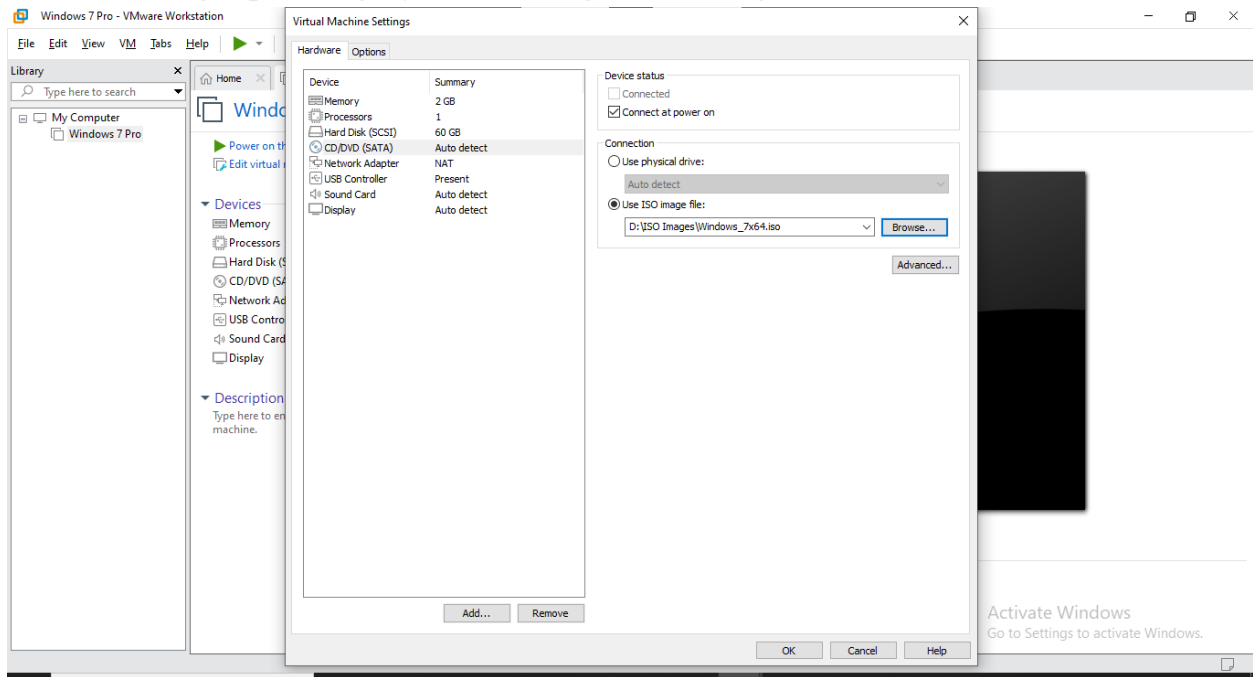


A Virtual Machine is Created

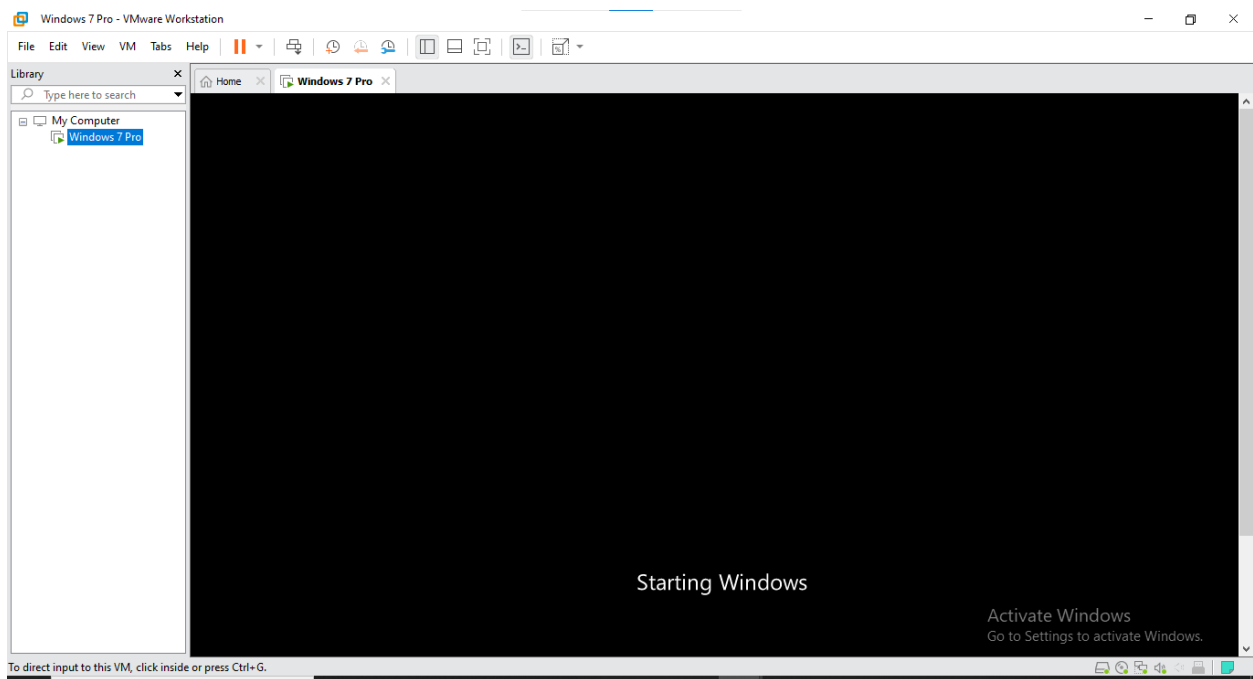


VIRTUALIZED DATA CENTER & DOMAIN SETUP

8.7. Installing Operating System through ISO Image

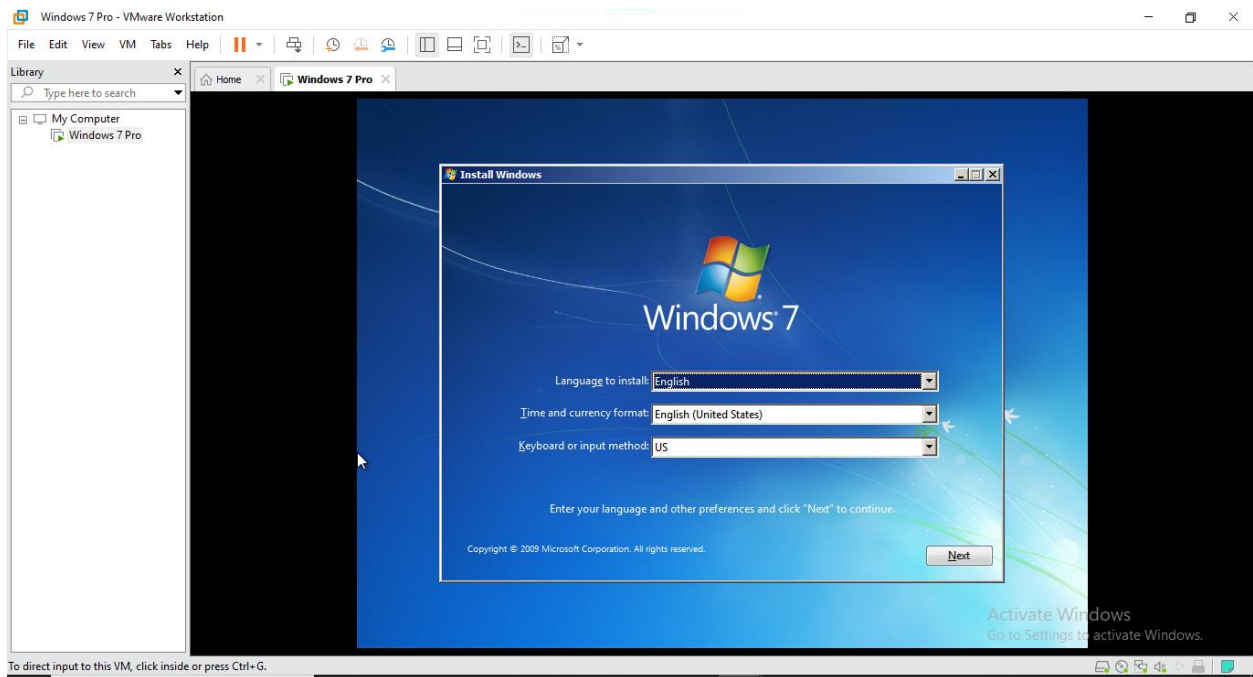


8.8. Power On Virtual Machine to Finish Installation

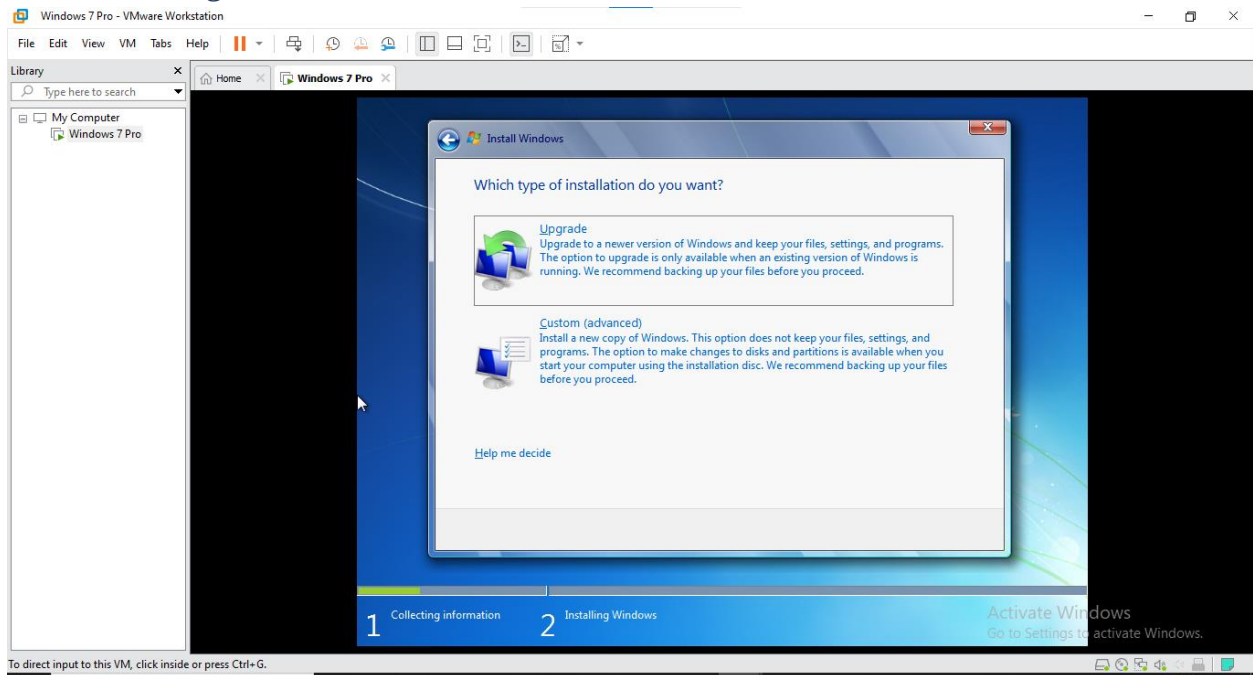


VIRTUALIZED DATA CENTER & DOMAIN SETUP

8.8.1. Click on Next to Proceed

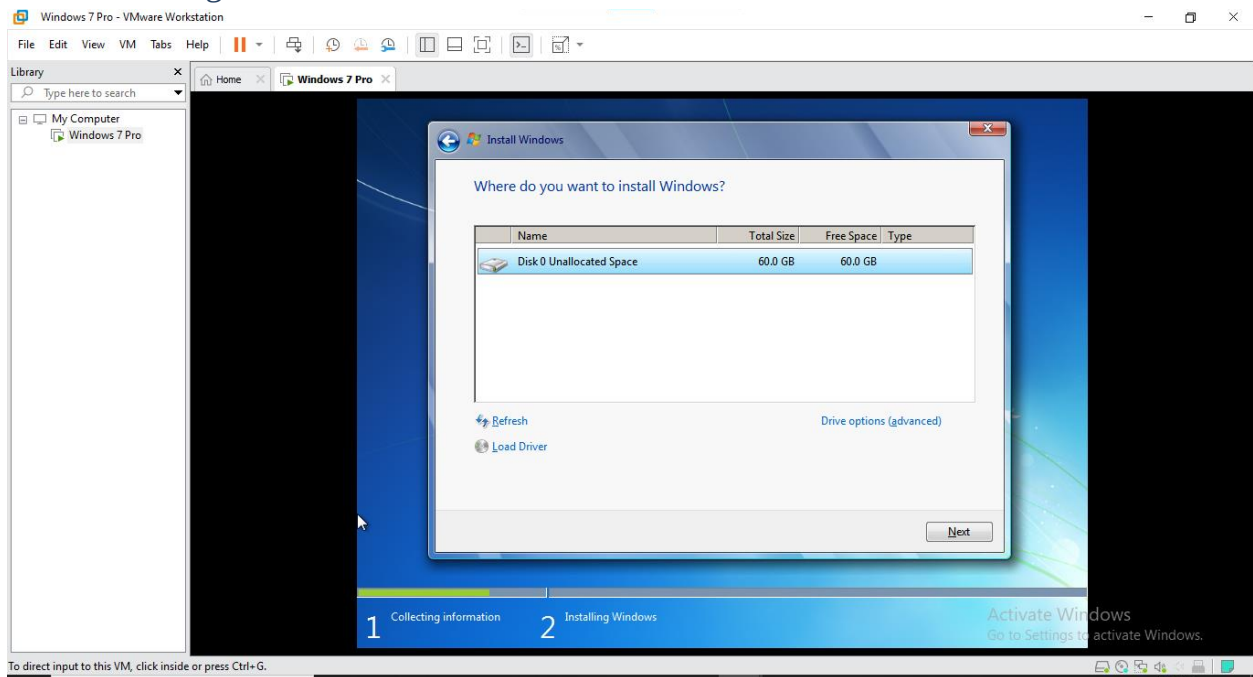


8.8.2. Selecting Custom to Proceed

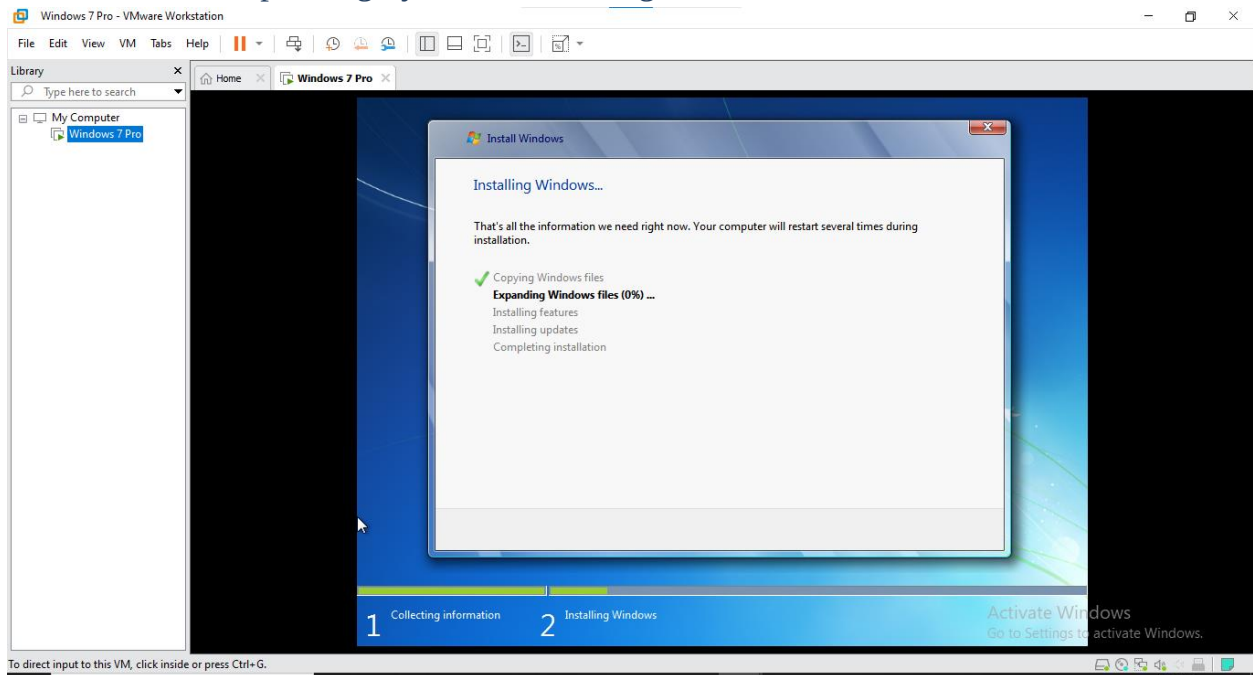


VIRTUALIZED DATA CENTER & DOMAIN SETUP

8.8.3. Selecting Allocated Disk

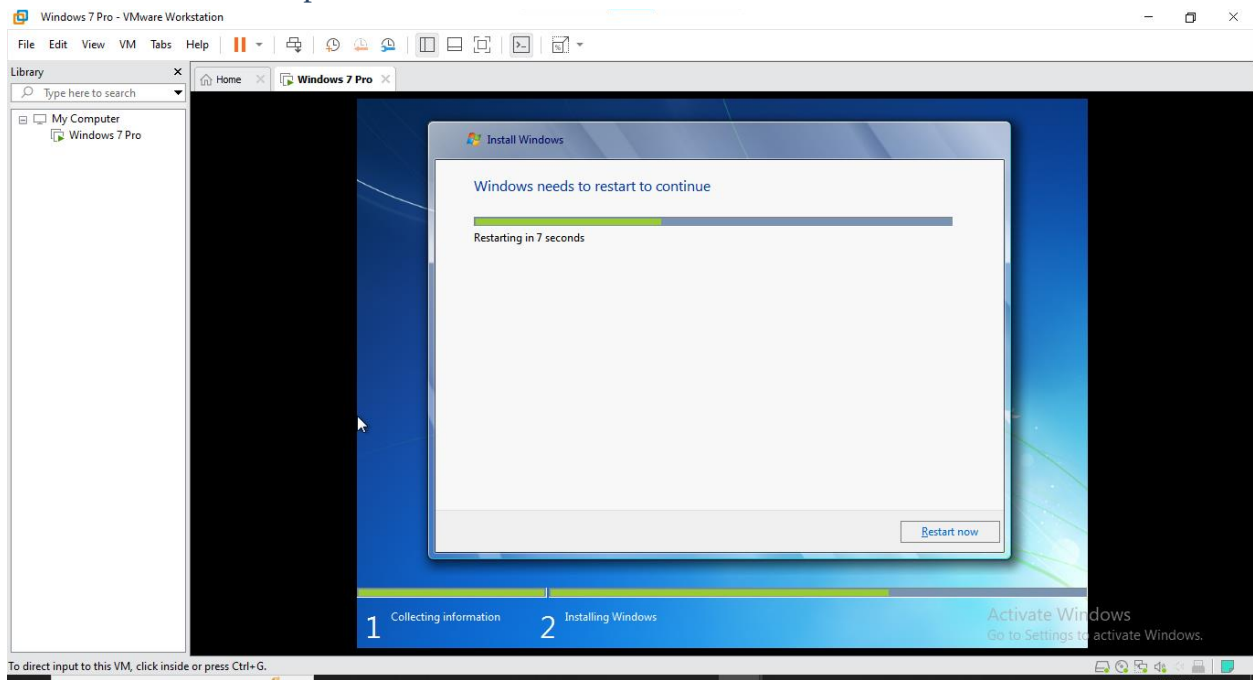


8.8.4. Windows Operating System is Installing

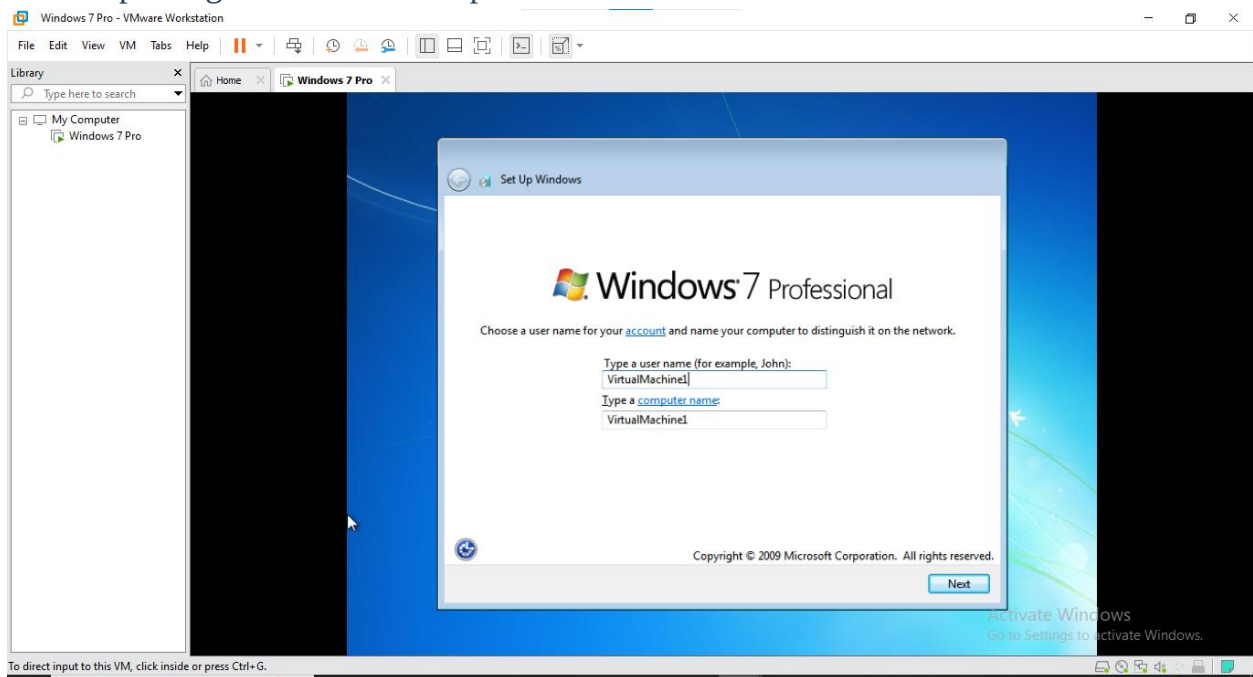


VIRTUALIZED DATA CENTER & DOMAIN SETUP

8.8.5. Restart to Complete Installation

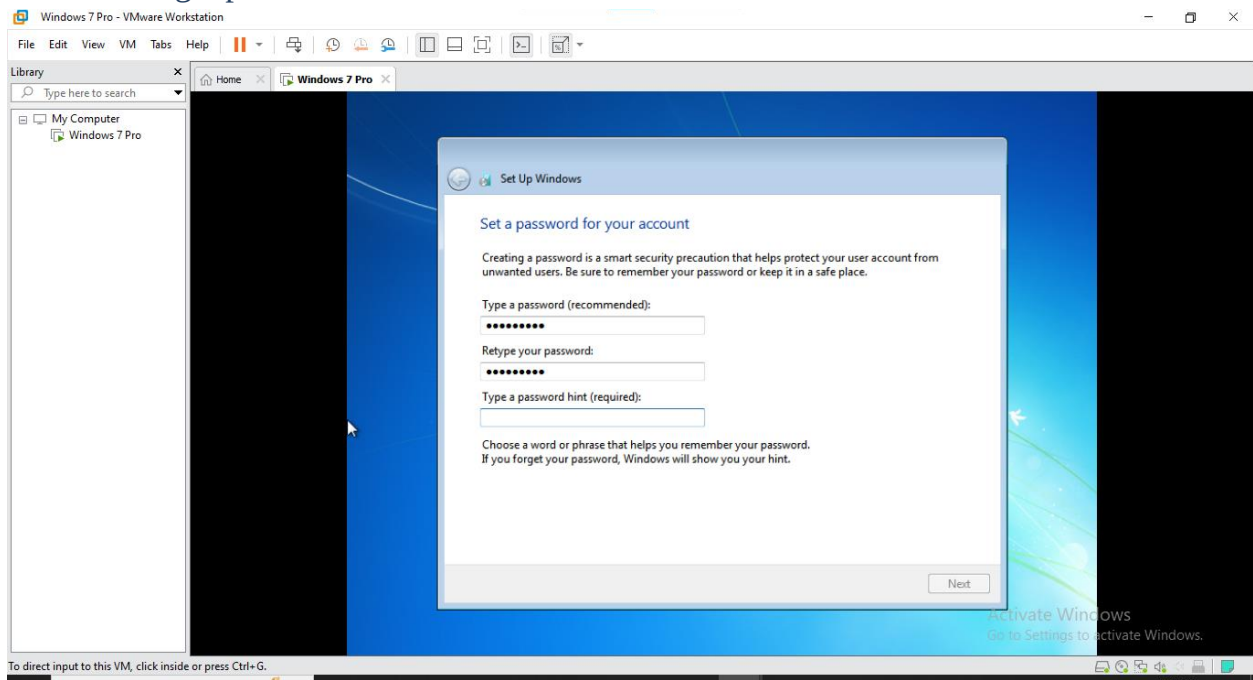


8.8.6. Updating Name for Desktop

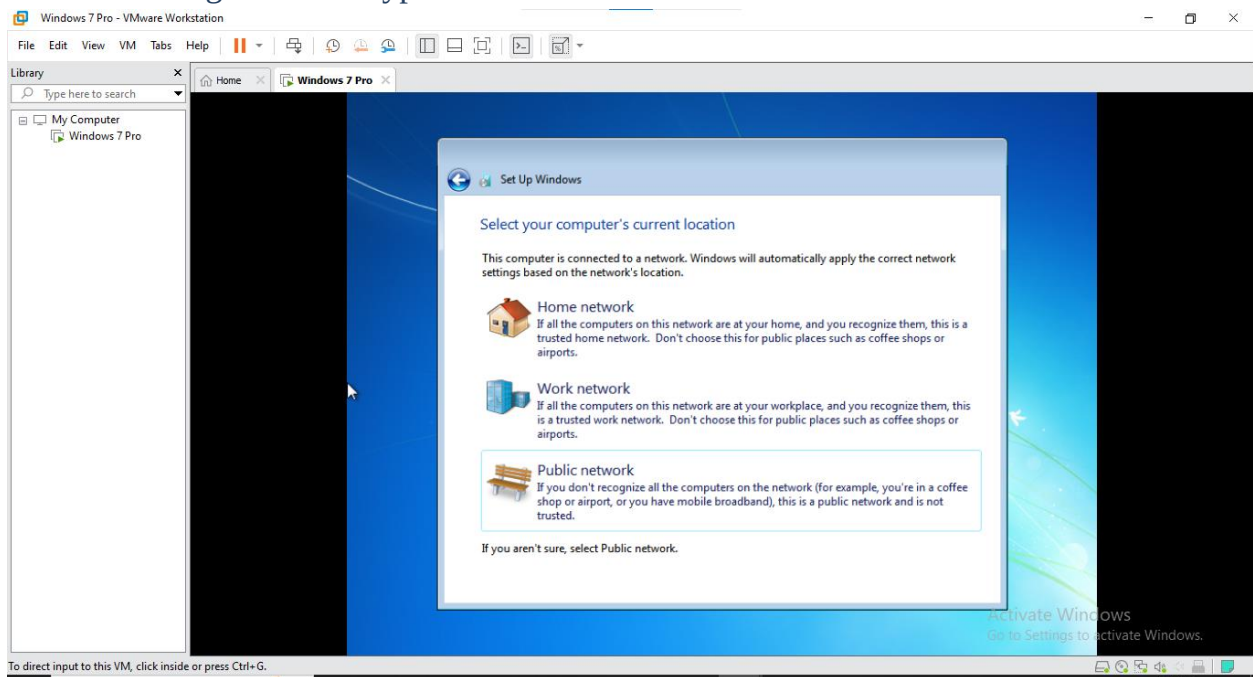


VIRTUALIZED DATA CENTER & DOMAIN SETUP

8.8.7. Setting Up Password

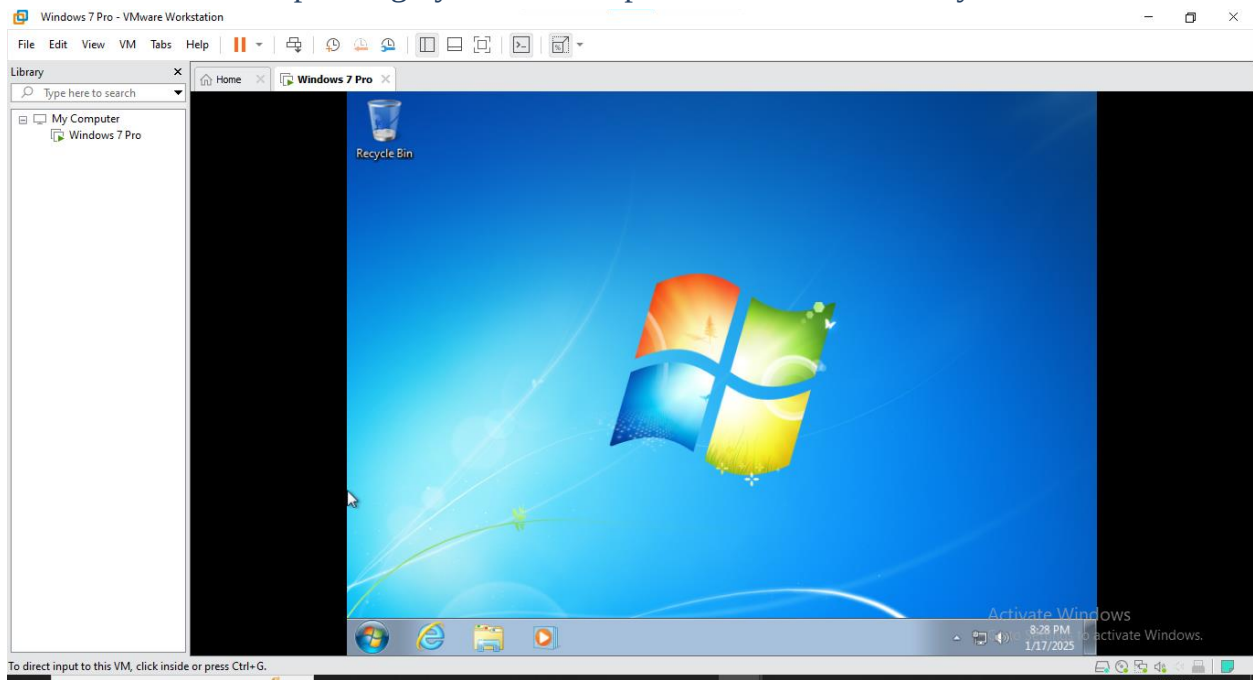


8.8.8. Selecting Network Type

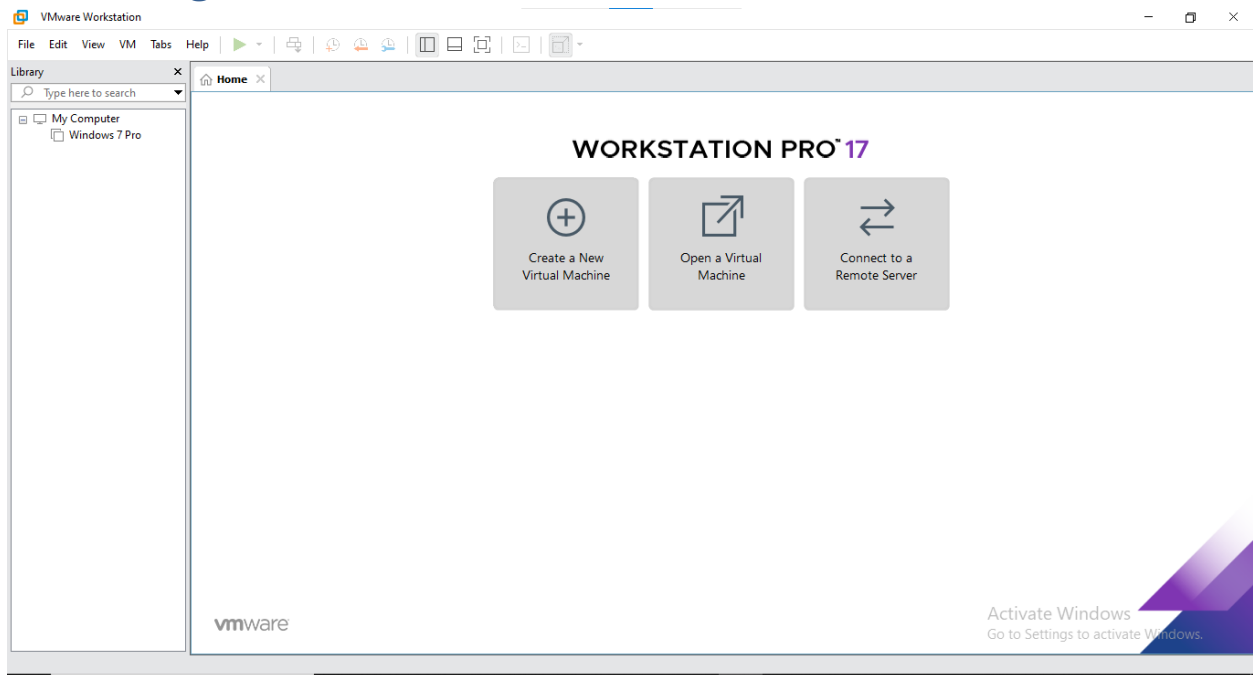


VIRTUALIZED DATA CENTER & DOMAIN SETUP

8.8.9. A Windows Operating System Desktop is Created Successfully

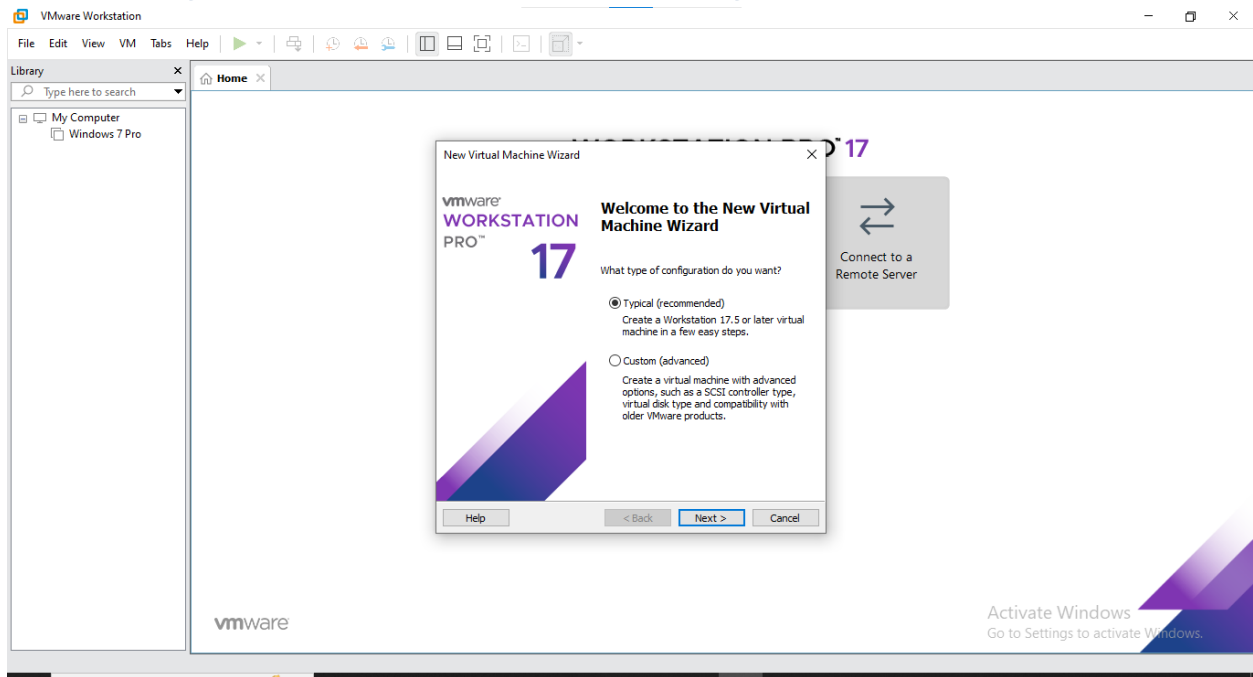


9. Creating a 2nd Virtual Machine

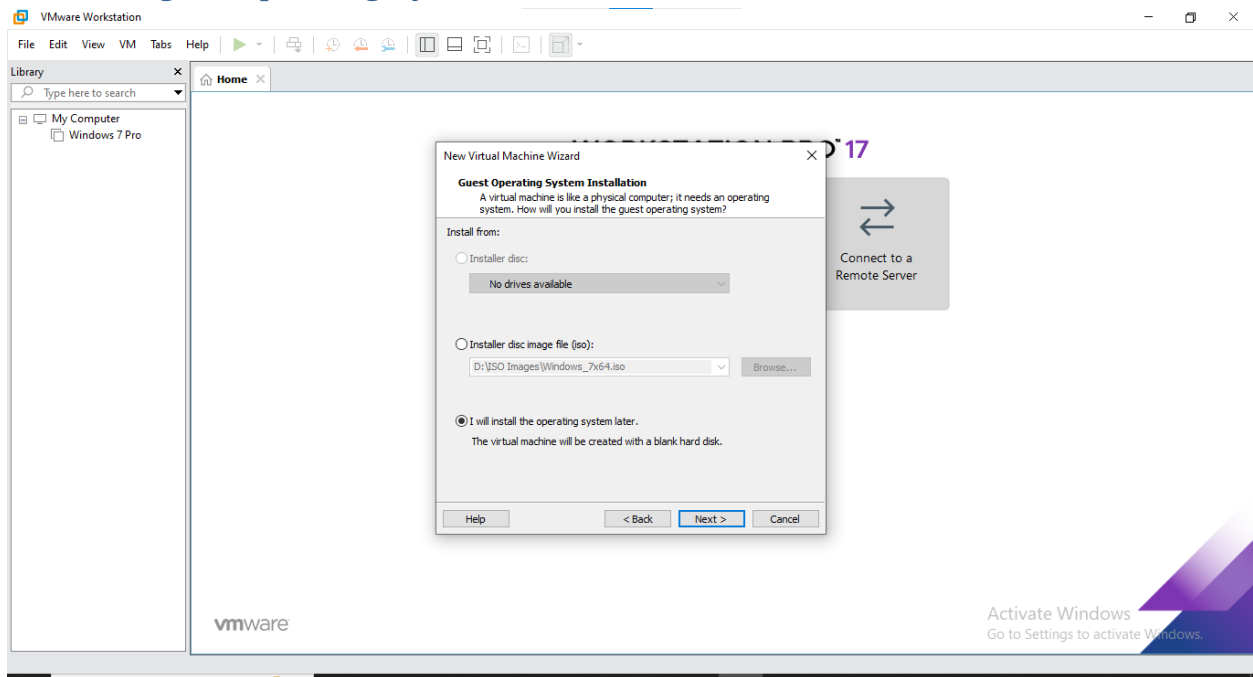


VIRTUALIZED DATA CENTER & DOMAIN SETUP

9.1. Creating a 2nd Virtual Machine and Installing Windows 10 Pro

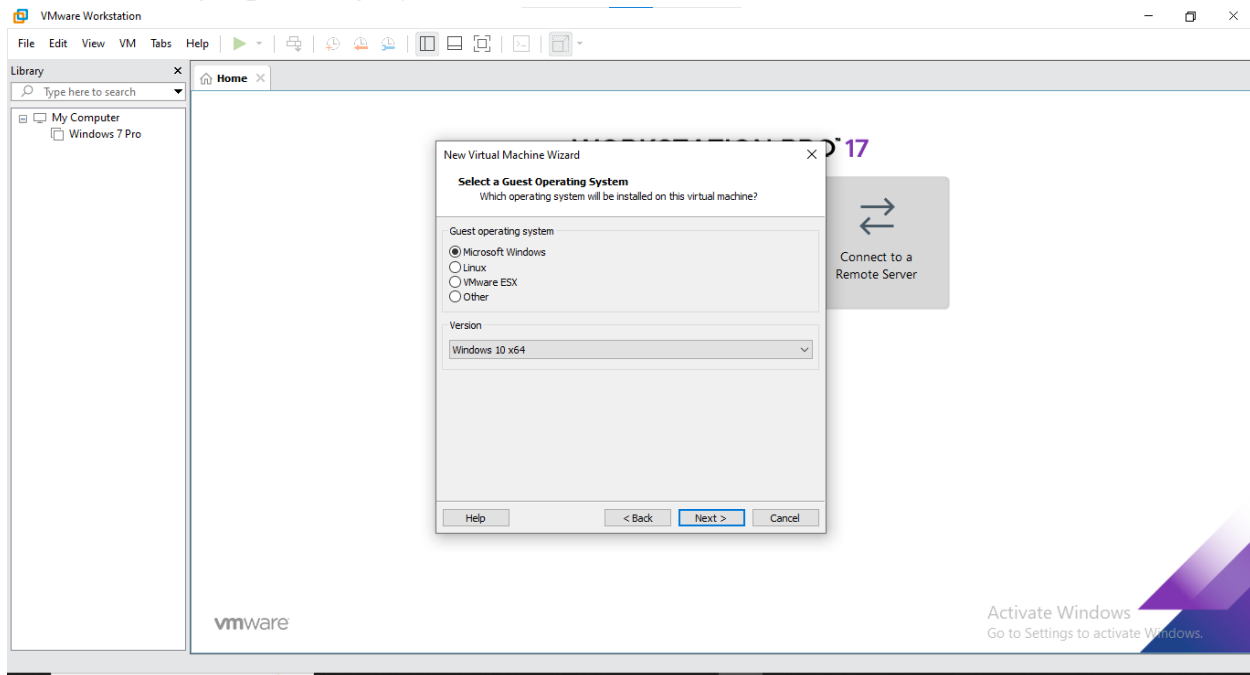


9.2. Configure Operating System Information

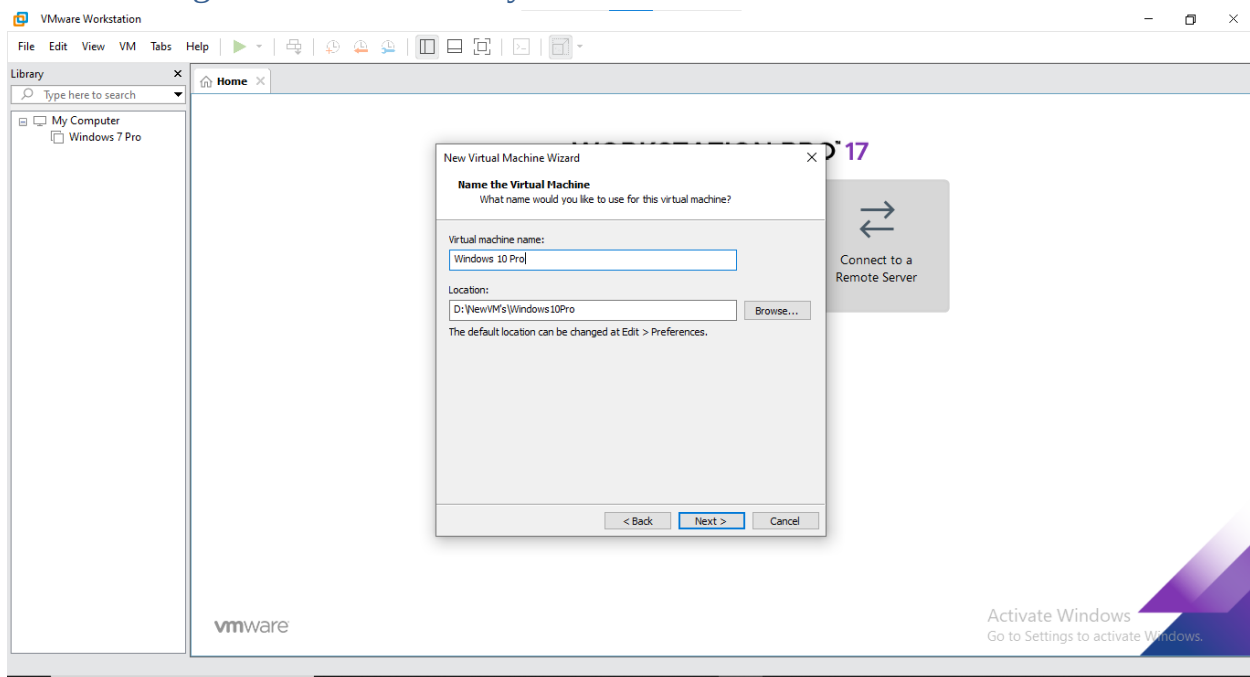


VIRTUALIZED DATA CENTER & DOMAIN SETUP

9.3. Selecting Operating System

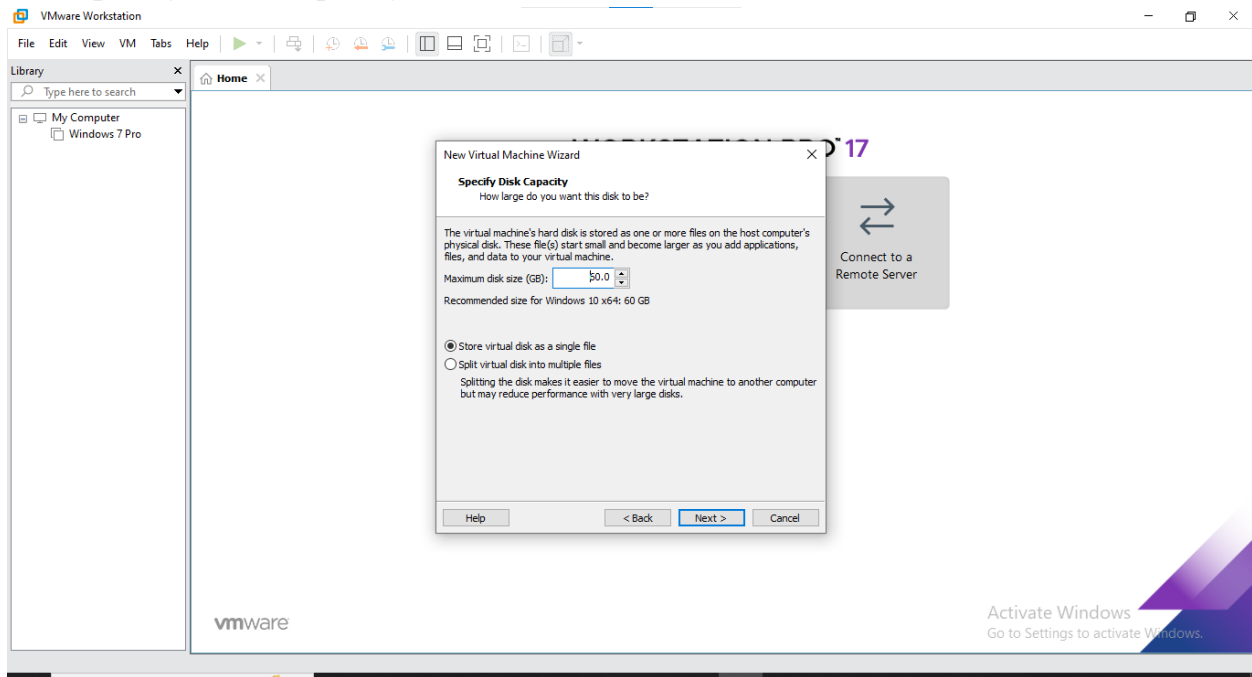


9.4. Selecting Name and Directory to Store VM

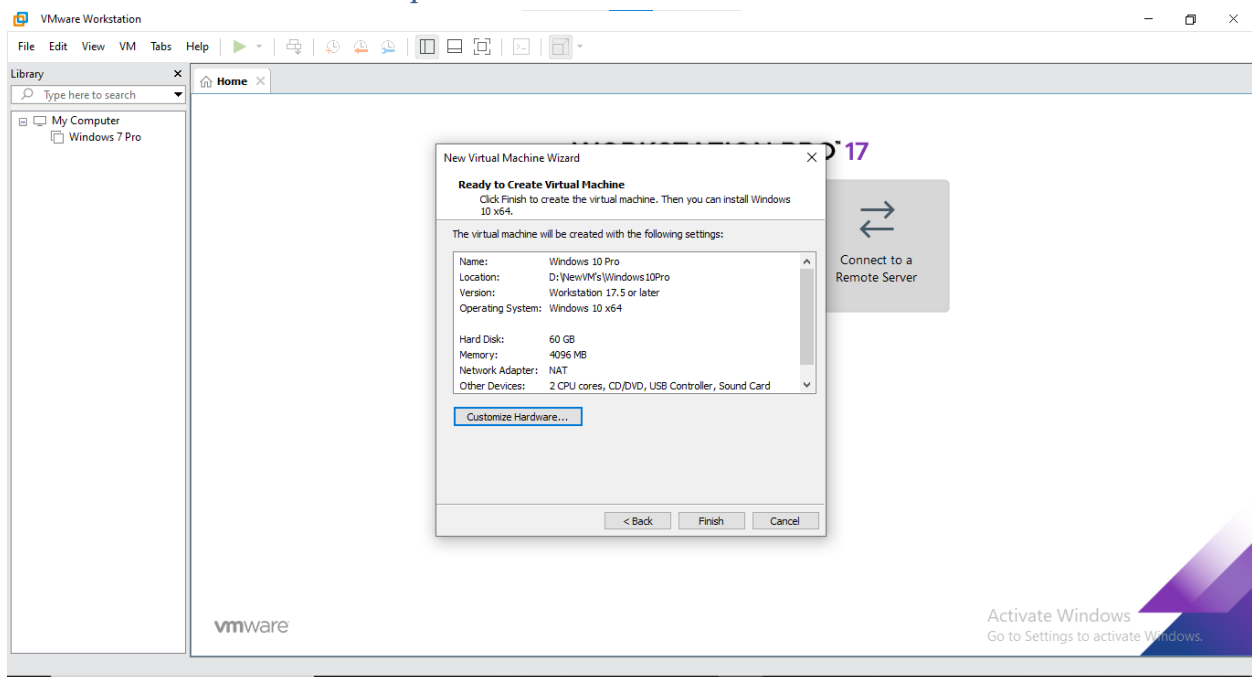


VIRTUALIZED DATA CENTER & DOMAIN SETUP

9.5. Specify Disk Capacity

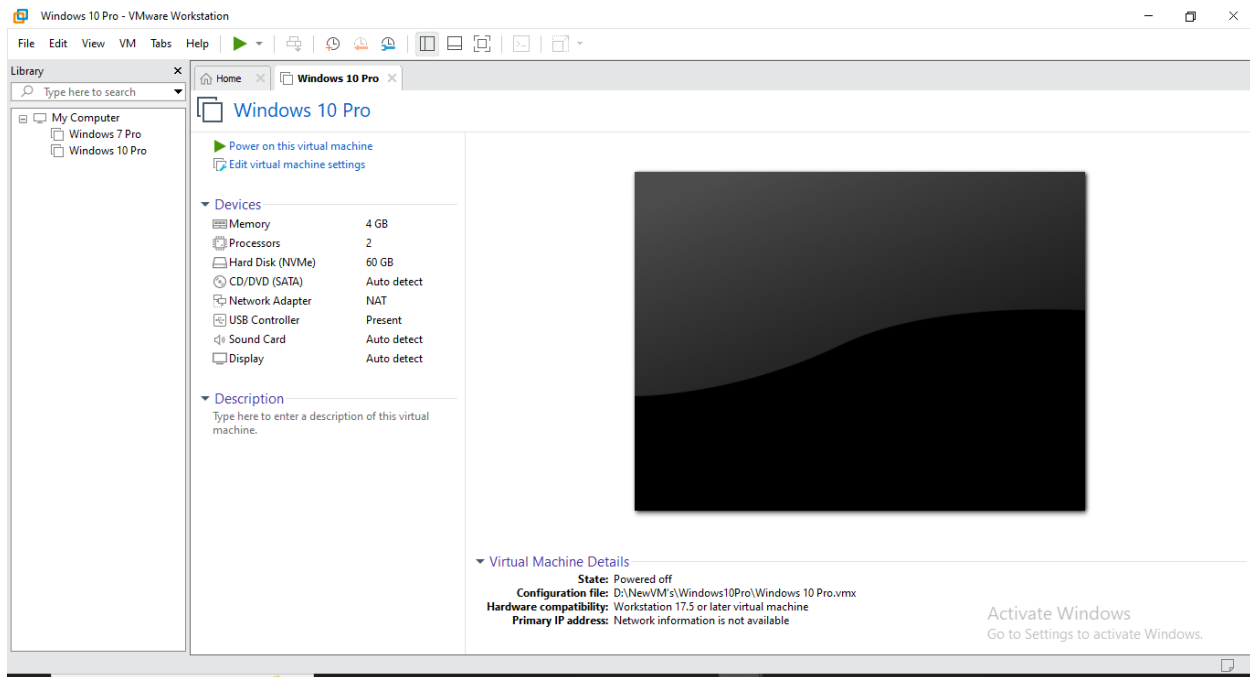


9.6. Finalize Hardware Requirement

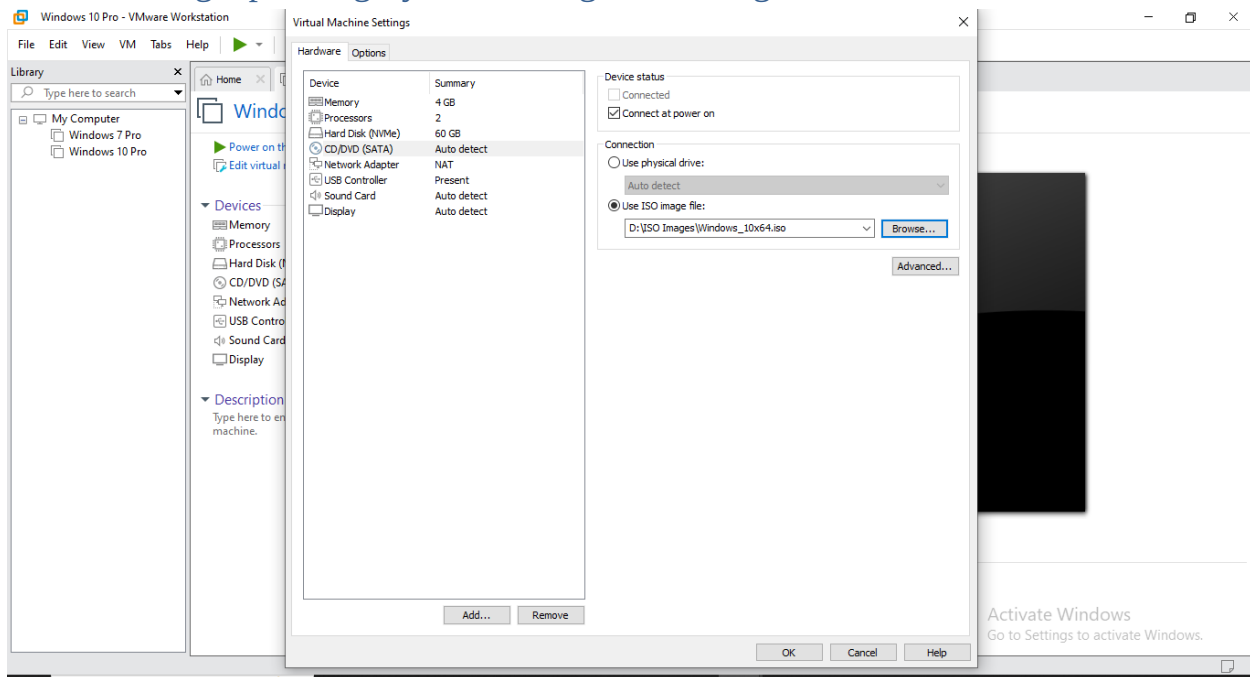


VIRTUALIZED DATA CENTER & DOMAIN SETUP

A Virtual Machine is Created

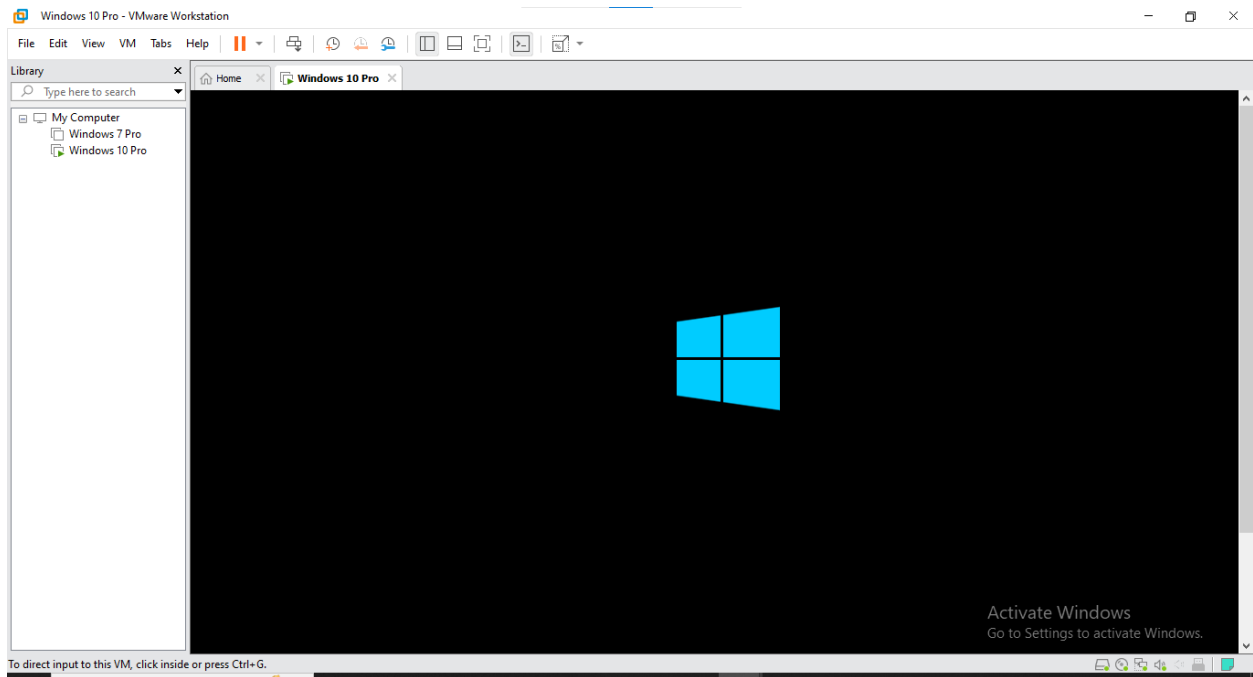


9.7. Installing Operating System through ISO Image

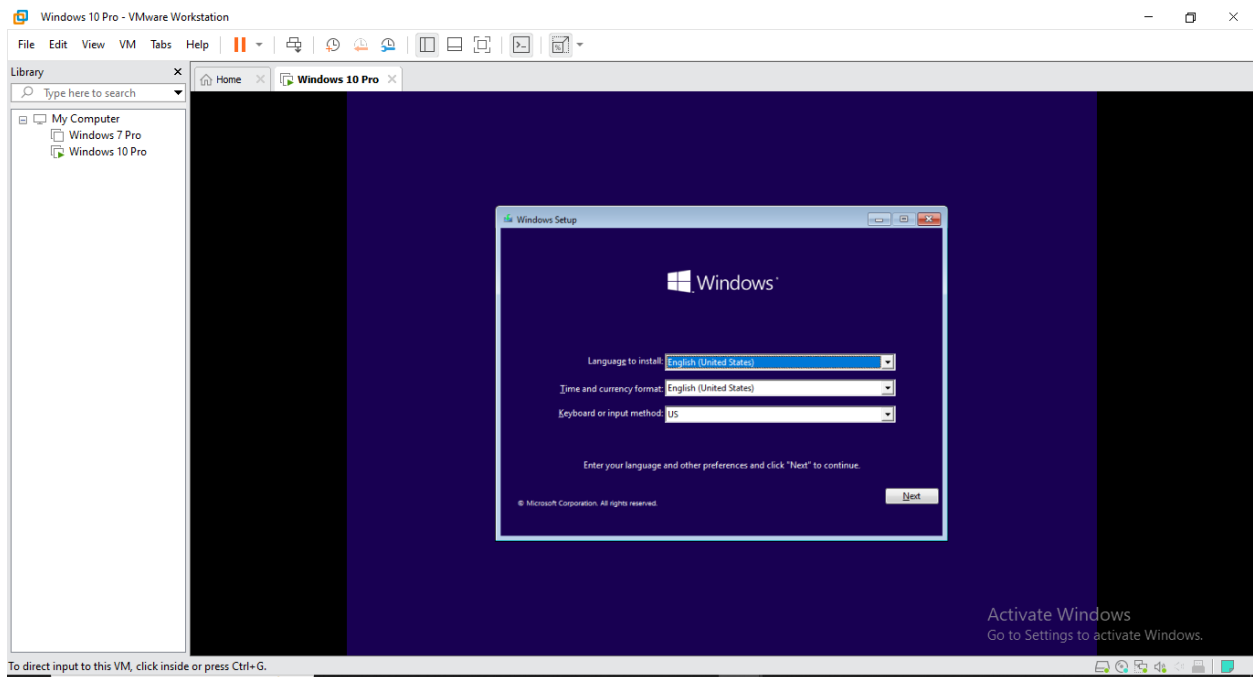


VIRTUALIZED DATA CENTER & DOMAIN SETUP

9.8. Power On Virtual Machine to Finish Installation

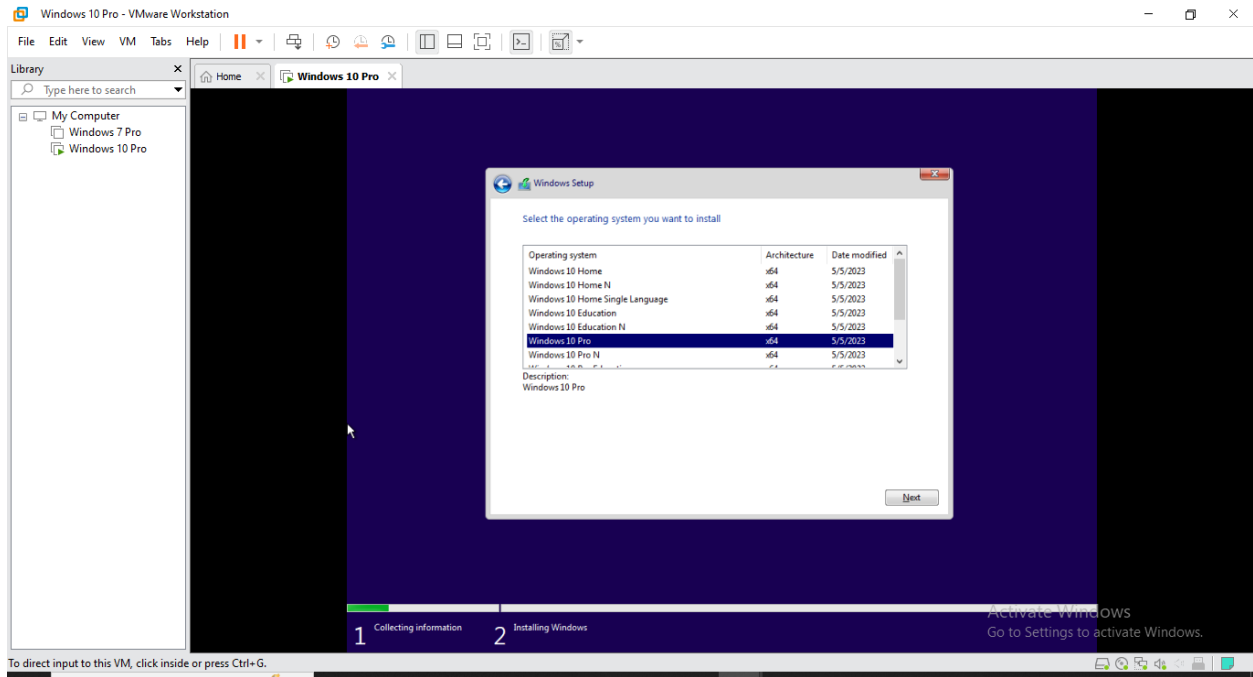


9.8.1. Click on Next to Proceed

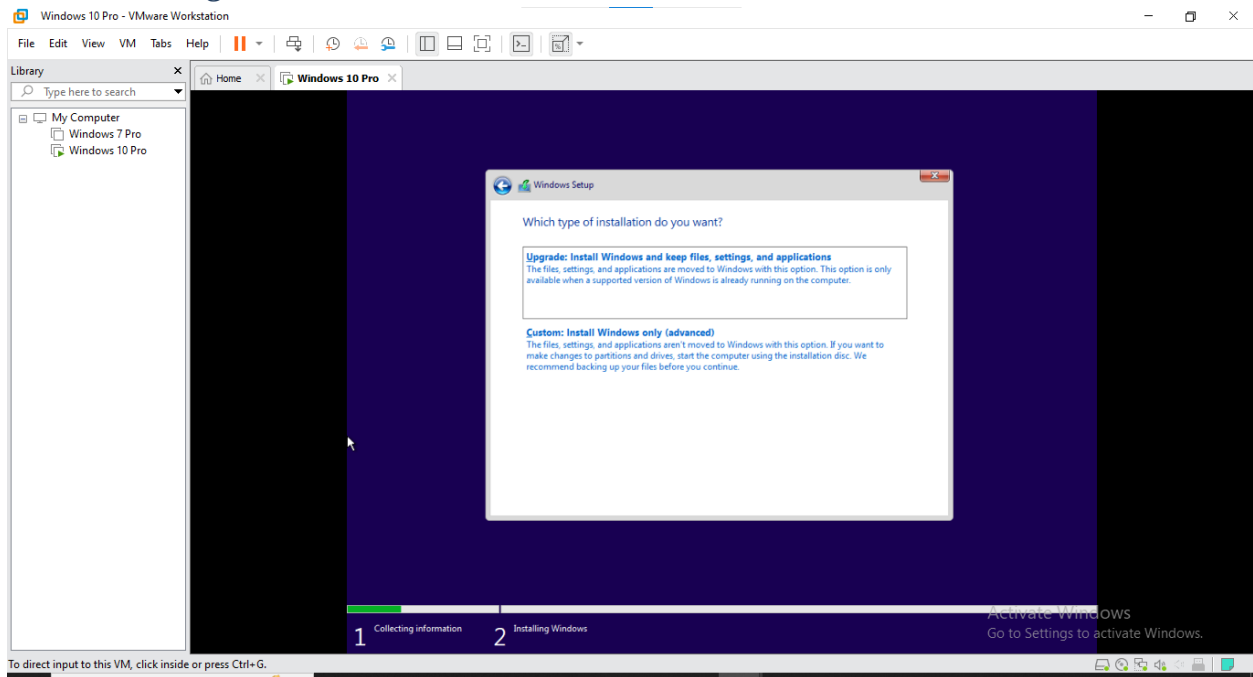


VIRTUALIZED DATA CENTER & DOMAIN SETUP

9.8.2. Selecting Operating System Version (Pro x64 bits)

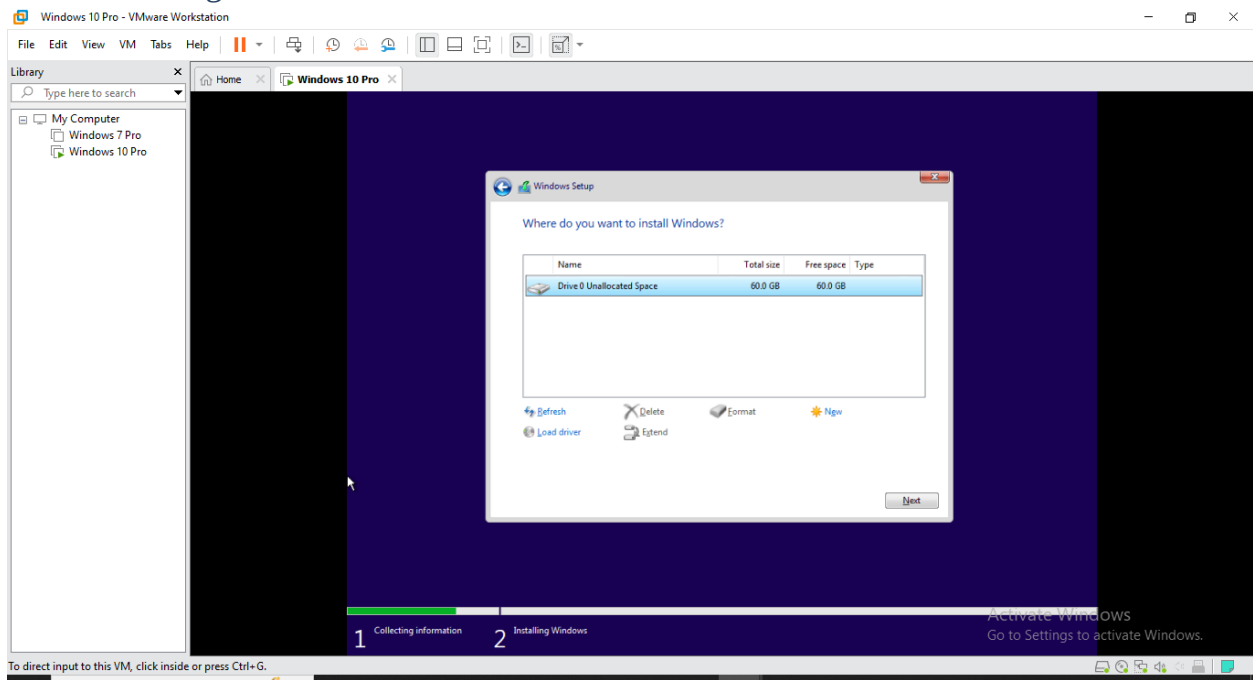


9.8.3. Selecting Custom to Proceed

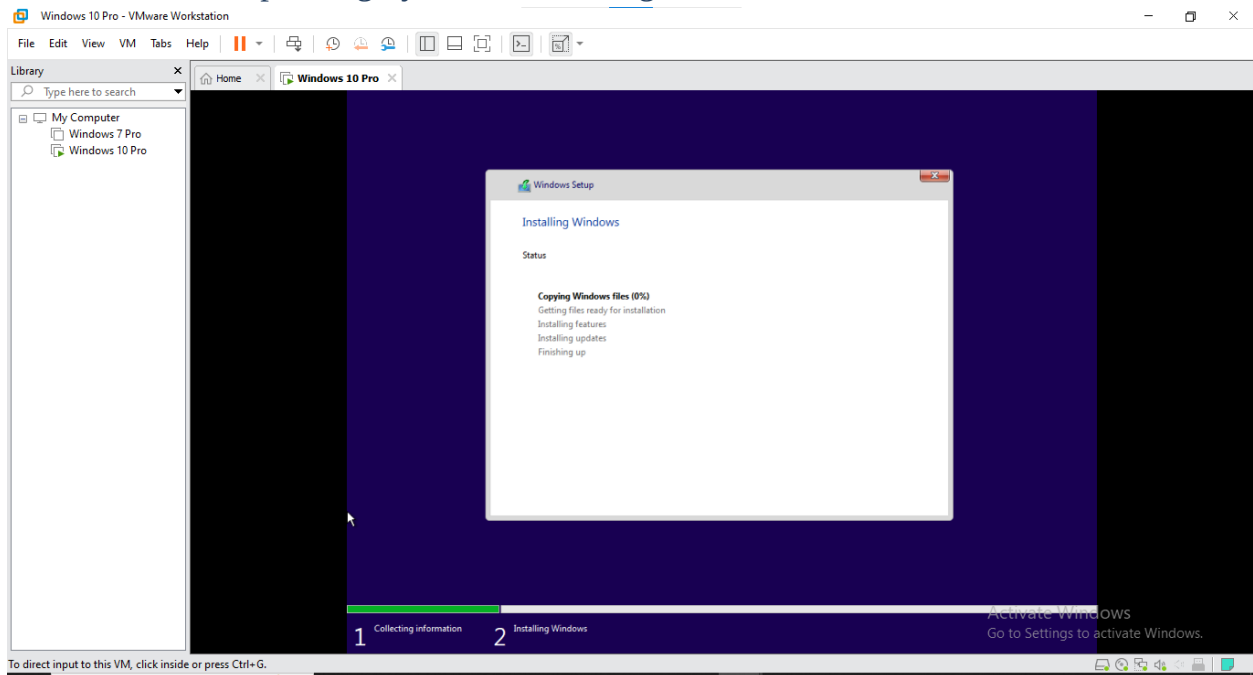


VIRTUALIZED DATA CENTER & DOMAIN SETUP

9.8.4. Selecting Allocated Disk

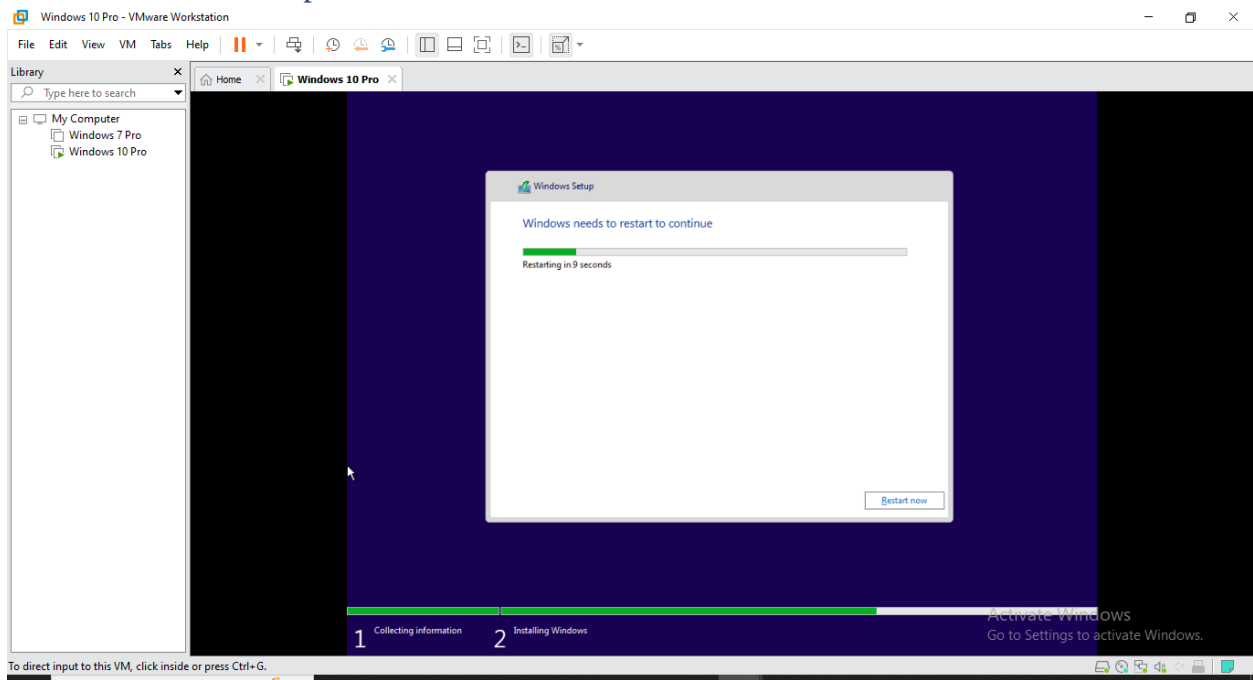


9.8.5. Windows Operating System is Installing

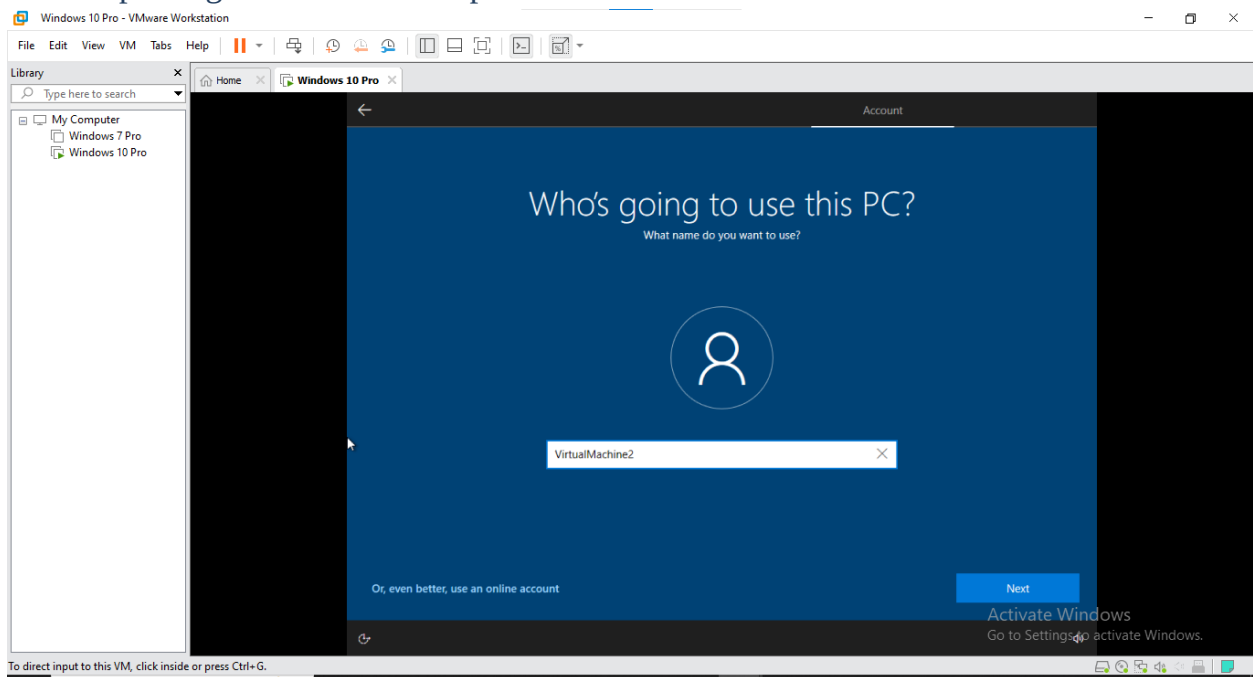


VIRTUALIZED DATA CENTER & DOMAIN SETUP

9.8.6. Restart to Complete Installation

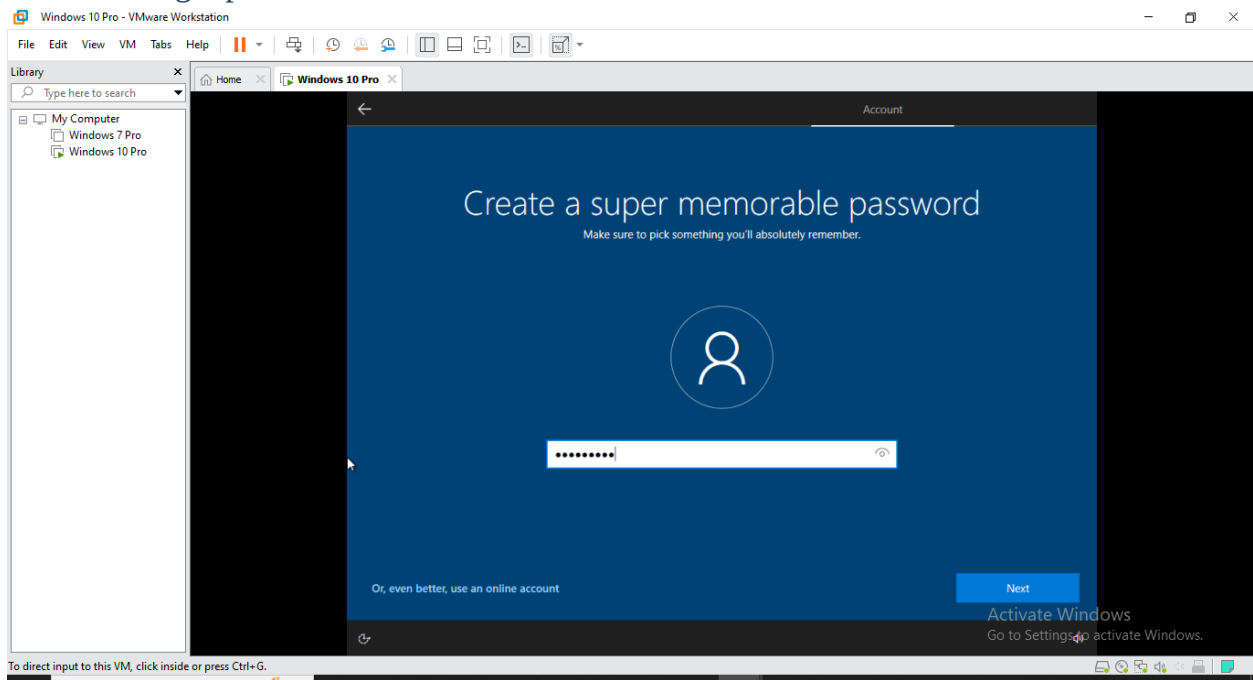


9.8.7. Updating Name for Desktop

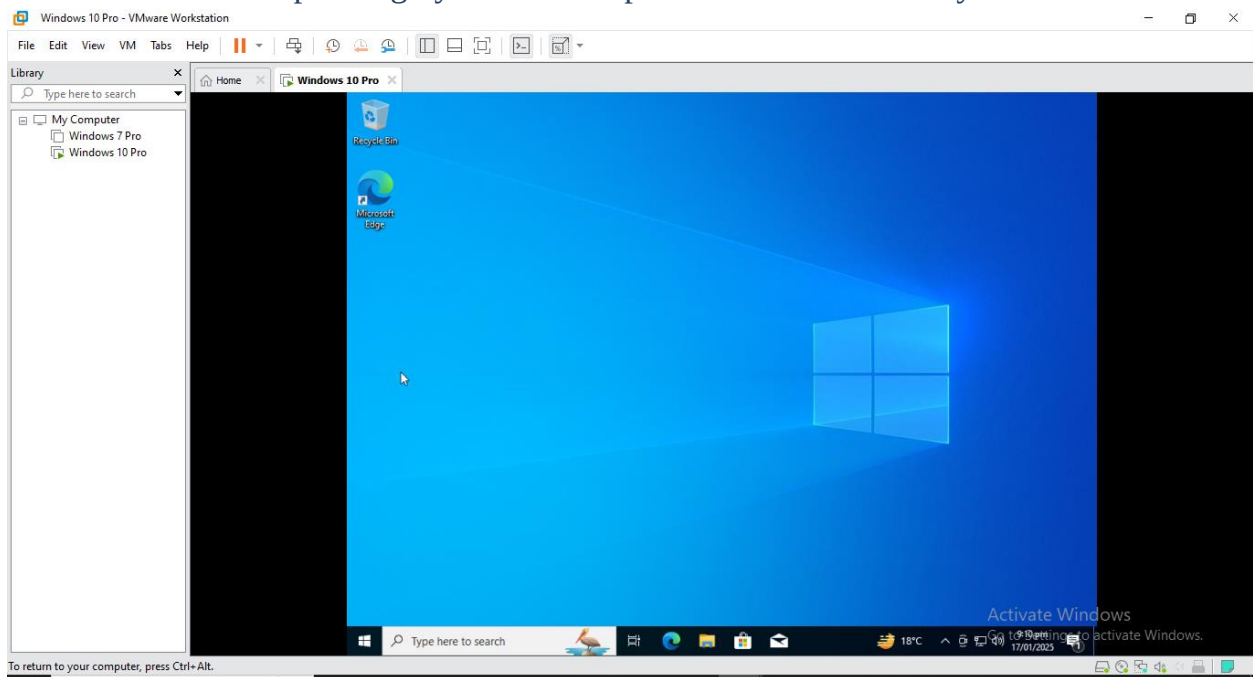


VIRTUALIZED DATA CENTER & DOMAIN SETUP

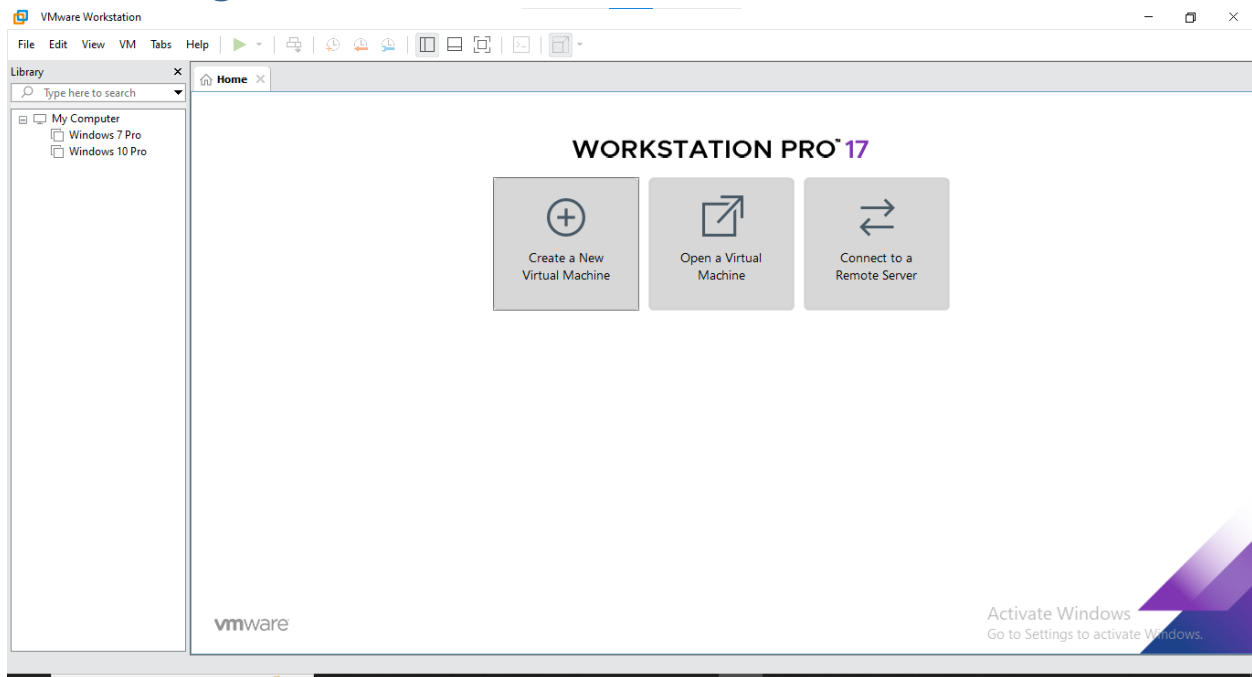
9.8.8. Setting Up Password



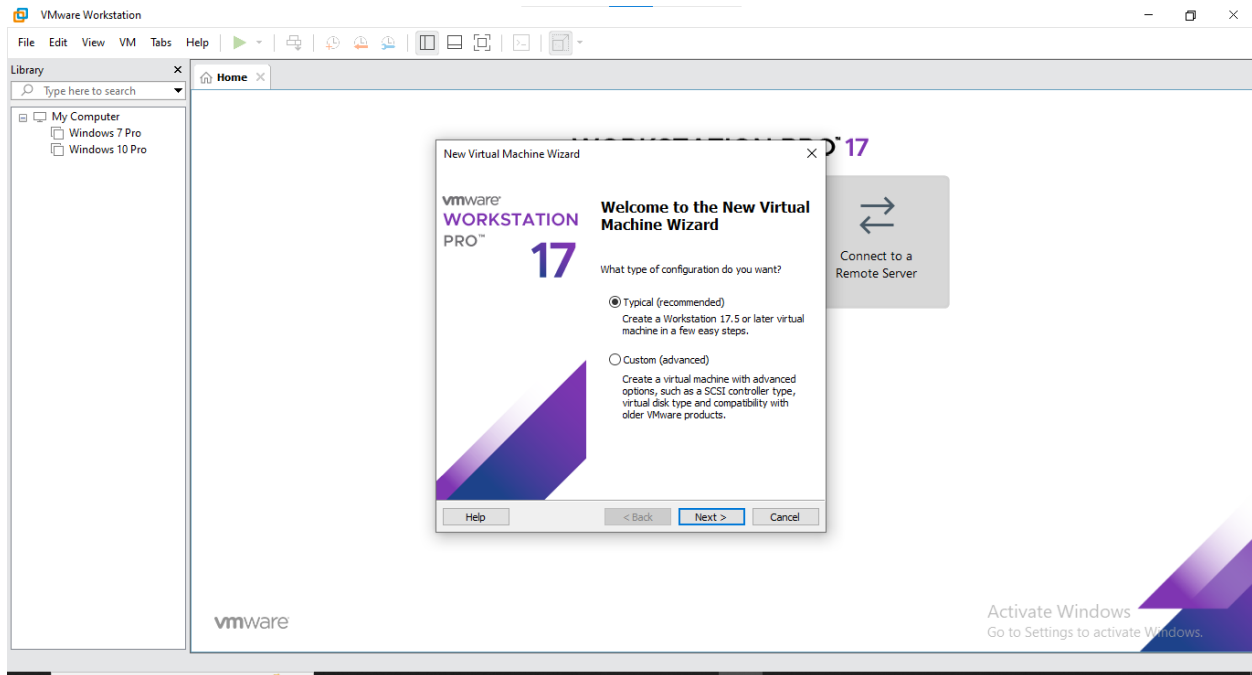
9.8.9. A Windows Operating System Desktop is Created Successfully



10. Creating a 3rd Virtual Machine

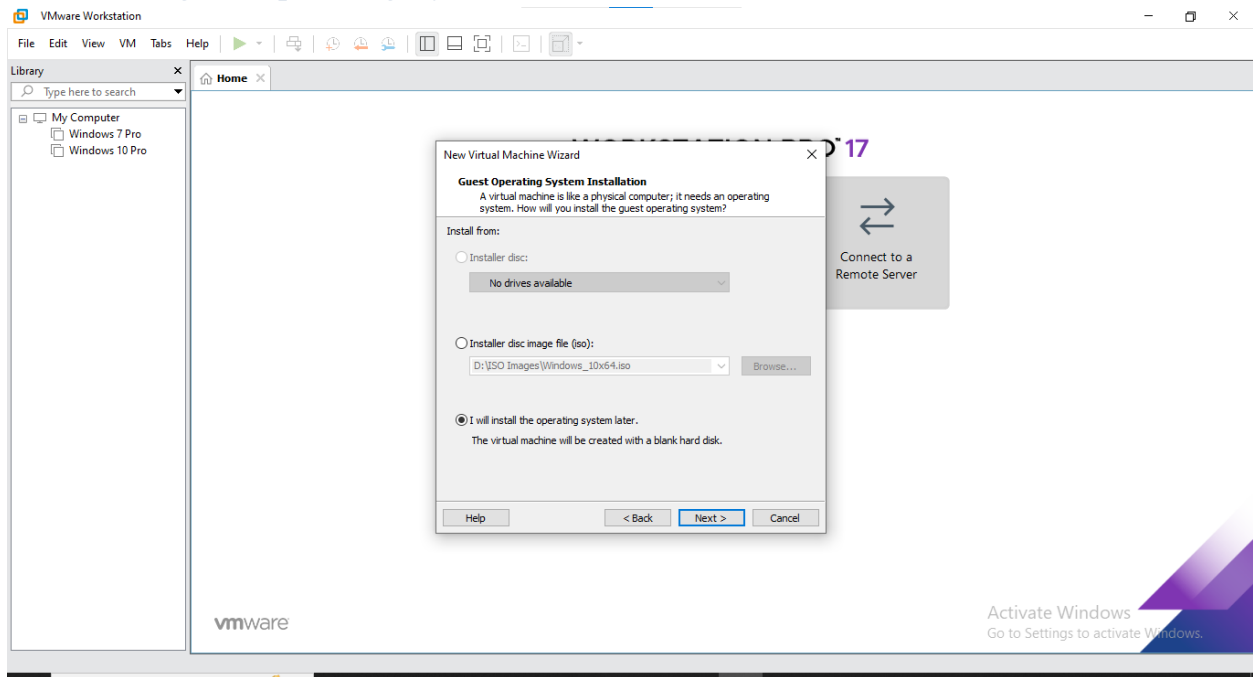


10.1. Creating a 3rd Virtual Machine and Installing ESXi Server

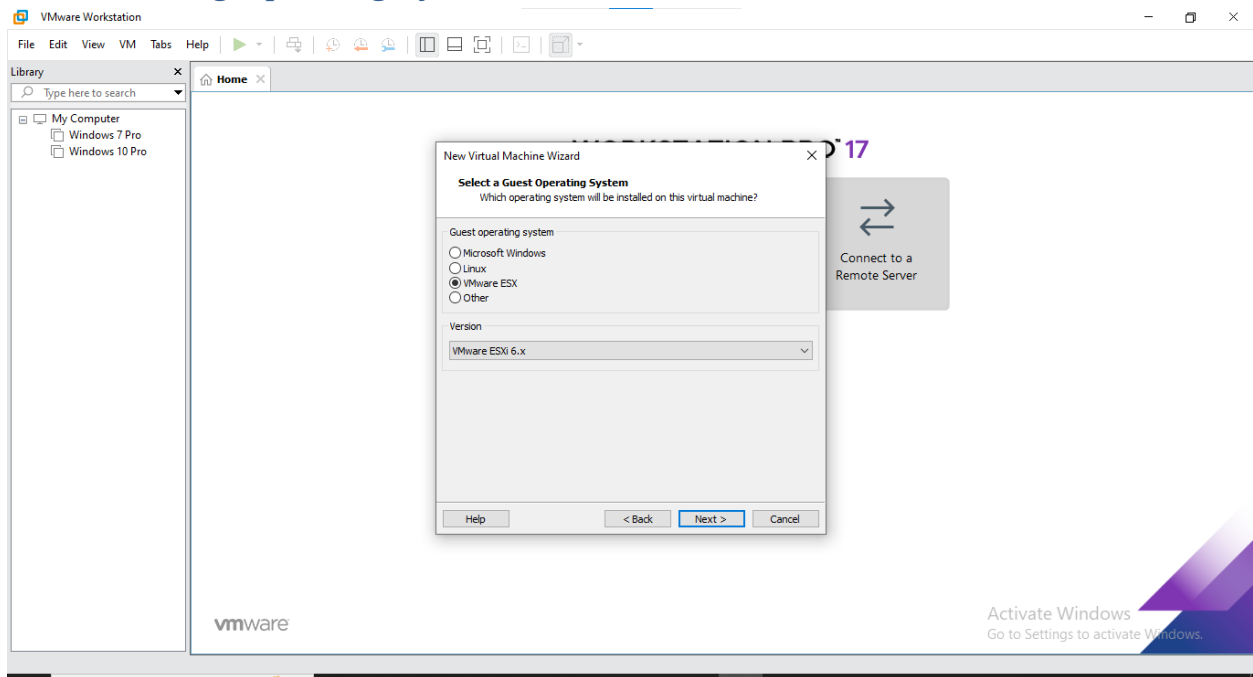


VIRTUALIZED DATA CENTER & DOMAIN SETUP

10.2. Configure Operating System Information

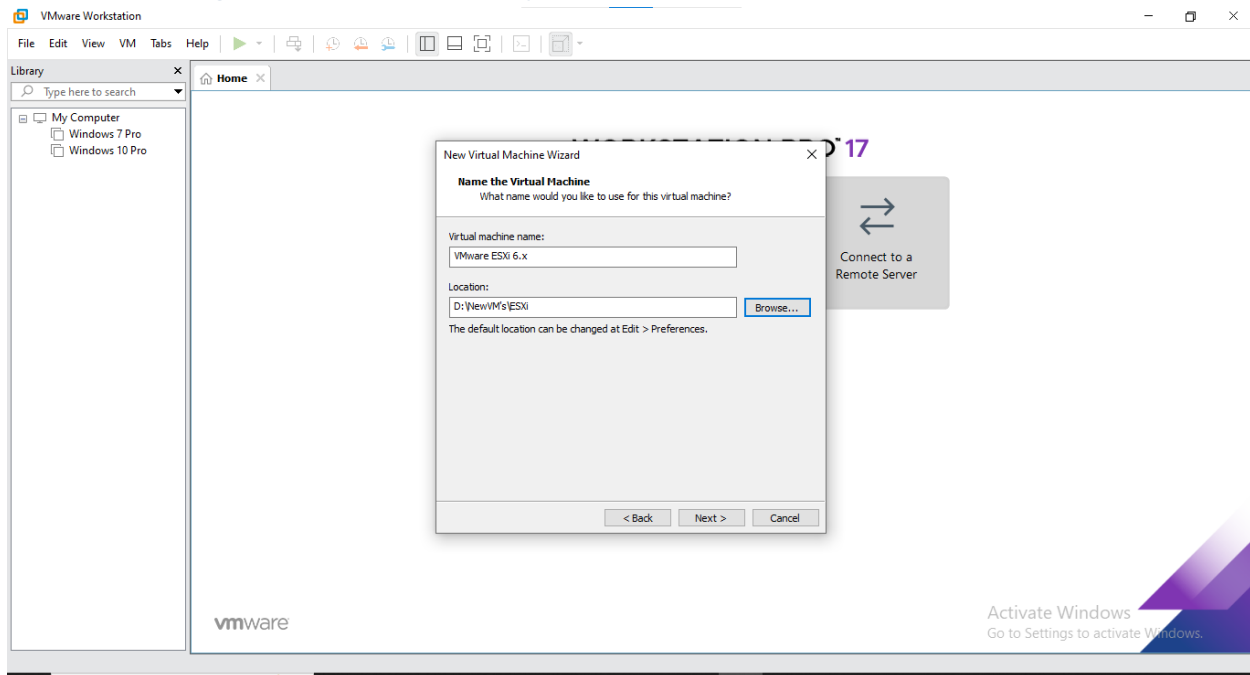


10.3. Selecting Operating System

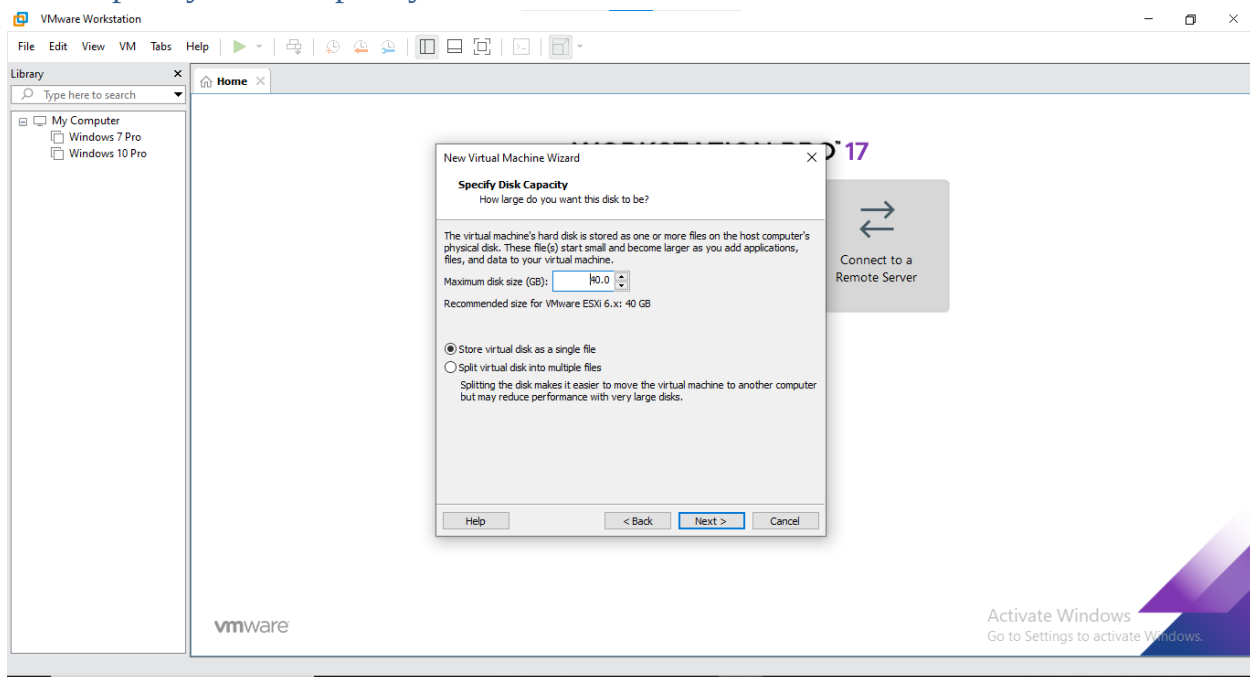


VIRTUALIZED DATA CENTER & DOMAIN SETUP

10.4. Selecting Name and Directory to Store VM

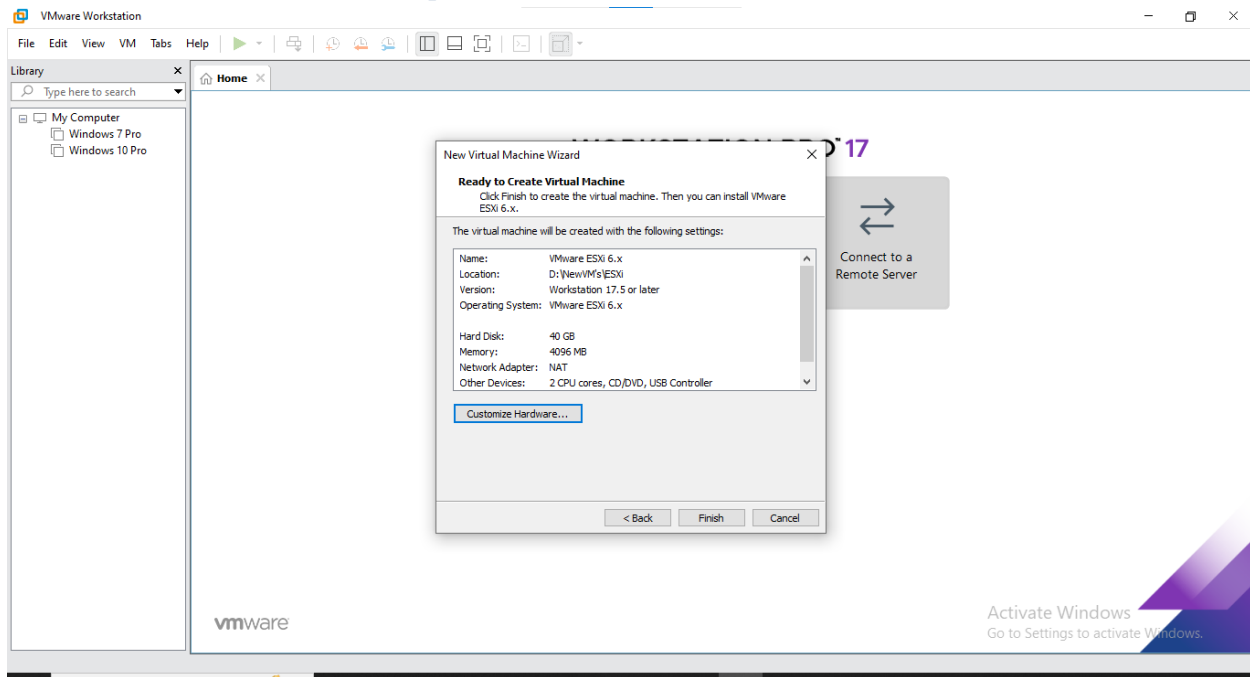


10.5. Specify Disk Capacity

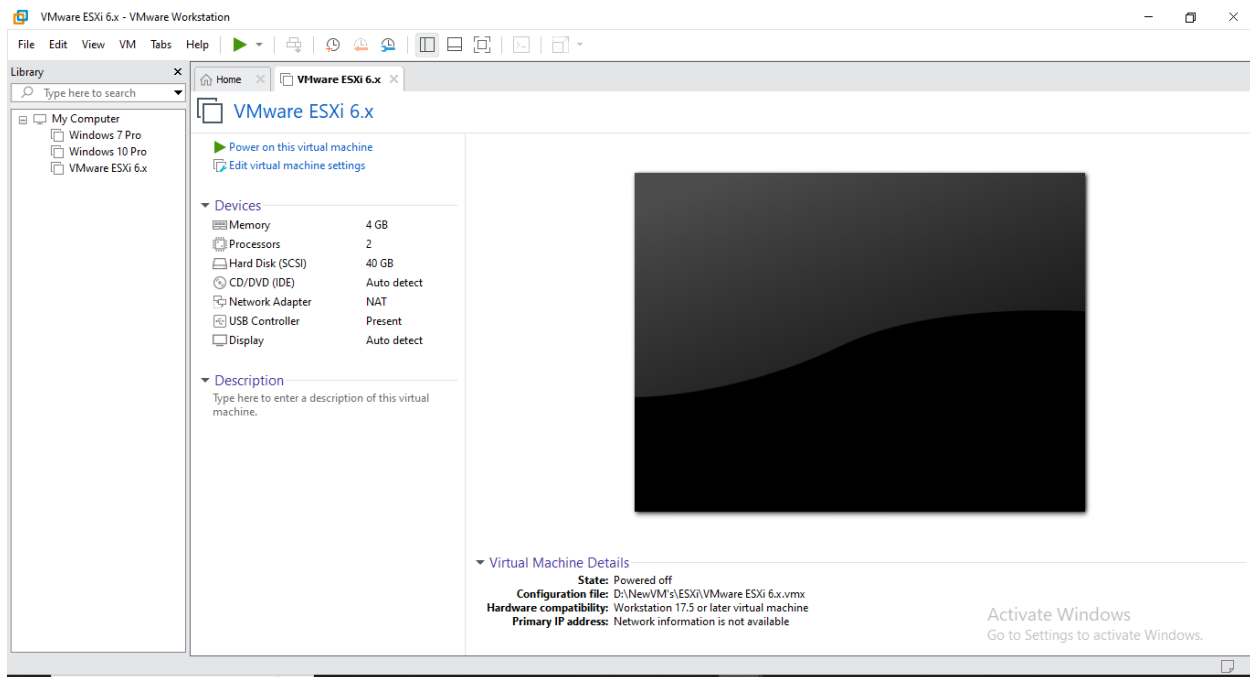


VIRTUALIZED DATA CENTER & DOMAIN SETUP

10.6. Finalize Hardware Requirement

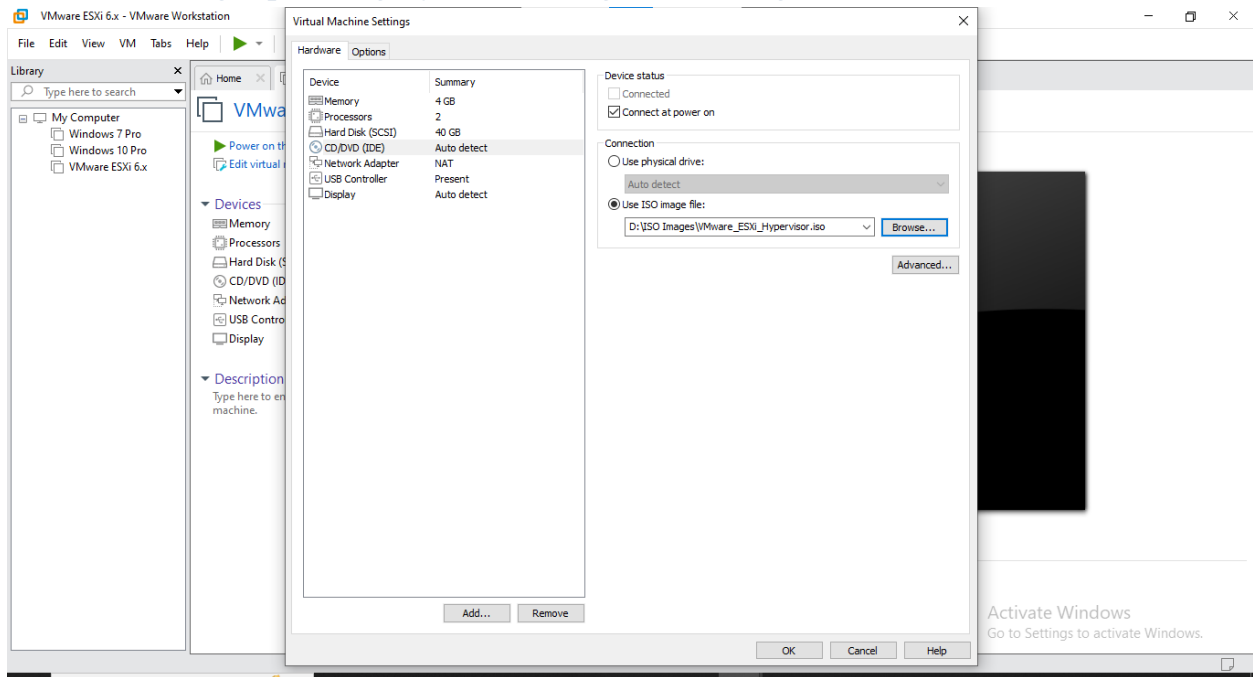


A Virtual Machine is Created

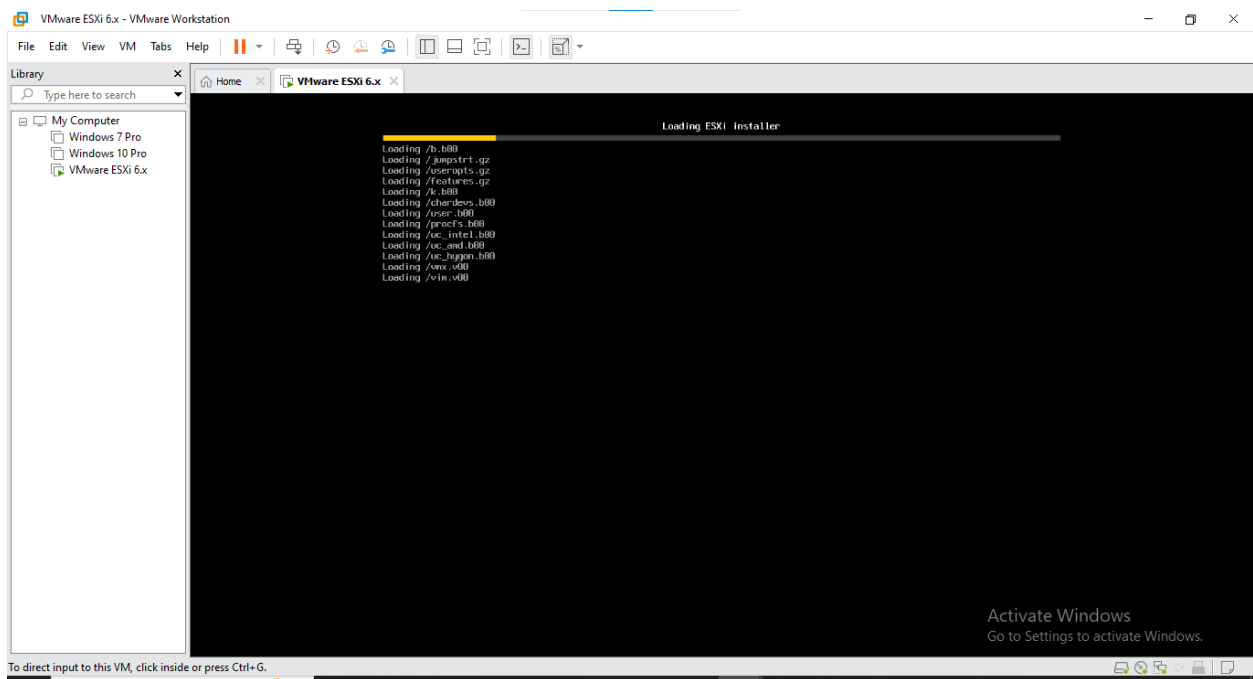


VIRTUALIZED DATA CENTER & DOMAIN SETUP

10.7. Installing Operating System through ISO Image

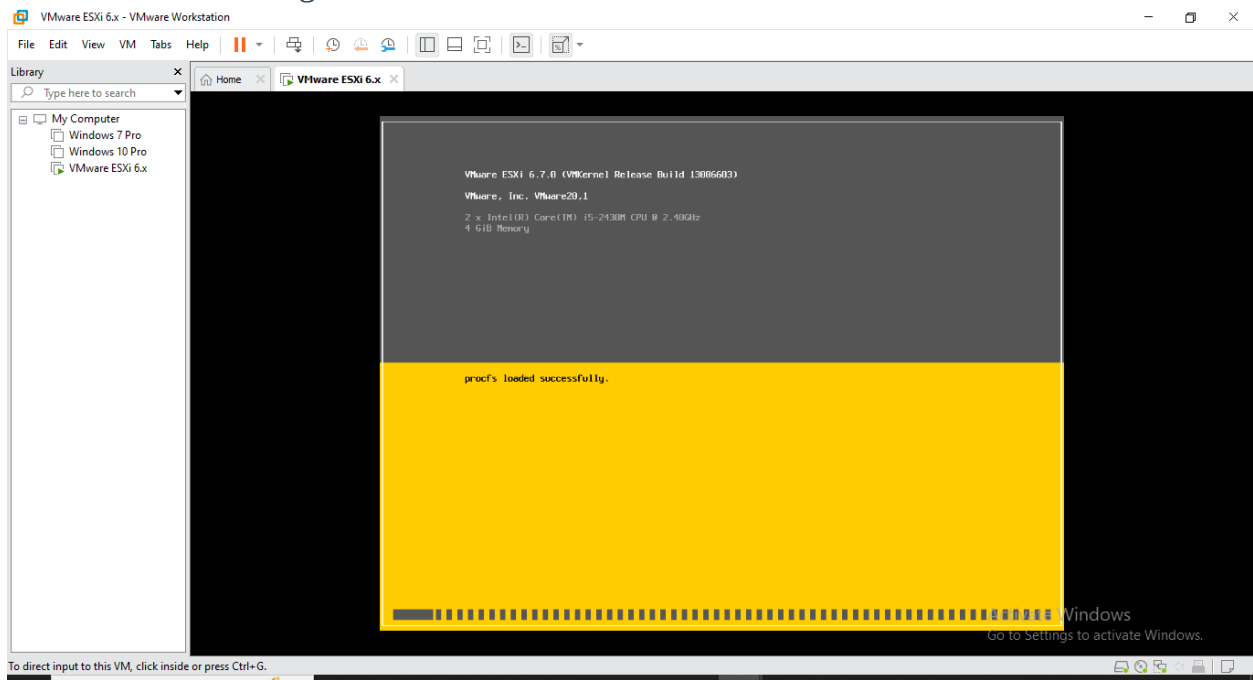


10.8. Power on Virtual Machine to Finish Installation

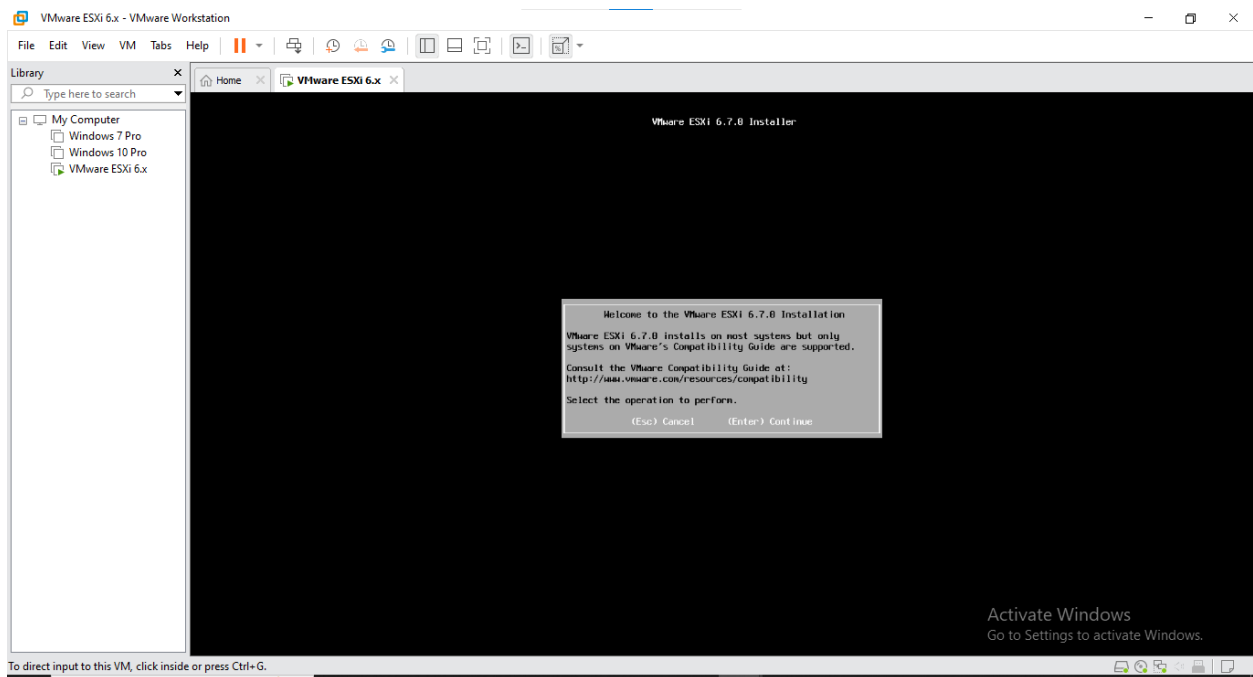


VIRTUALIZED DATA CENTER & DOMAIN SETUP

10.8.1. Start Installing ESXi Server

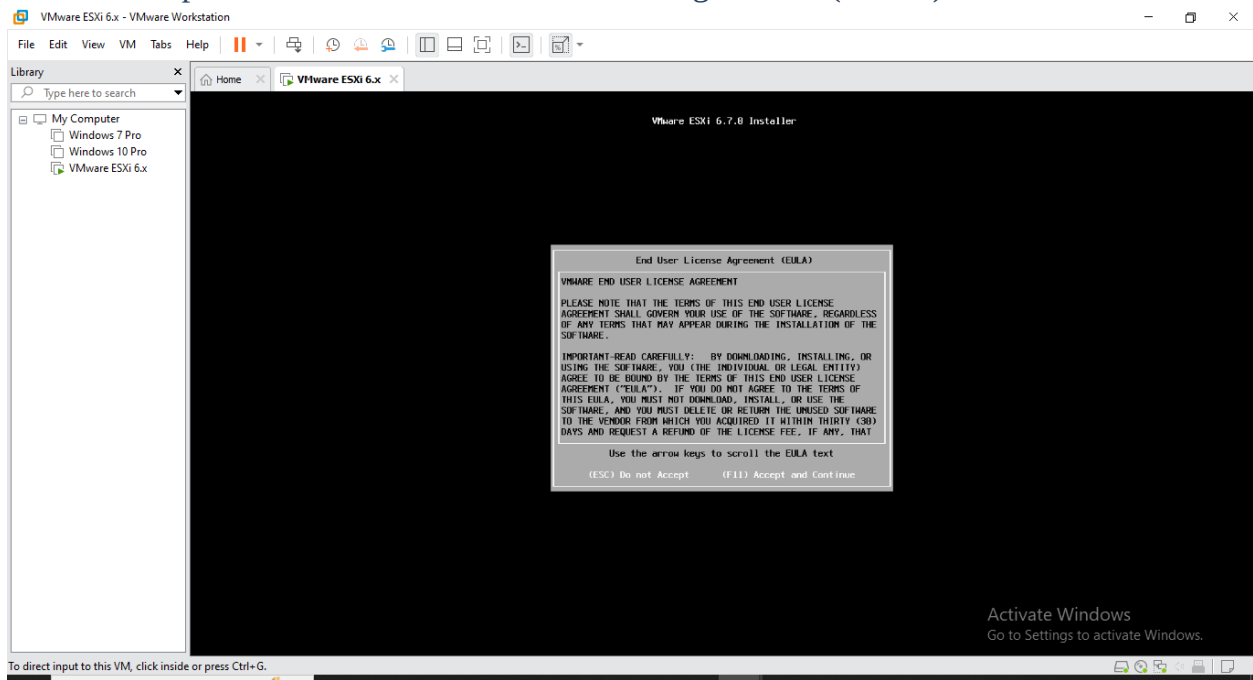


10.8.2. Click Enter to Continue Installation

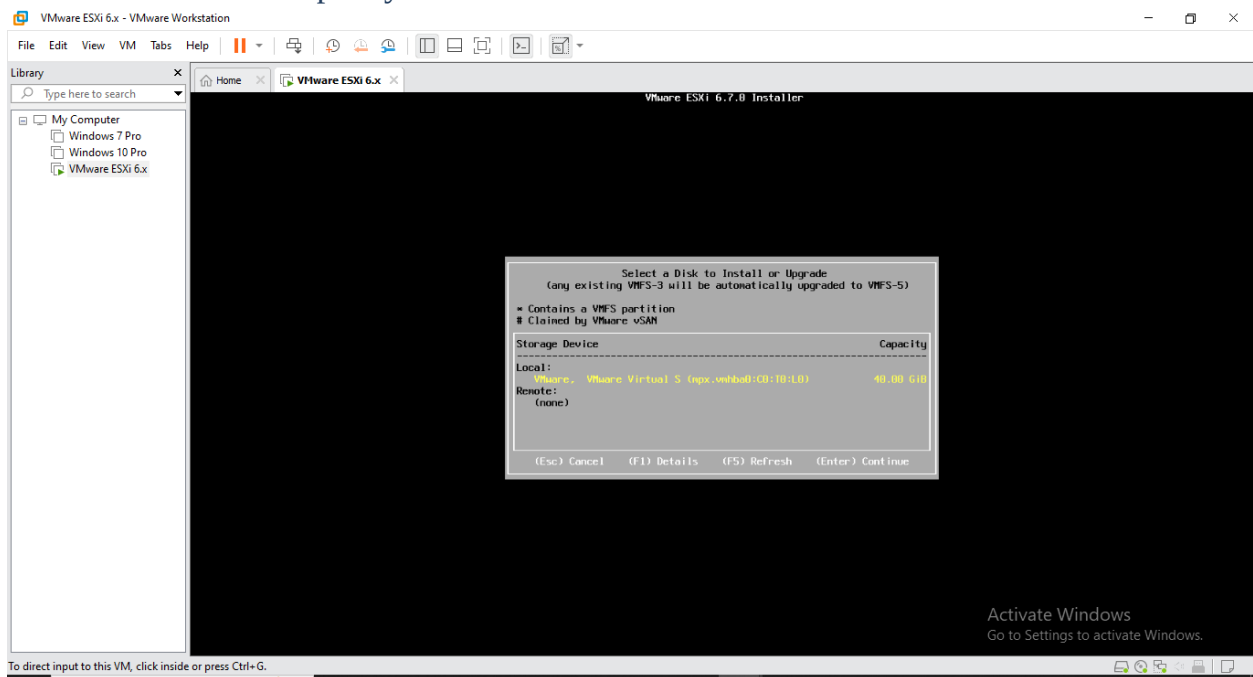


VIRTUALIZED DATA CENTER & DOMAIN SETUP

10.8.3. Accept and Continue End User License Agreement (EULA)

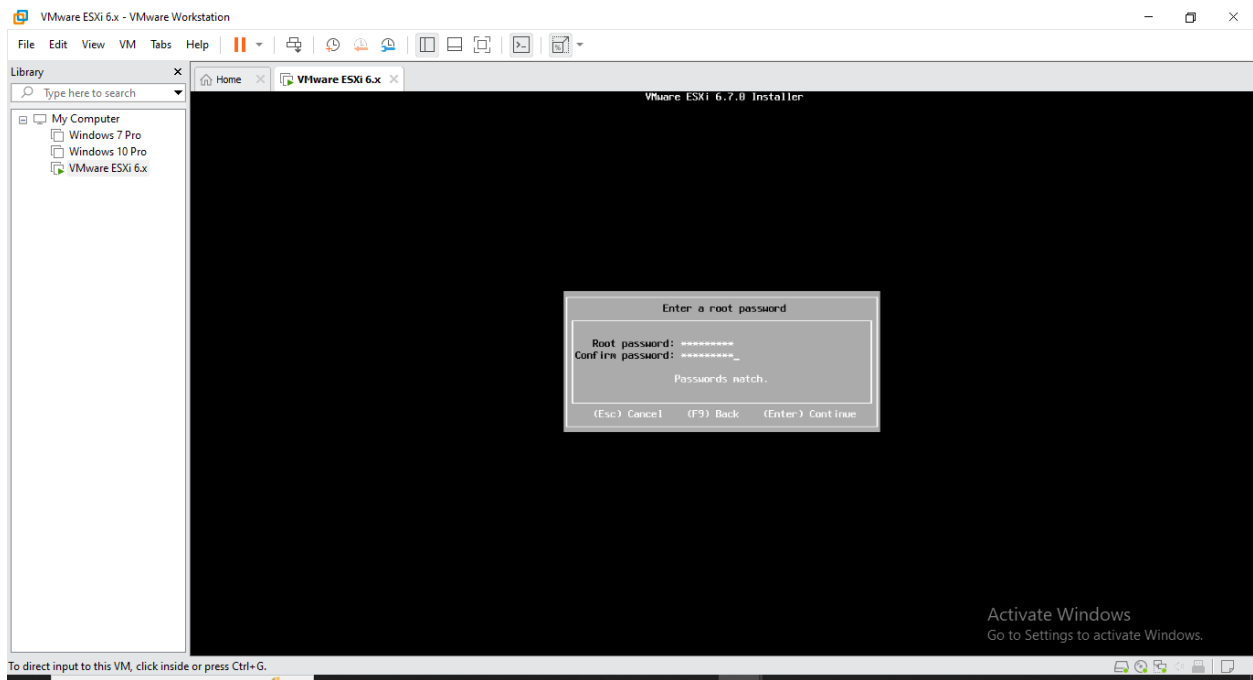


10.8.4. Select Disk Capacity

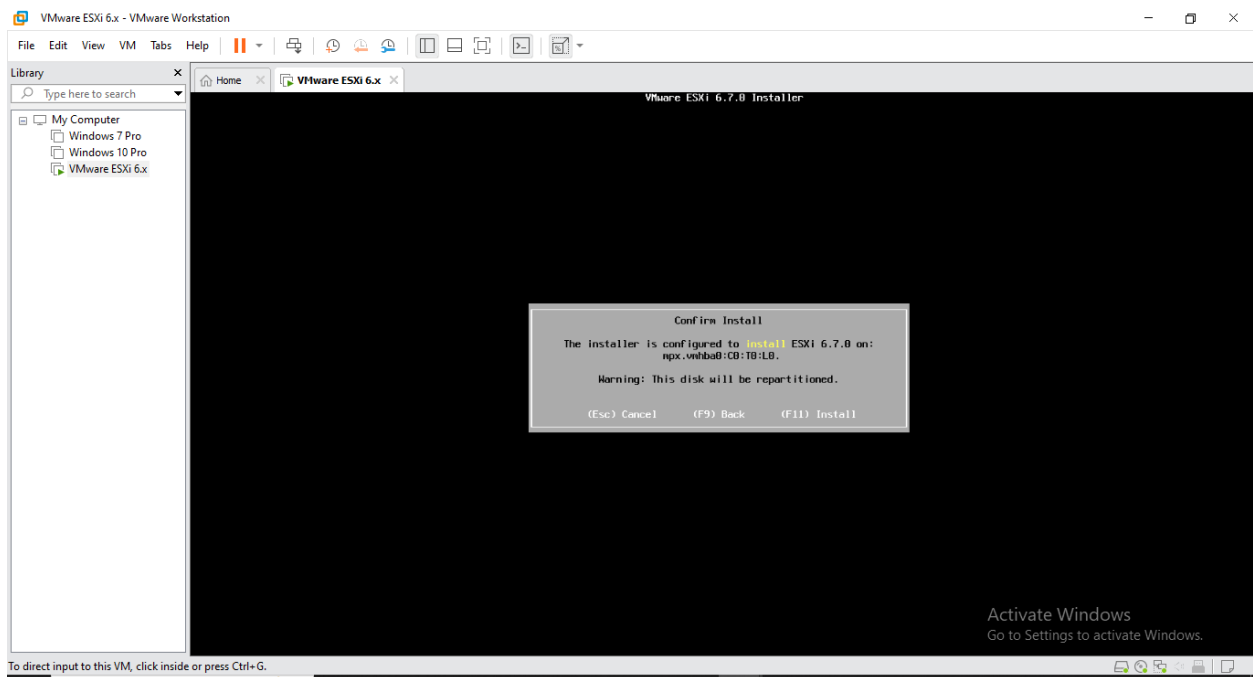


VIRTUALIZED DATA CENTER & DOMAIN SETUP

10.8.5. Enter Root Password

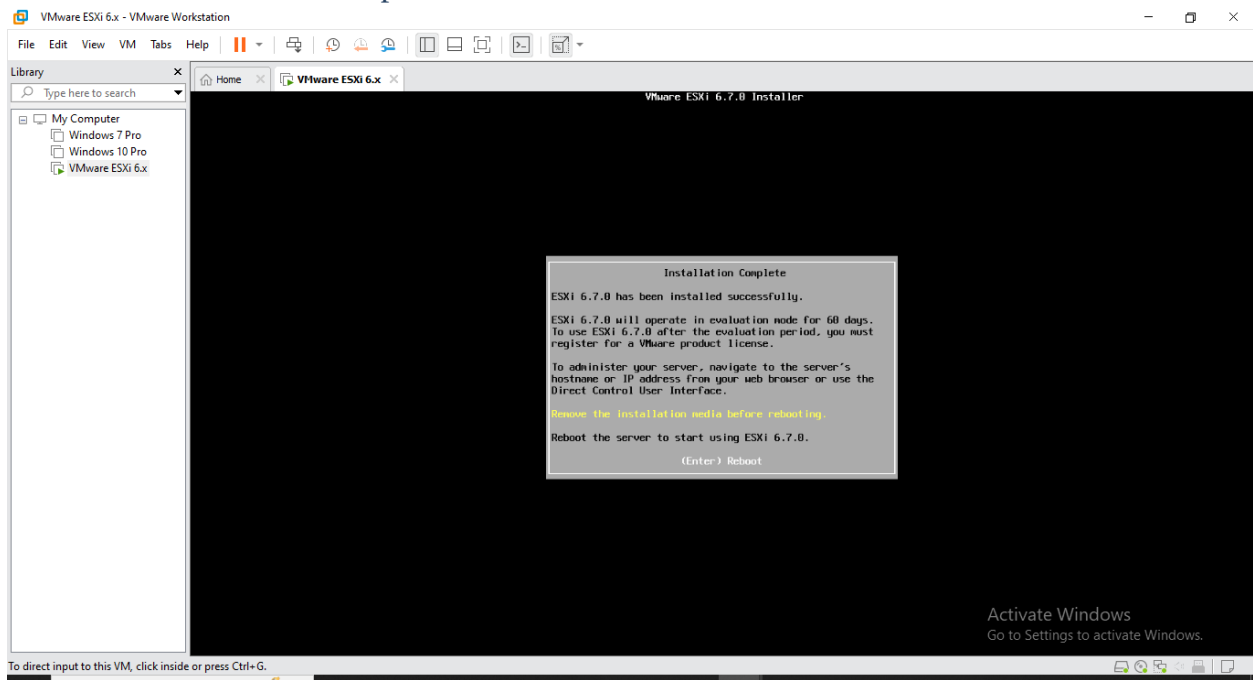


10.8.6. Install ESXi Server

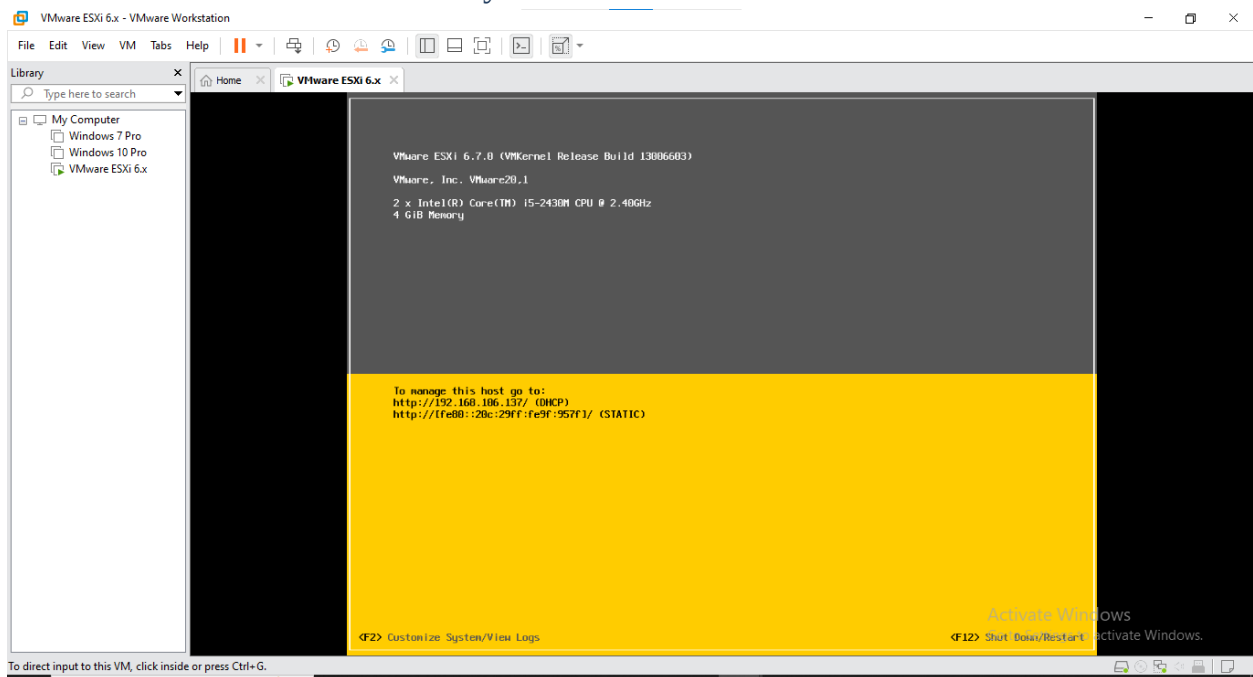


VIRTUALIZED DATA CENTER & DOMAIN SETUP

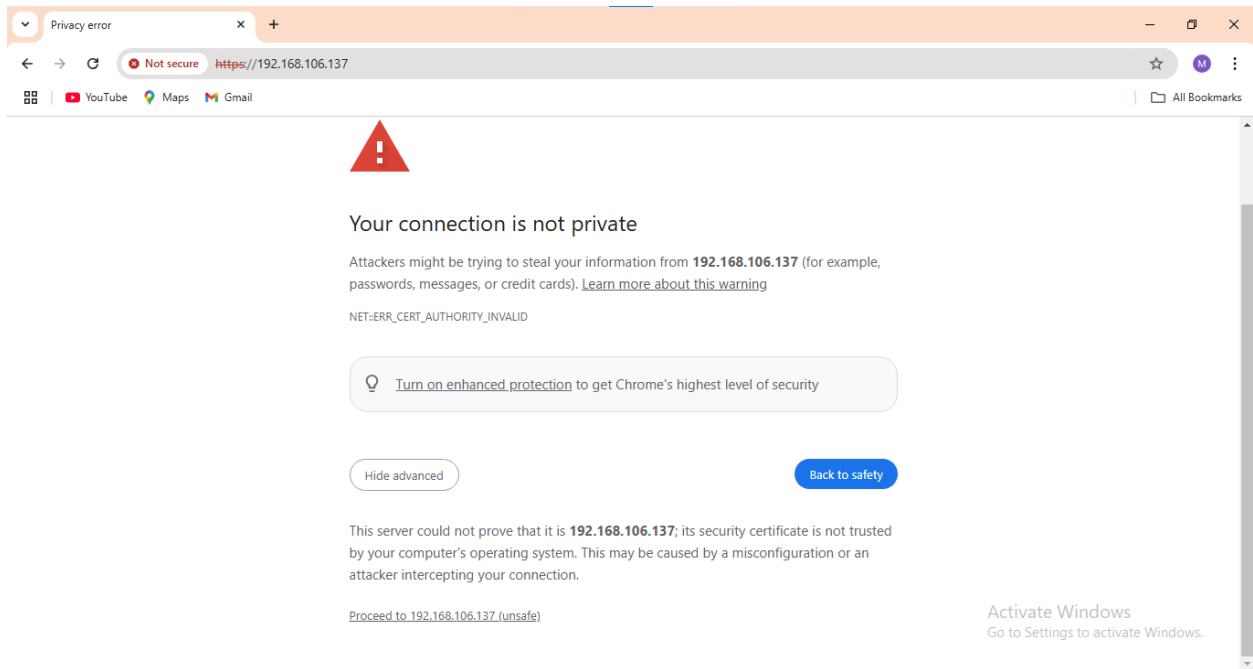
10.8.7. Click Enter to Complete Installation



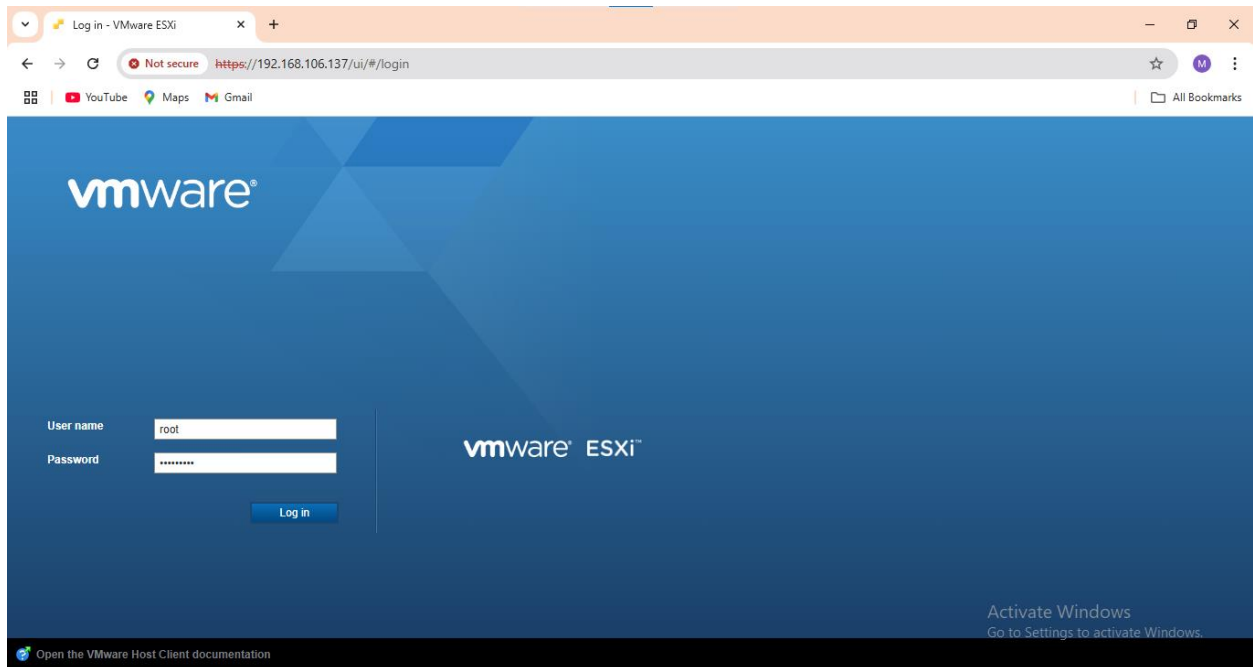
10.8.8. ESXi Server is Successfully Installed



11. Accessing ESXi Server through Web Server (Chrome Brower)



11.1. Enter Username and Password to Login



VIRTUALIZED DATA CENTER & DOMAIN SETUP

11.2. Dashboard of ESXi Server

The screenshot shows the VMware ESXi web interface for the host 'localhost.localdomain'. The left sidebar contains a 'Navigator' with links to Host, Virtual Machines, Storage, and Networking. The main content area displays host information, including version (6.7.0 Update 2), state (Normal), and uptime (0 days). It also shows resource usage for CPU, Memory, and Storage. A notification bar indicates the system is in evaluation mode. Below this, there are sections for Hardware (Manufacturer: VMware, Inc.) and Configuration (Image profile: ESXi-6.7.0-20190402001-standard). A 'Recent tasks' table is visible at the bottom.

Task	Target	Initiator	Queued	Started	Result	Completed
Update Options	localhost.localdomain	root	01/17/2025 21:55:13	01/17/2025 21:55:13	Completed successfully	01/17/2025 21:55:13
Auto Start Power On	localhost.localdomain	root	01/17/2025 21:50:40	01/17/2025 21:50:40	Completed successfully	01/17/2025 21:50:40

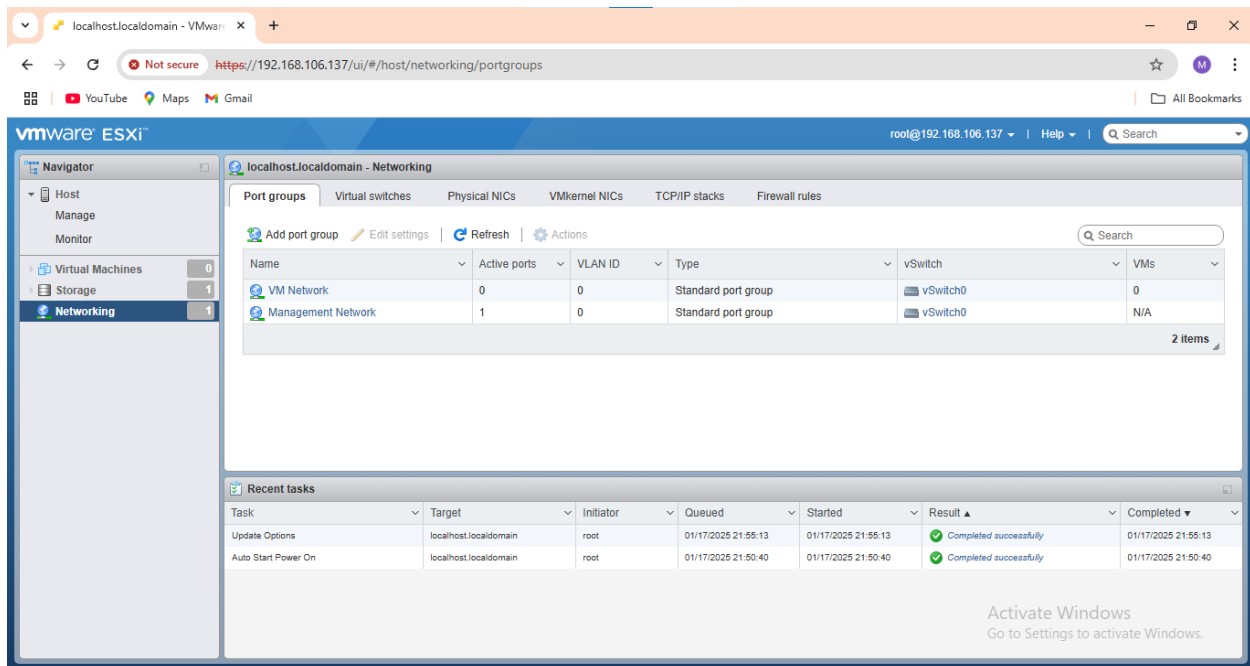
11.3. ESXi Server Datastore Information

The screenshot shows the 'Storage' page in the VMware ESXi web interface. The left sidebar is the same as in the previous screenshot. The main content area displays a table of datastores. The 'Recent tasks' table is also visible at the bottom.

Name	Drive Type	Capacity	Provisioned	Free	Type	Thin provisio...	Access
datastore1	Non-SSD	32.5 GB	1.41 GB	31.09 GB	VMFS6	Supported	Single

VIRTUALIZED DATA CENTER & DOMAIN SETUP

11.4. ESXi Server Network Information

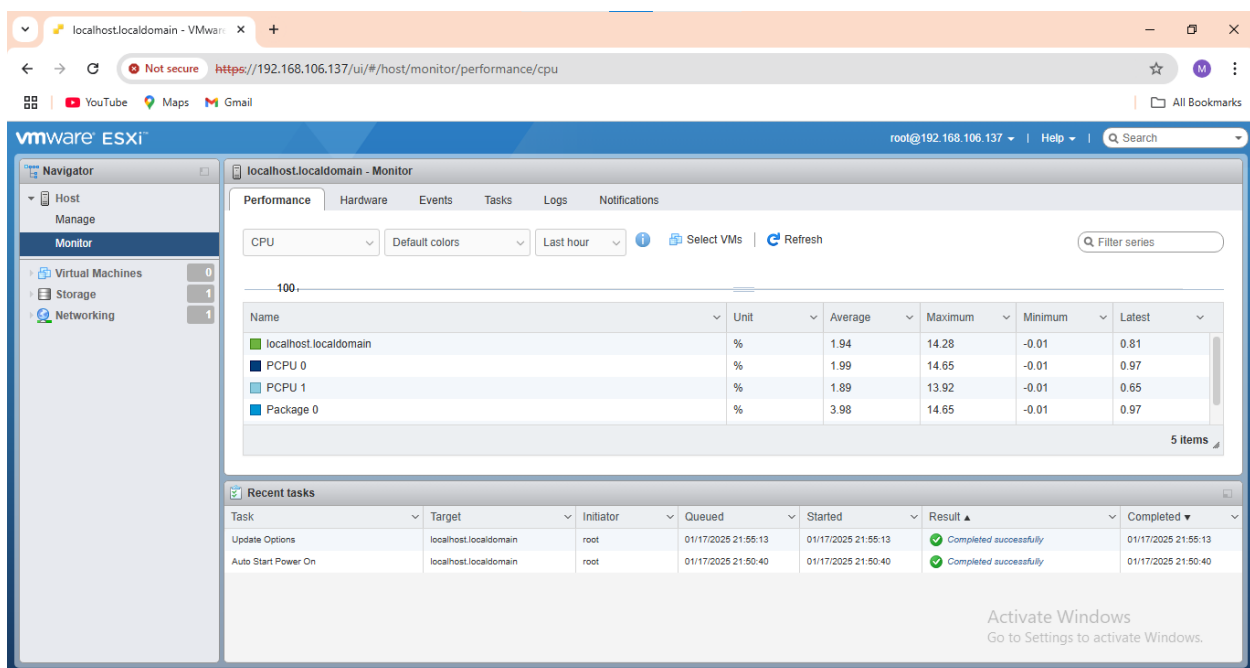


The screenshot shows the VMware ESXi web interface for the host 'localhost.localdomain'. The 'Networking' tab is selected in the left-hand menu. The main content area displays the 'Port groups' section, which includes a table of configured port groups. Below this, the 'Recent tasks' section shows a list of completed tasks. An 'Activate Windows' watermark is visible in the bottom right corner.

Name	Active ports	VLAN ID	Type	vSwitch	VMs
VM Network	0	0	Standard port group	vSwitch0	0
Management Network	1	0	Standard port group	vSwitch0	N/A

Task	Target	Initiator	Queued	Started	Result	Completed
Update Options	localhost.localdomain	root	01/17/2025 21:55:13	01/17/2025 21:55:13	Completed successfully	01/17/2025 21:55:13
Auto Start Power On	localhost.localdomain	root	01/17/2025 21:50:40	01/17/2025 21:50:40	Completed successfully	01/17/2025 21:50:40

11.5. Monitor ESXi Server



The screenshot shows the VMware ESXi web interface for the host 'localhost.localdomain', specifically the 'Monitor' tab. The 'Performance' section is active, displaying a table of CPU performance metrics for the host and its components. Below this, the 'Recent tasks' section shows a list of completed tasks. An 'Activate Windows' watermark is visible in the bottom right corner.

Name	Unit	Average	Maximum	Minimum	Latest
localhost.localdomain	%	1.94	14.28	-0.01	0.81
PCPU 0	%	1.99	14.65	-0.01	0.97
PCPU 1	%	1.89	13.92	-0.01	0.65
Package 0	%	3.98	14.65	-0.01	0.97

Task	Target	Initiator	Queued	Started	Result	Completed
Update Options	localhost.localdomain	root	01/17/2025 21:55:13	01/17/2025 21:55:13	Completed successfully	01/17/2025 21:55:13
Auto Start Power On	localhost.localdomain	root	01/17/2025 21:50:40	01/17/2025 21:50:40	Completed successfully	01/17/2025 21:50:40

VIRTUALIZED DATA CENTER & DOMAIN SETUP

11.6. Manage ESXi Server

The screenshot shows the VMware ESXi Advanced Settings page. The left sidebar contains a 'Navigator' with 'Host' selected, and 'Manage' and 'Monitor' sub-sections. Under 'Manage', 'Virtual Machines' (0), 'Storage' (1), and 'Networking' (1) are listed. The main content area is titled 'localhost.localdomain - Manage' and has tabs for 'System', 'Hardware', 'Licensing', 'Packages', 'Services', and 'Security & users'. The 'Advanced settings' section is active, showing a table of settings. Below this is a 'Recent tasks' section with a table of tasks.

Key	Name	Value	Default	Overridden
Autostart				
Swap				
Time & date				
Annotations.WelcomeMessage	A welcome message in the initial screen o...			False
BufferCache.FlushInterval	Flush at this interval (milliseconds)	30000	30000	False

1067 items

Task	Target	Initiator	Queued	Started	Result	Completed
Update Options	localhost.localdomain	root	01/17/2025 21:55:13	01/17/2025 21:55:13	Completed successfully	01/17/2025 21:55:13
Auto Start Power On	localhost.localdomain	root	01/17/2025 21:50:40	01/17/2025 21:50:40	Completed successfully	01/17/2025 21:50:40

Activate Windows
Go to Settings to activate Windows.

11.7. Virtual Machine Dashboard on ESXi Server

The screenshot shows the VMware ESXi Virtual Machines dashboard. The left sidebar is the same as in the previous screenshot. The main content area is titled 'localhost.localdomain - Virtual Machines' and has tabs for 'Create / Register VM', 'Console', 'Power on', 'Power off', 'Suspend', 'Refresh', and 'Actions'. Below this is a table of virtual machines. The 'Recent tasks' section is also present.

Virtual machine	Status	Used space	Guest OS	Host name	Host CPU	Host memory
No virtual machines						

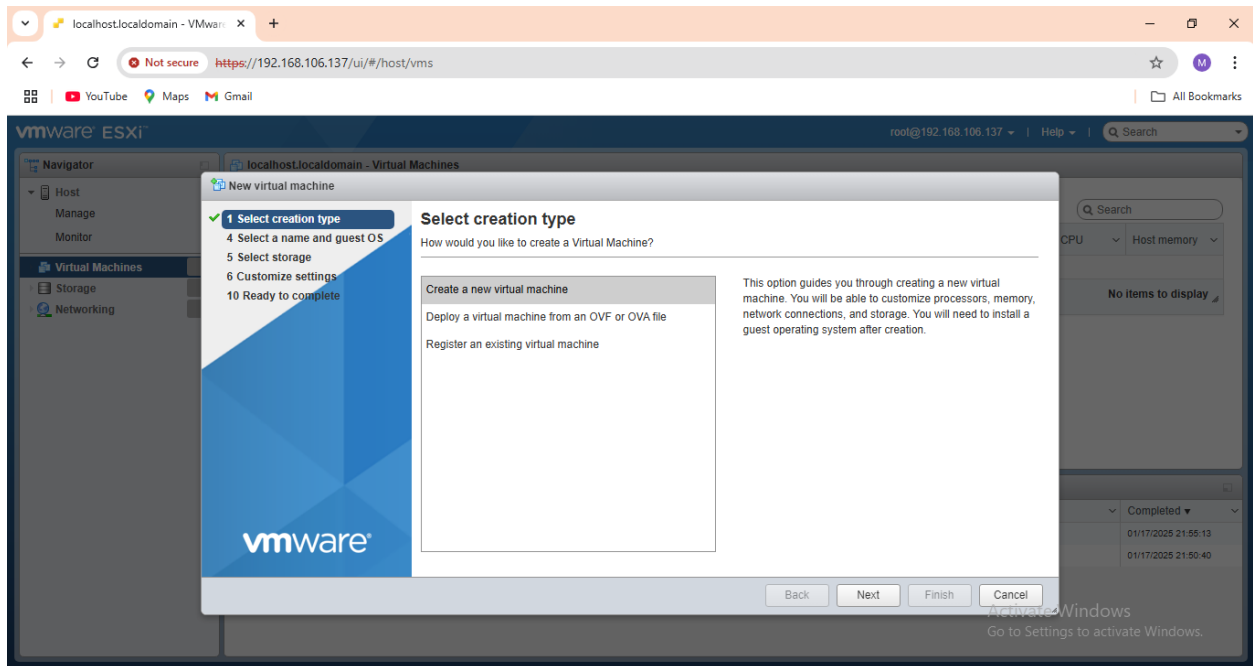
No items to display

Task	Target	Initiator	Queued	Started	Result	Completed
Update Options	localhost.localdomain	root	01/17/2025 21:55:13	01/17/2025 21:55:13	Completed successfully	01/17/2025 21:55:13
Auto Start Power On	localhost.localdomain	root	01/17/2025 21:50:40	01/17/2025 21:50:40	Completed successfully	01/17/2025 21:50:40

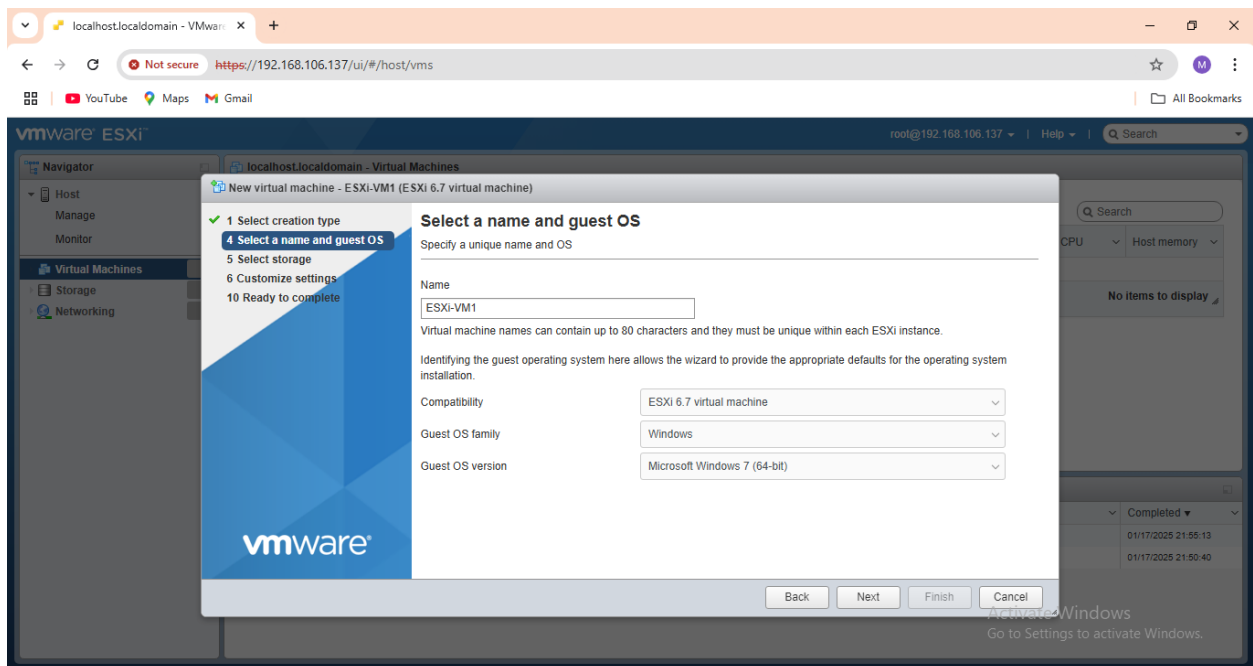
Activate Windows
Go to Settings to activate Windows.

VIRTUALIZED DATA CENTER & DOMAIN SETUP

11.7.1. Creating a Virtual Machine on ESXi Server

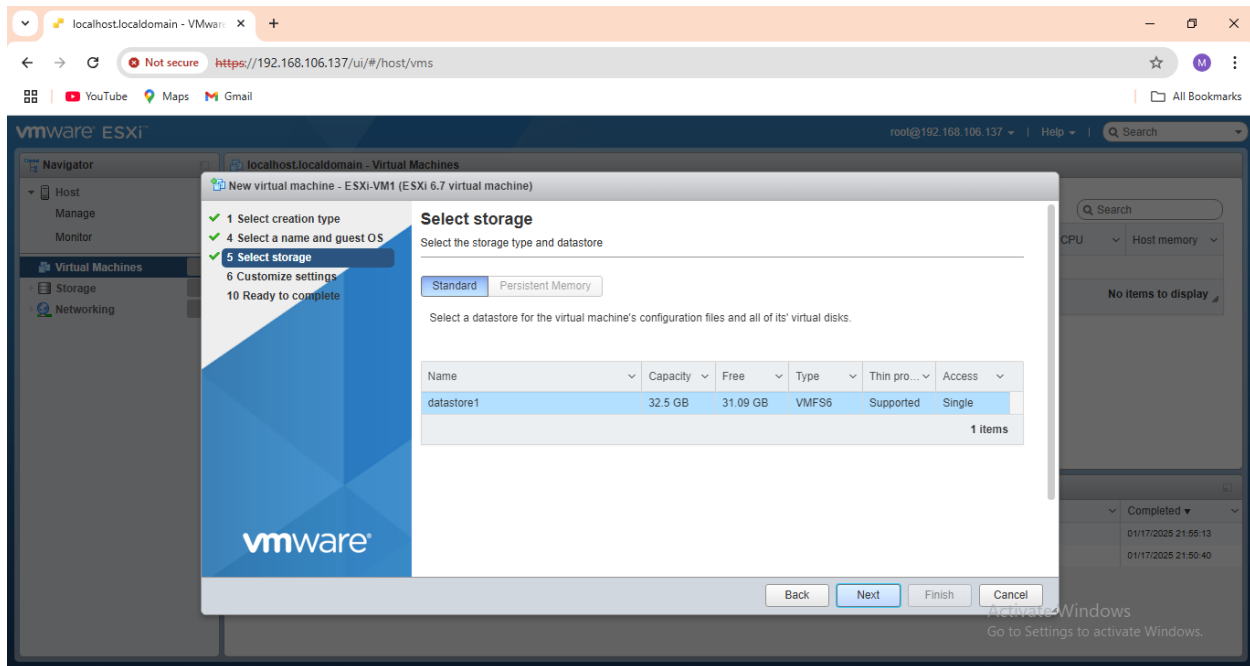


11.7.2. Selecting Name and Guest Operating System (Windows 7 Pro)

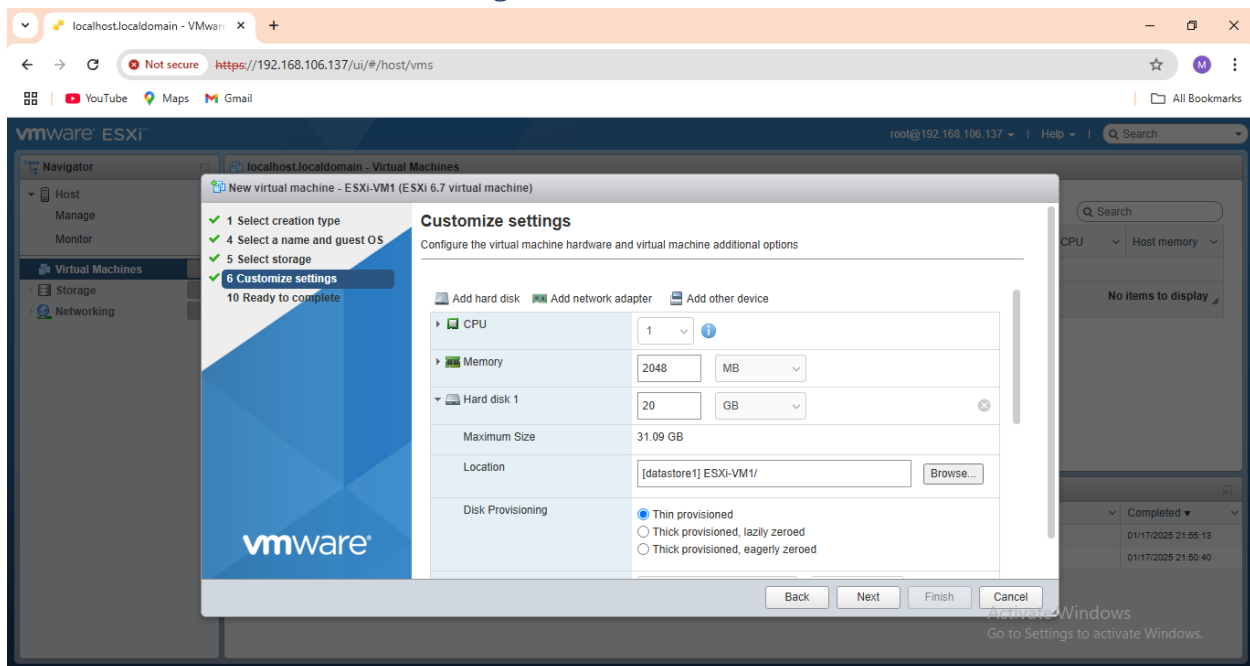


VIRTUALIZED DATA CENTER & DOMAIN SETUP

11.7.3. Selecting Datastore

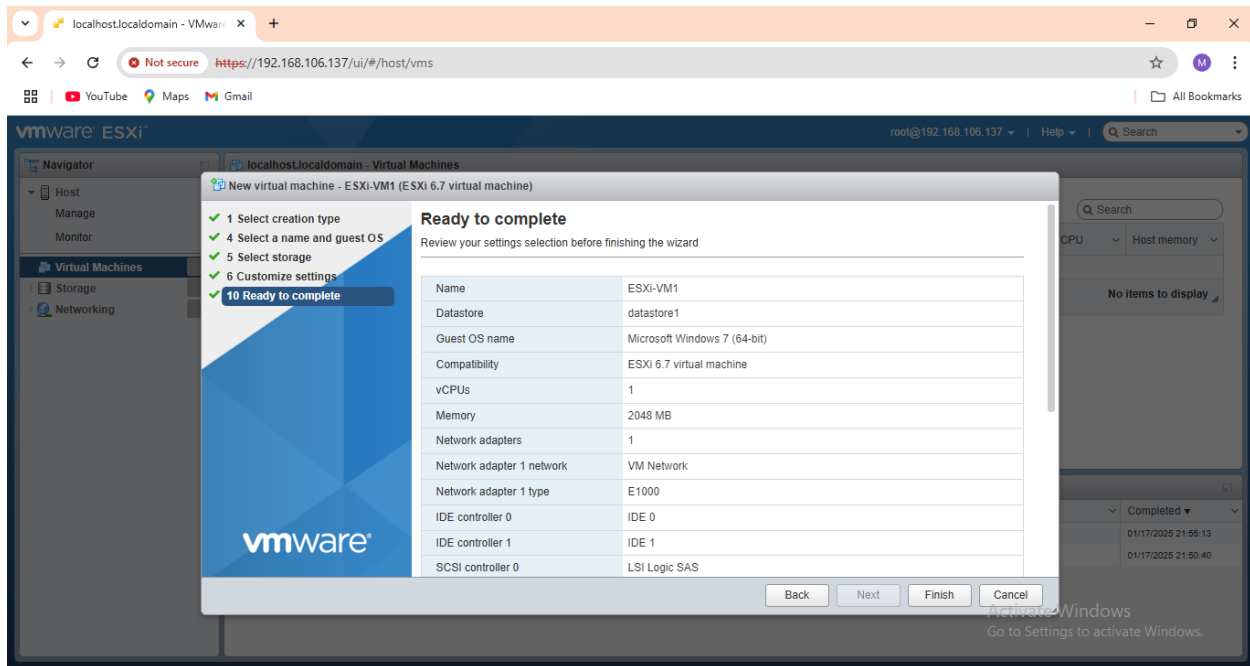


11.7.4. Customize Hardware Configuration

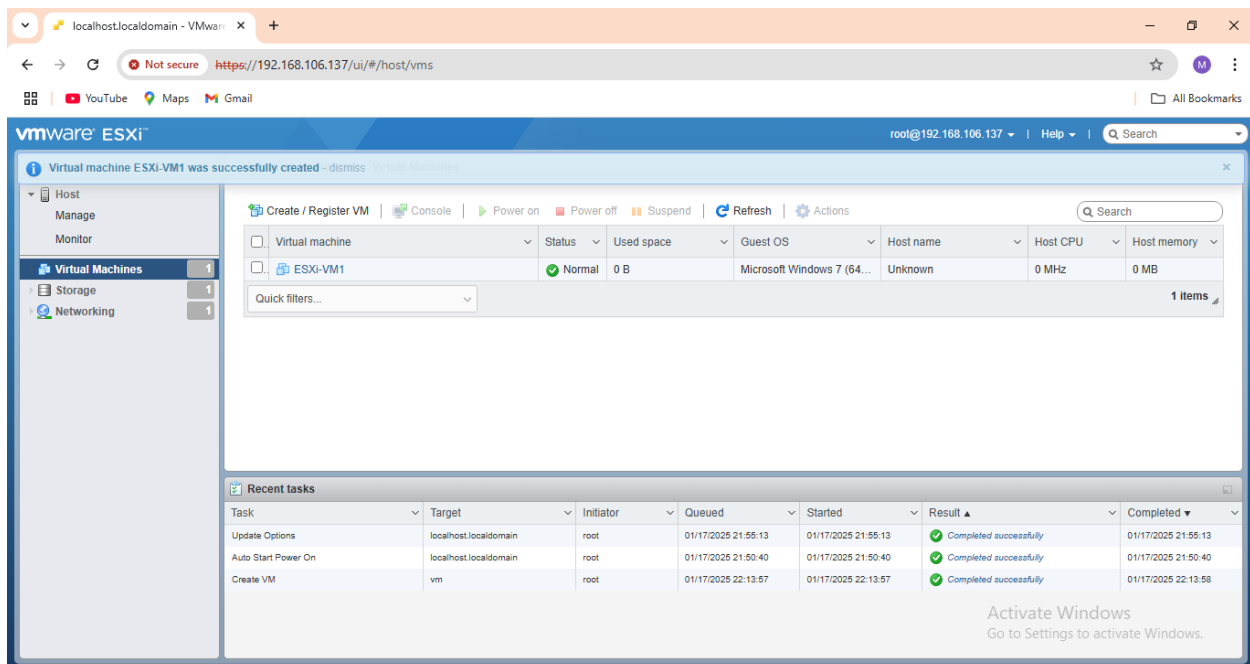


VIRTUALIZED DATA CENTER & DOMAIN SETUP

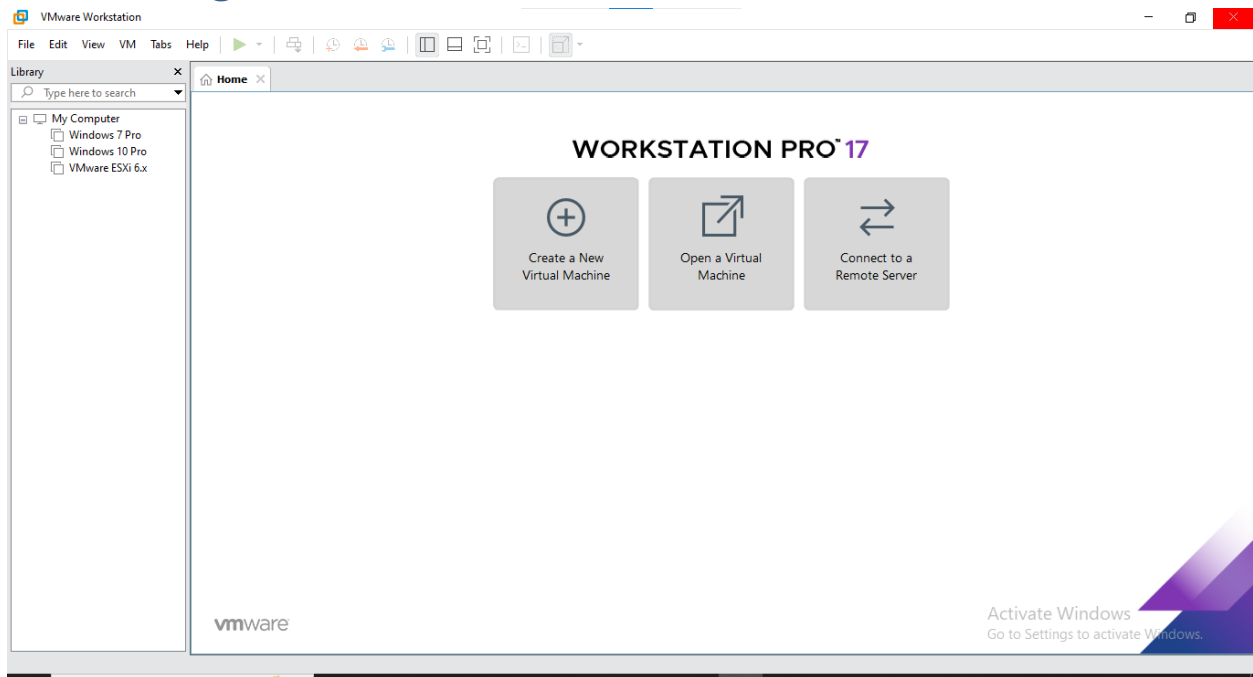
11.7.5. Ready to Complete



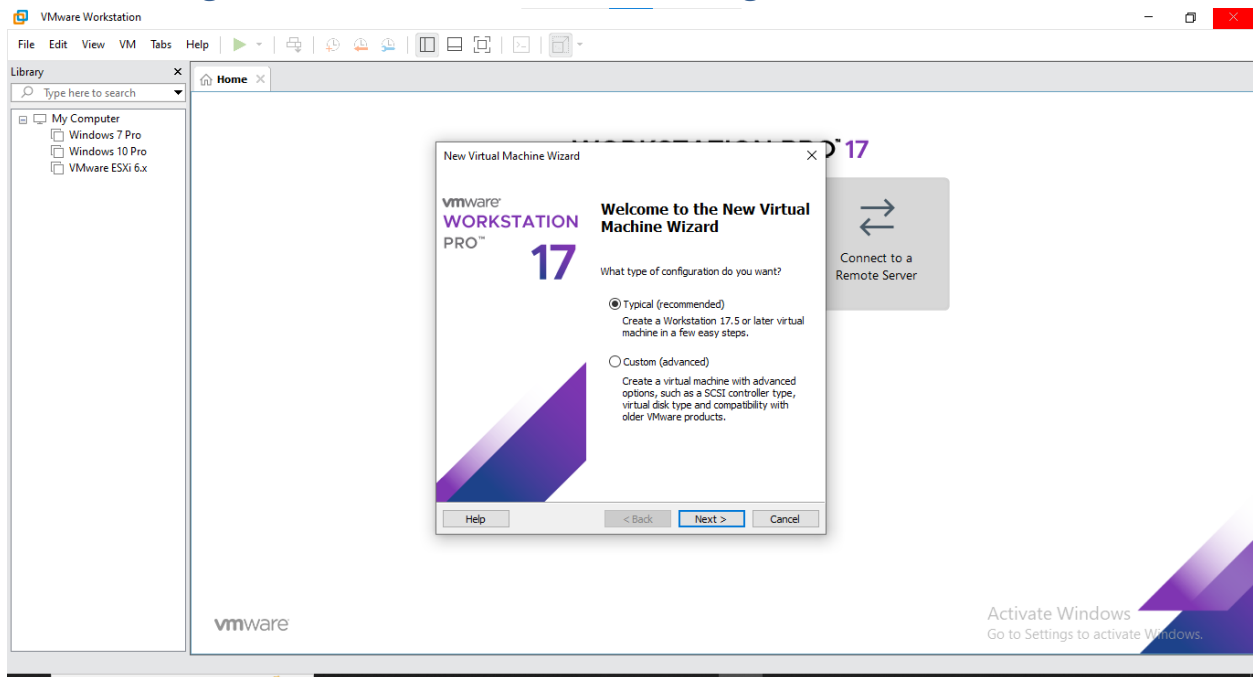
11.7.6. A Virtual Machine has been Created on ESXi Server



12. Creating a 4th Virtual Machine

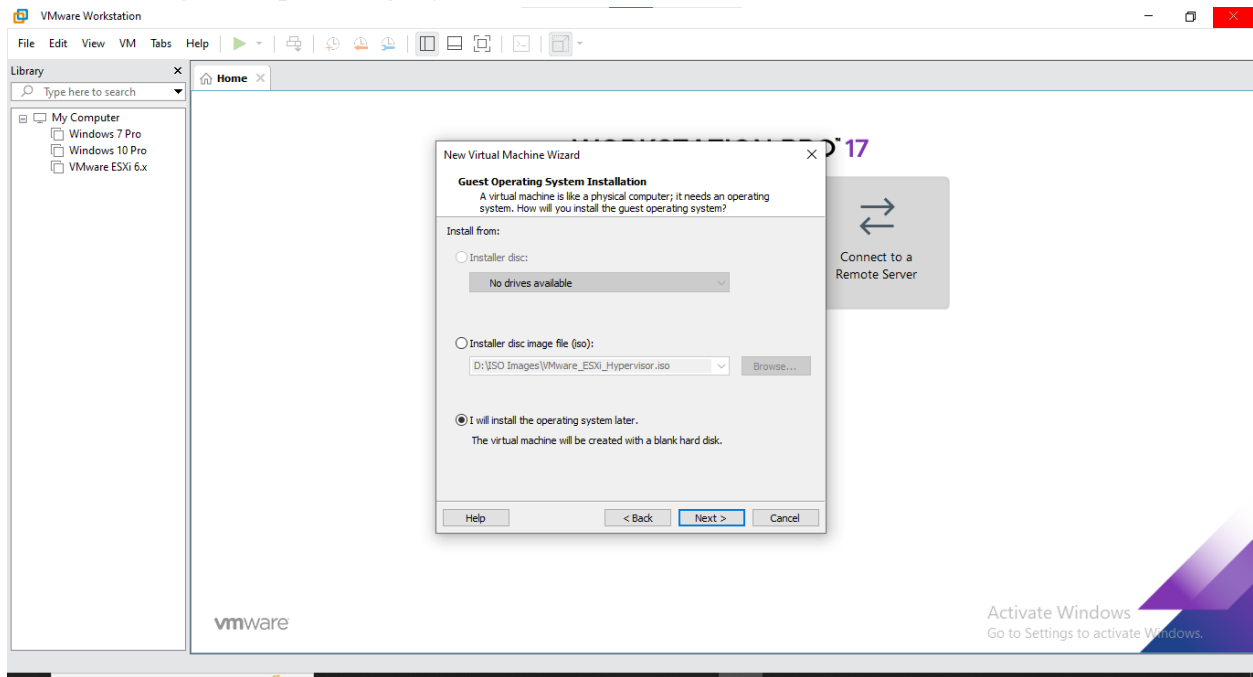


12.1. Creating a 4th Virtual Machine and Installing Windows Server

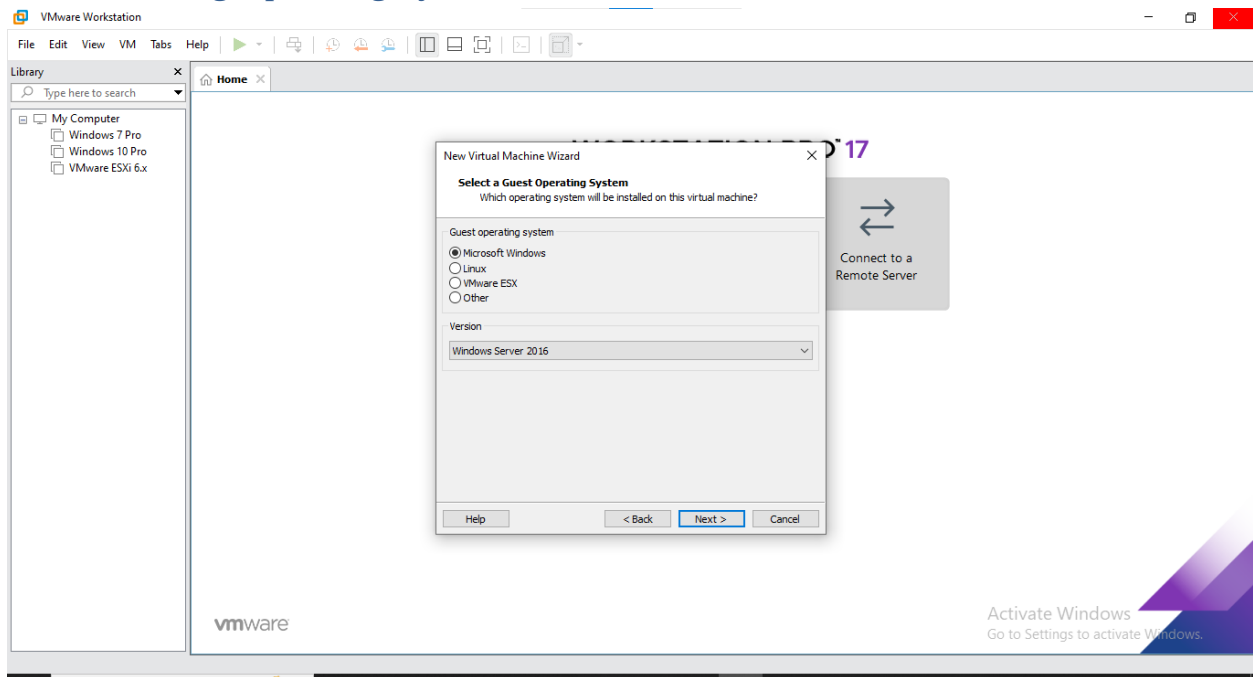


VIRTUALIZED DATA CENTER & DOMAIN SETUP

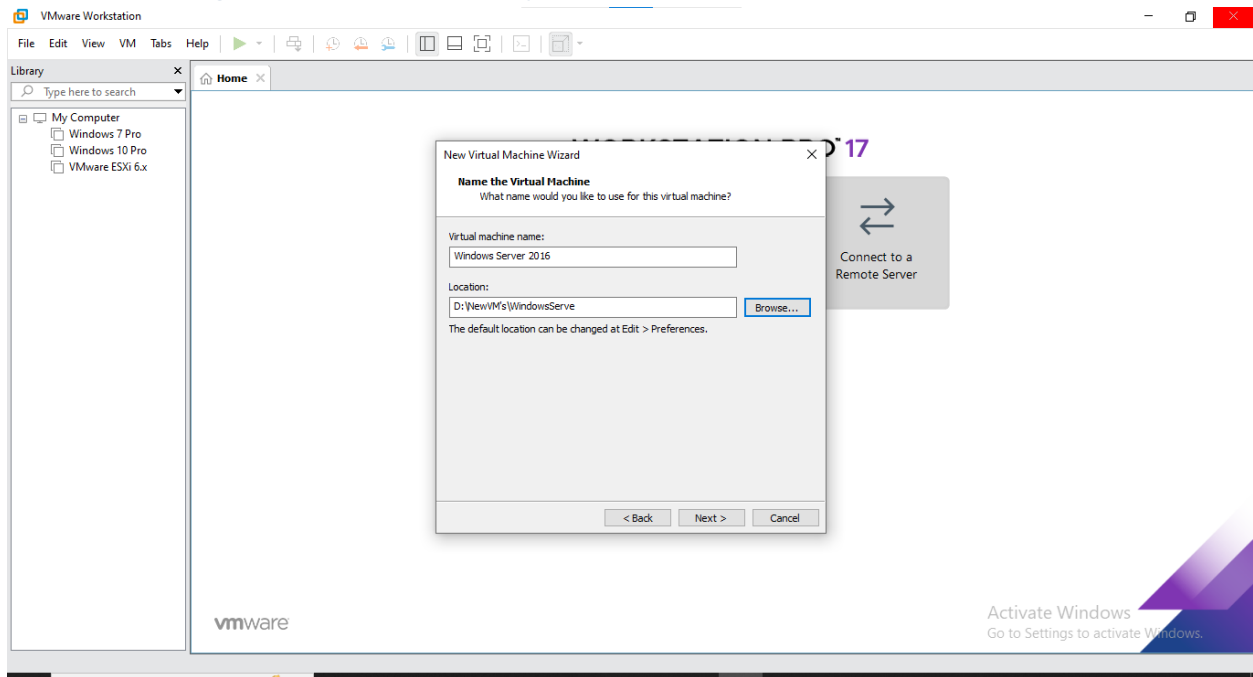
12.2. Configure Operating System Information



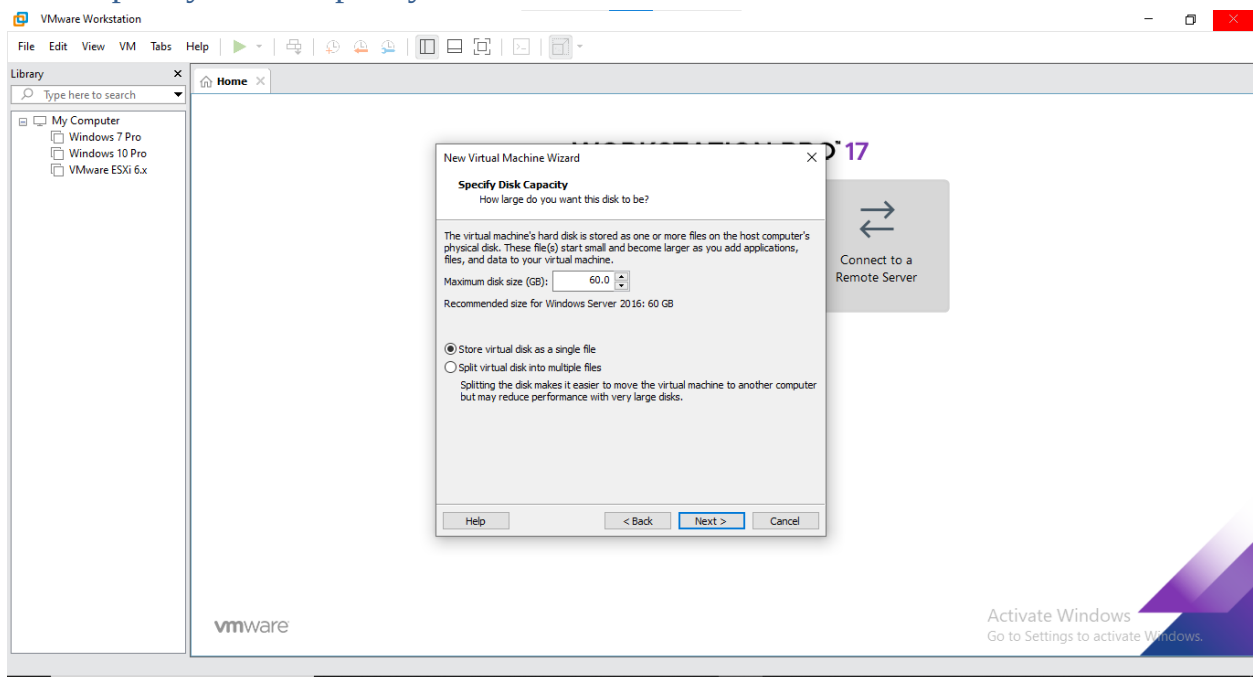
12.3. Selecting Operating System



12.4. Selecting Name and Directory to Store VM

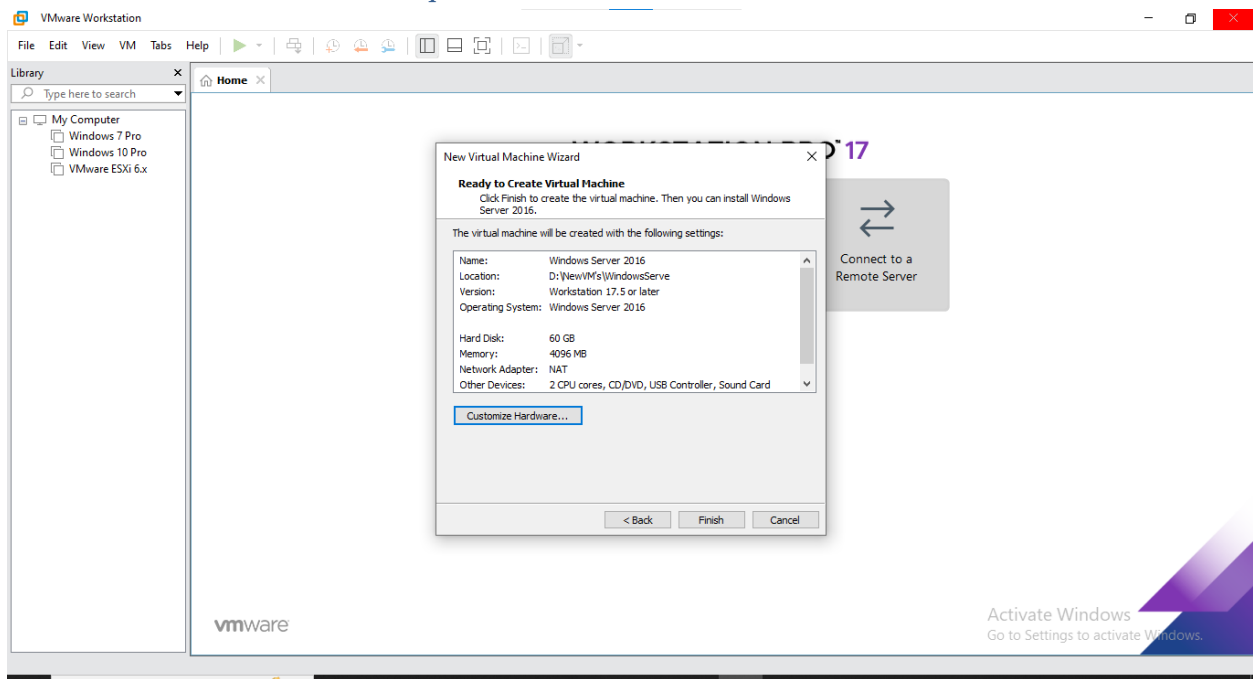


12.5. Specify Disk Capacity

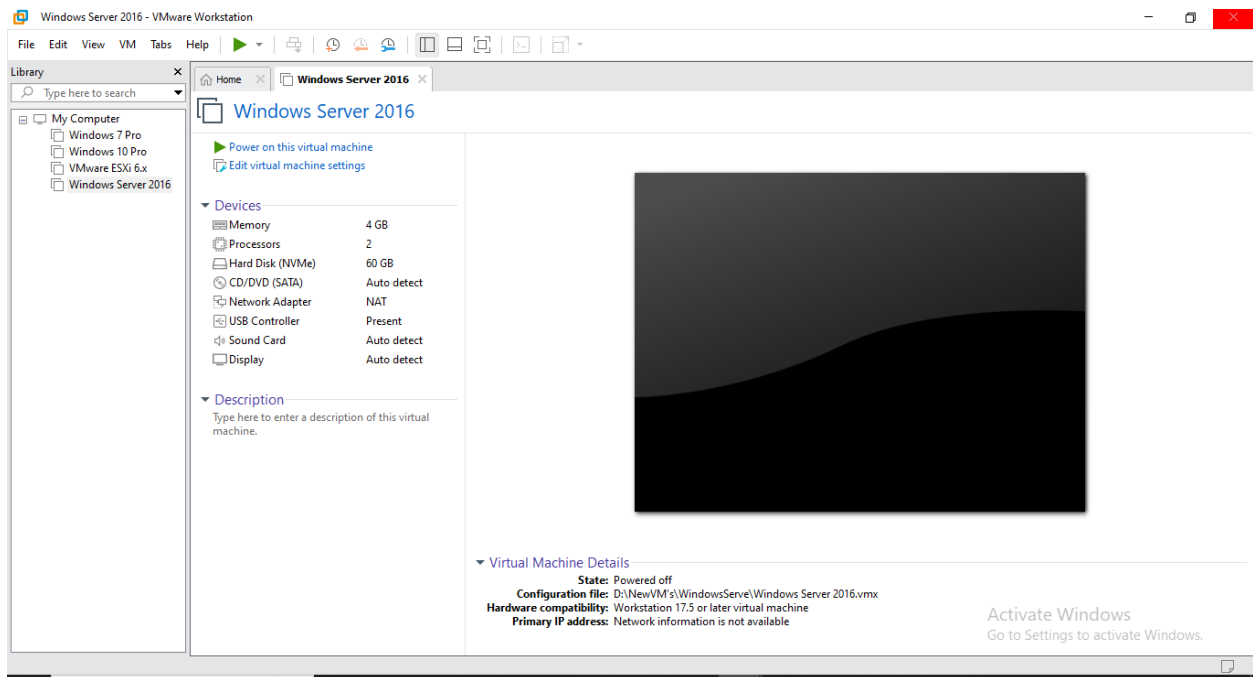


VIRTUALIZED DATA CENTER & DOMAIN SETUP

12.6. Finalize Hardware Requirement

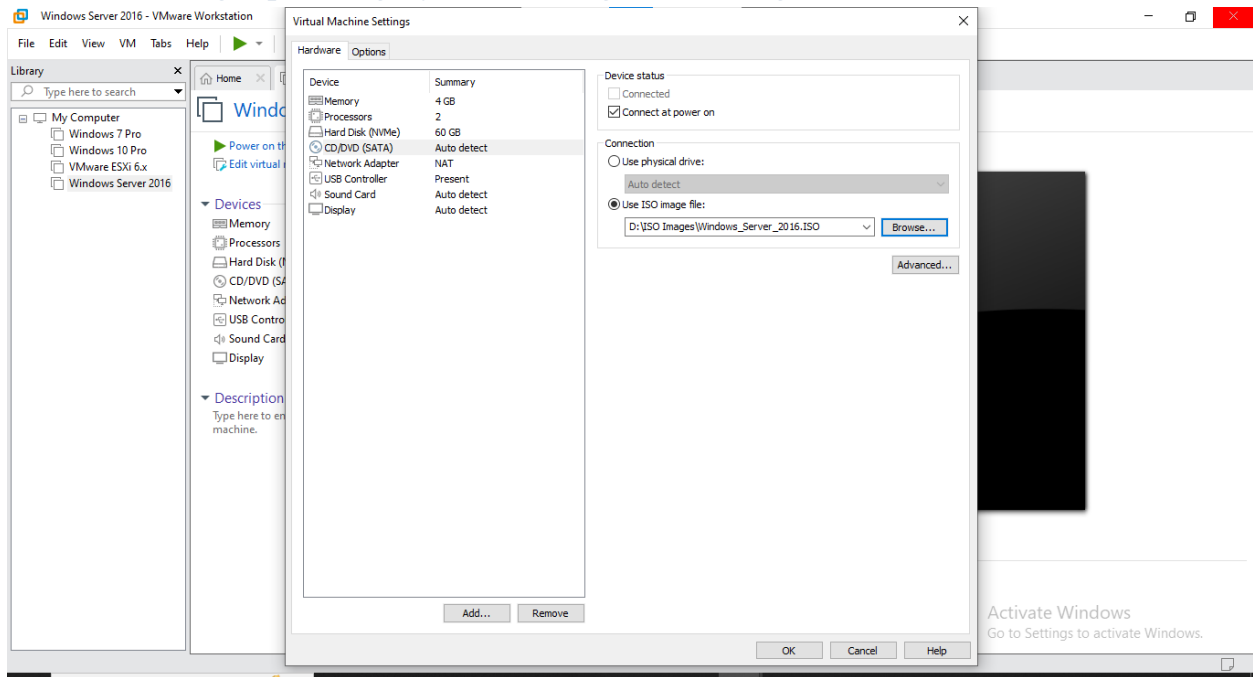


A Virtual Machine is Created

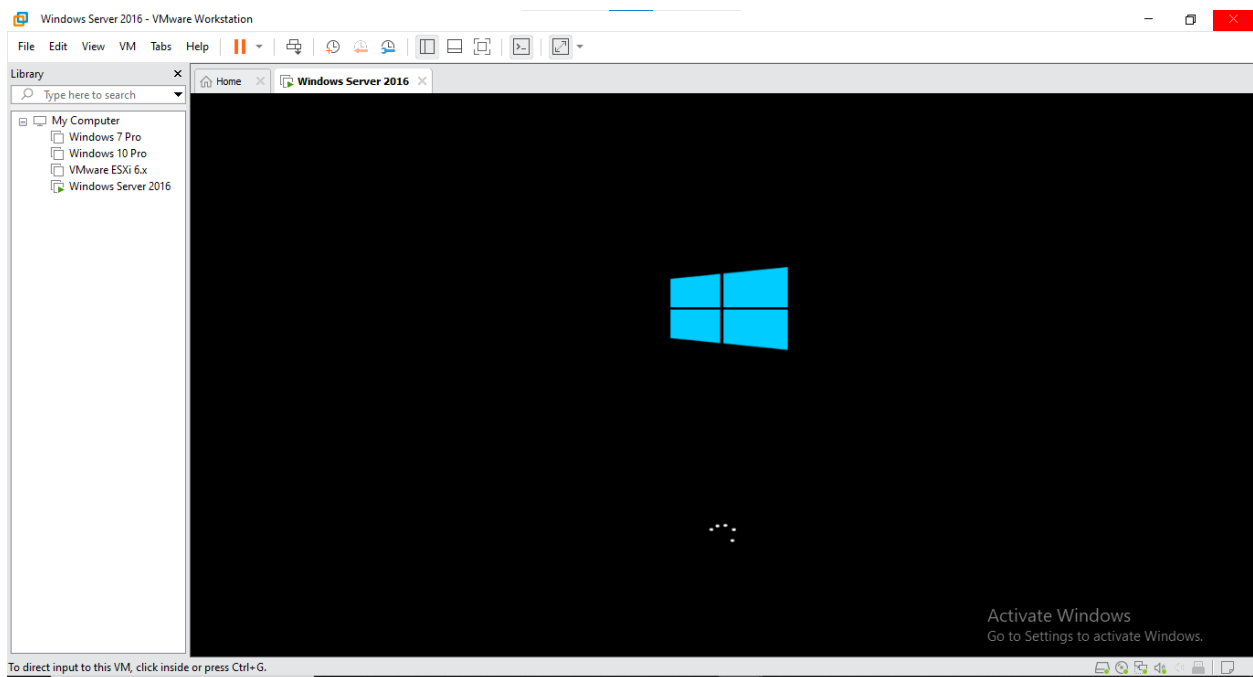


VIRTUALIZED DATA CENTER & DOMAIN SETUP

12.7. Installing Operating System through ISO Image

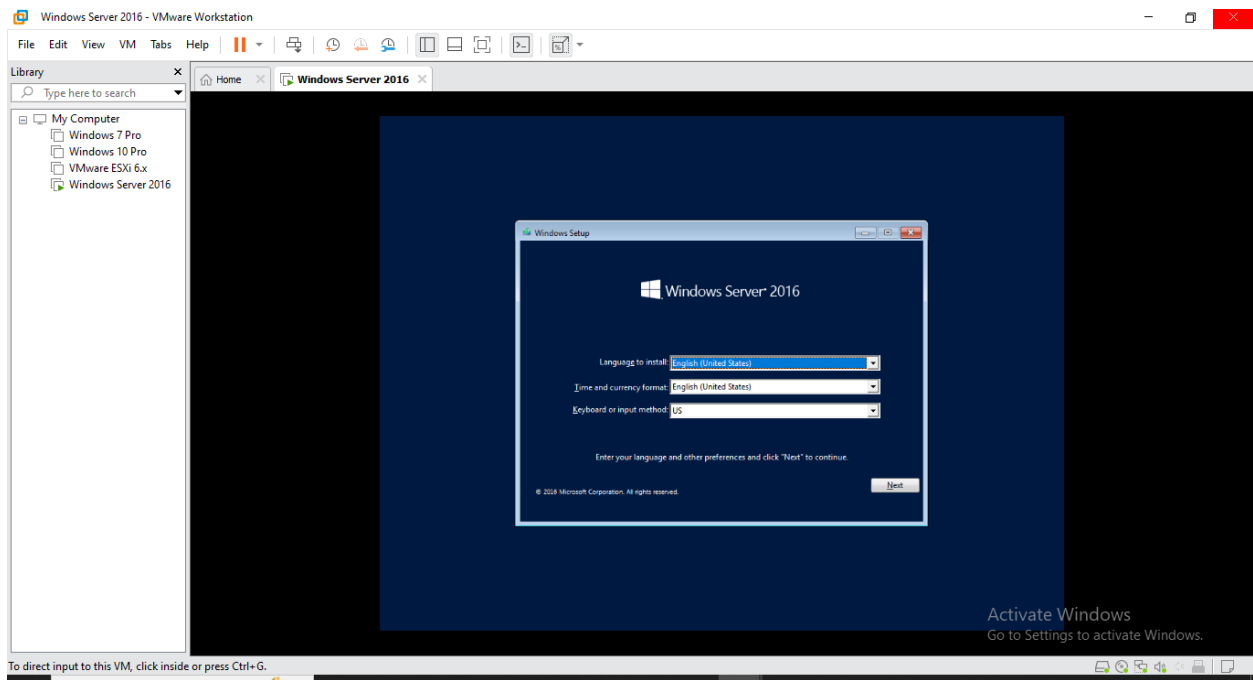


12.8. Power on Virtual Machine to Finish Installation

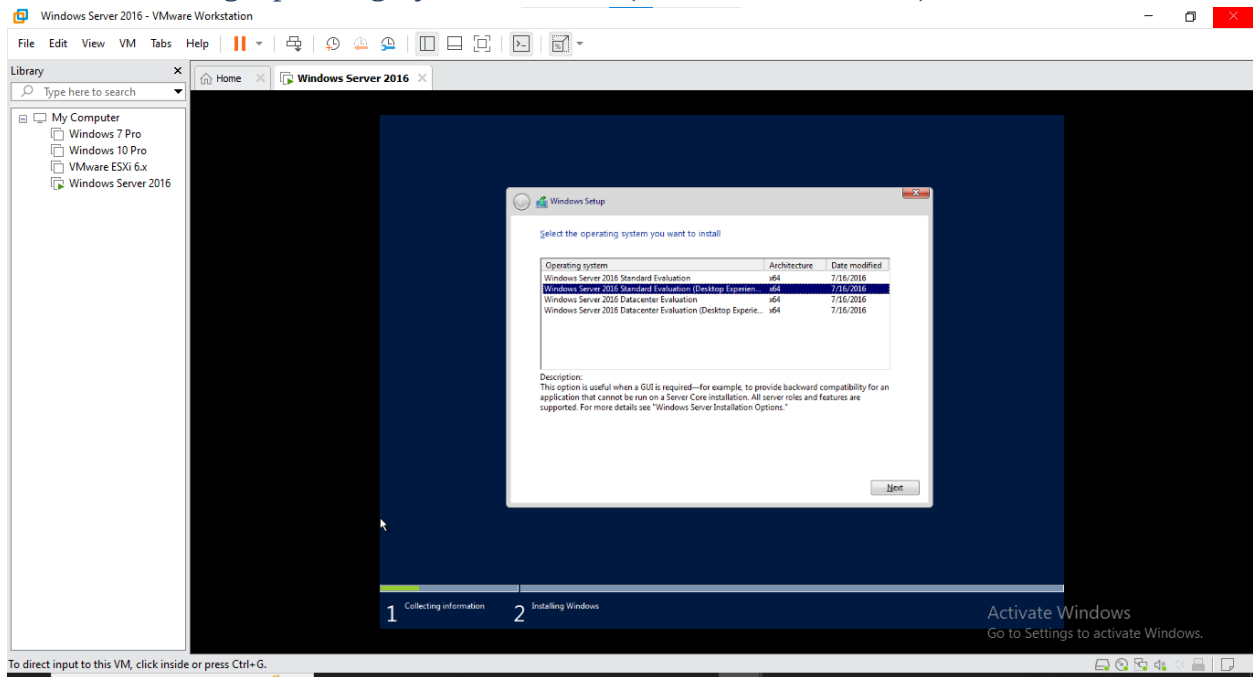


VIRTUALIZED DATA CENTER & DOMAIN SETUP

12.8.1. Click Next to Proceed

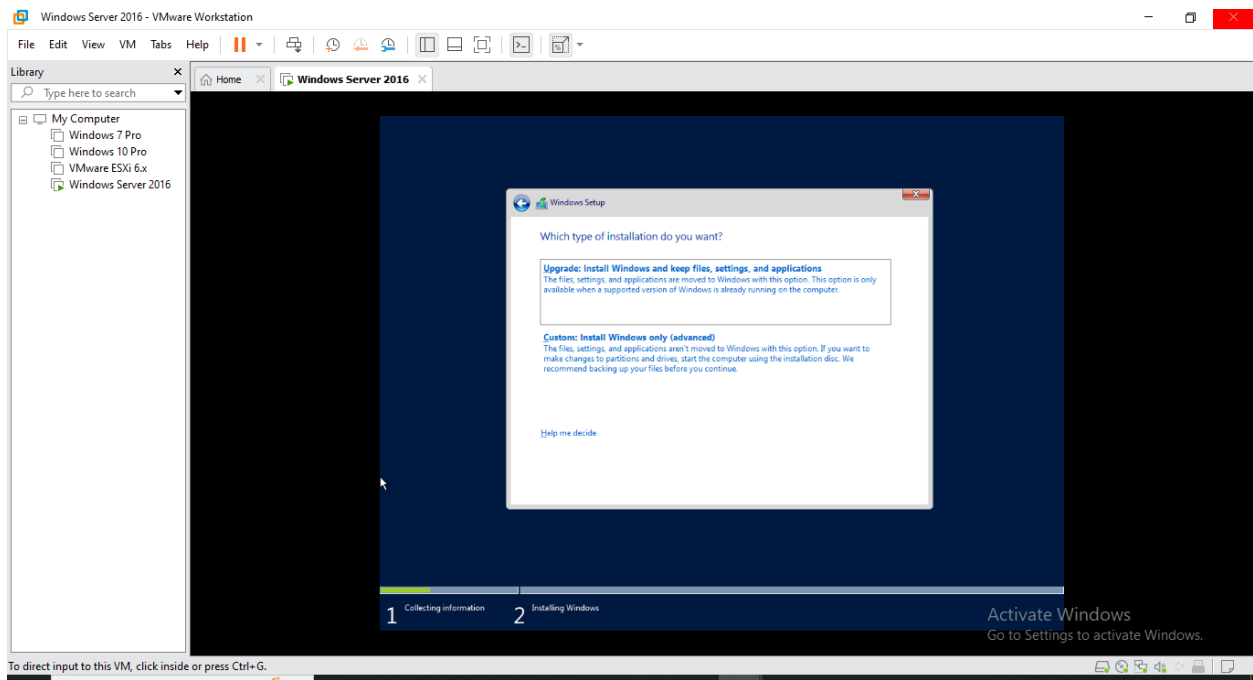


12.8.2. Selecting Operating System Version (Standard Evaluation)

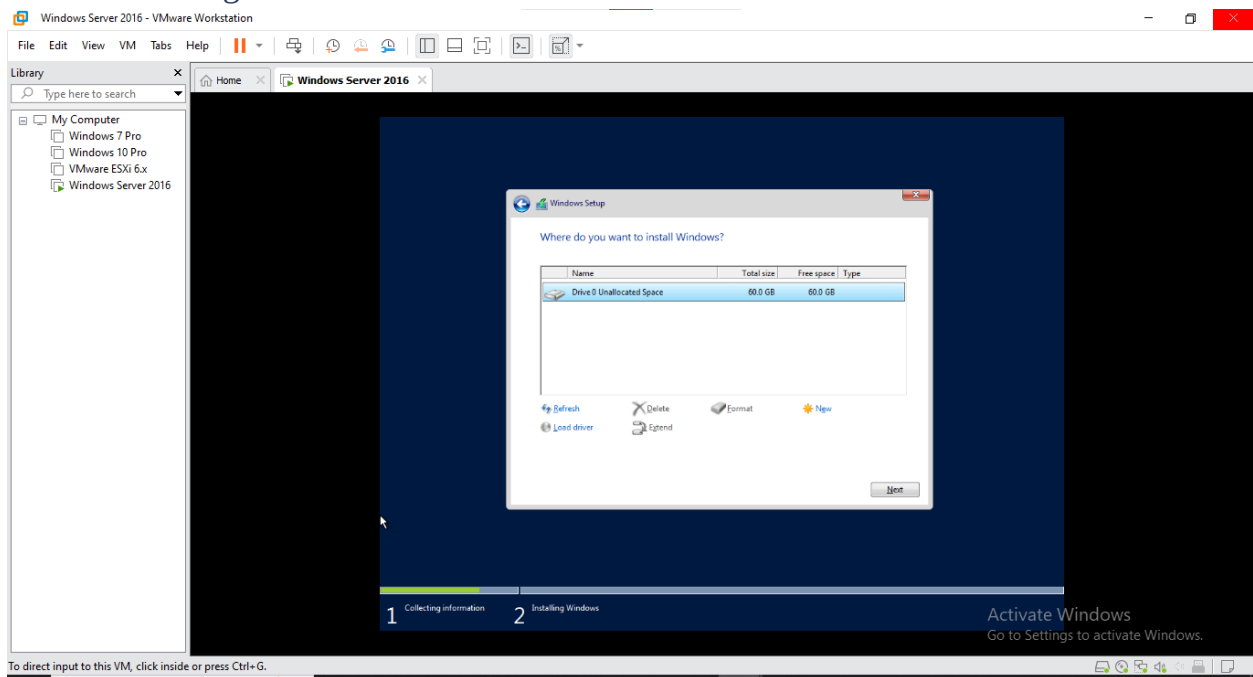


VIRTUALIZED DATA CENTER & DOMAIN SETUP

12.8.3. Select Custom to Proceed

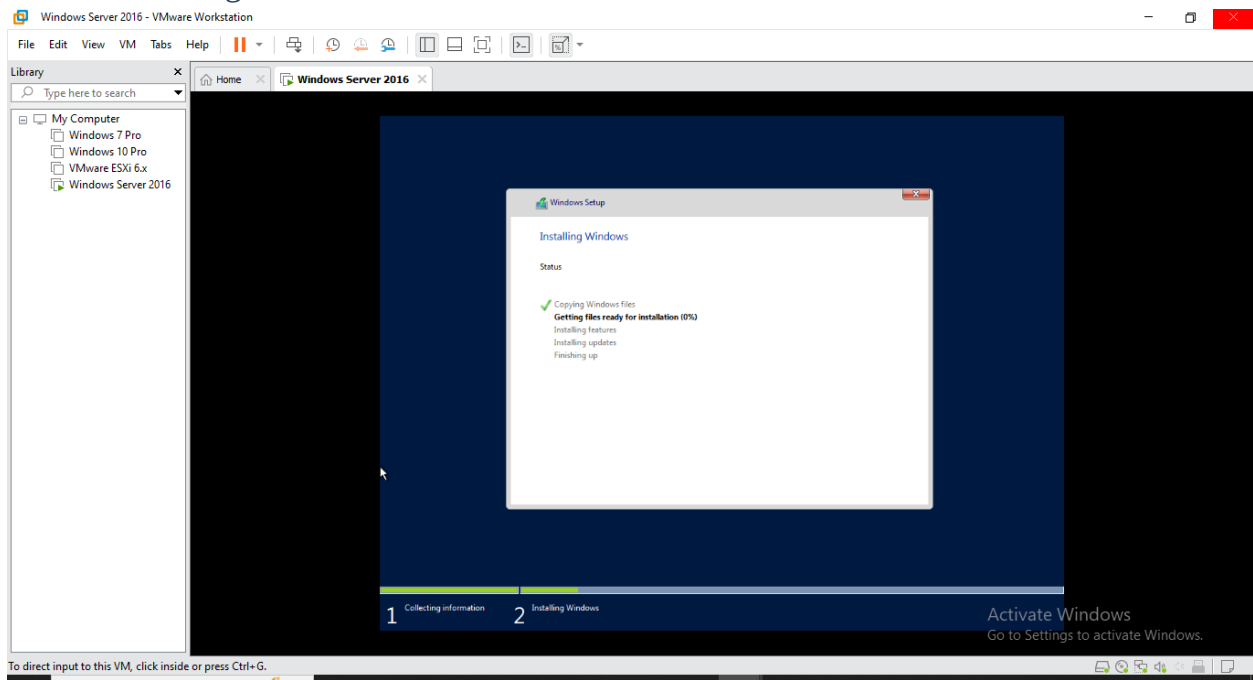


12.8.4. Selecting Disk Allocated

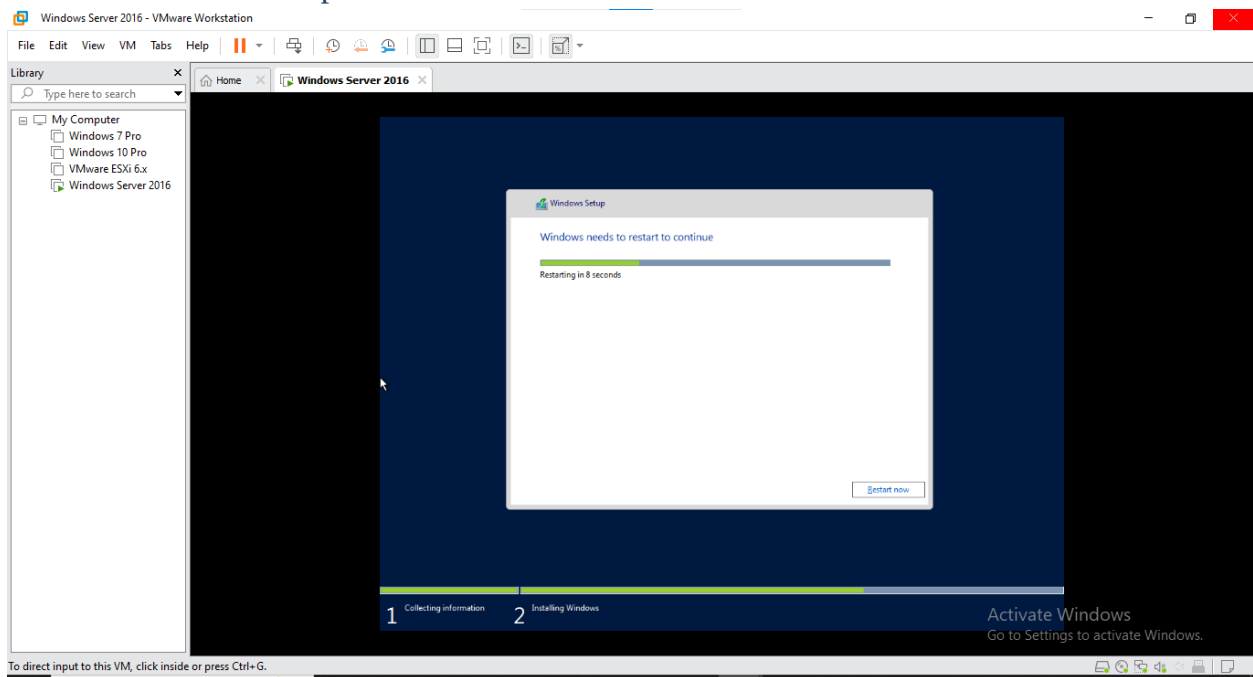


VIRTUALIZED DATA CENTER & DOMAIN SETUP

12.8.5. Installing Windows Server 2016

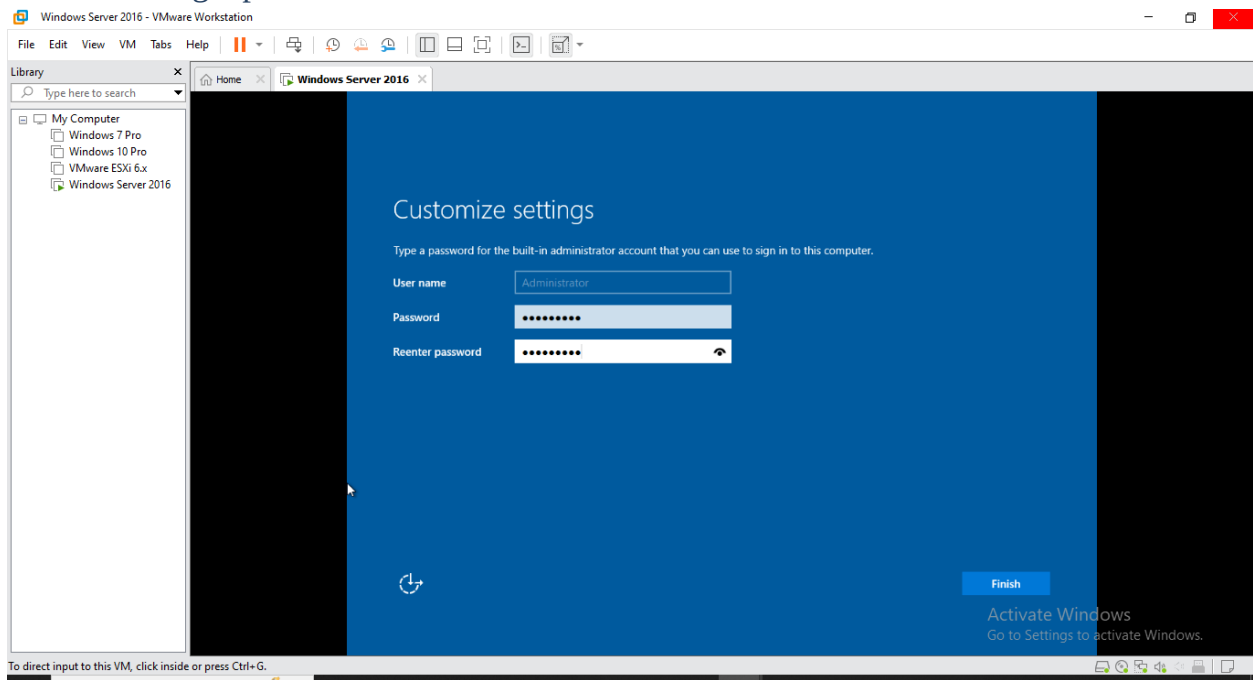


12.8.6. Restart to Complete Installation

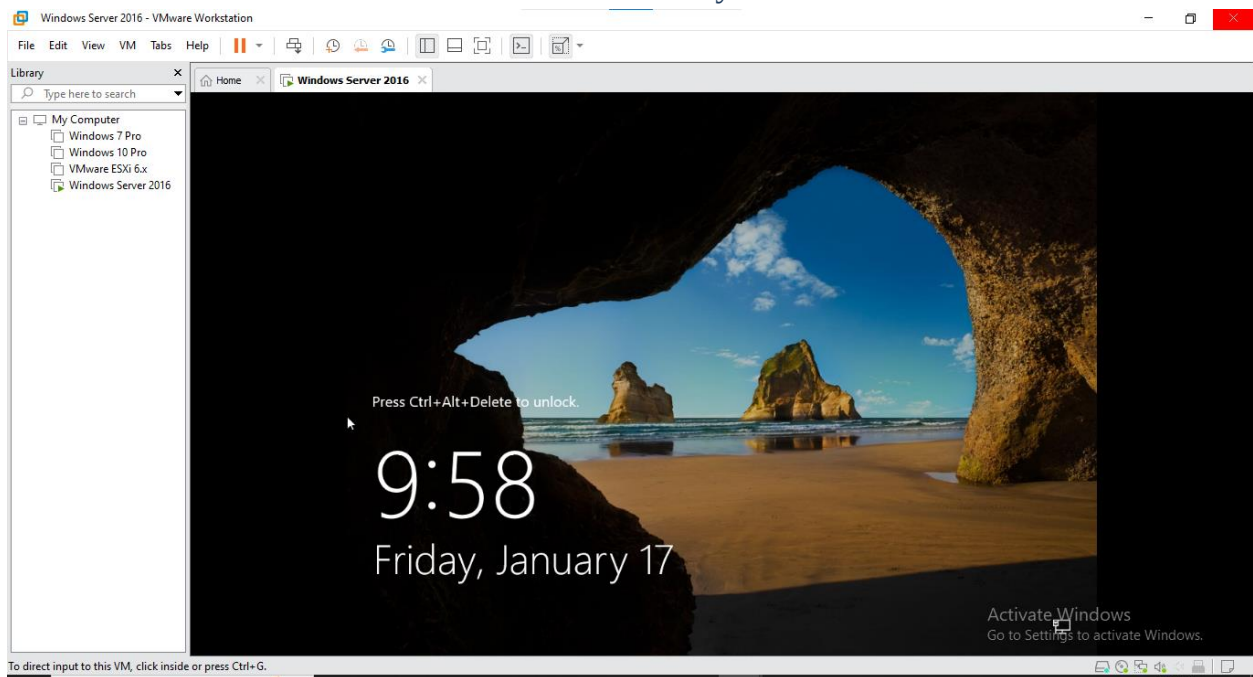


VIRTUALIZED DATA CENTER & DOMAIN SETUP

12.8.7. Setting Up Password

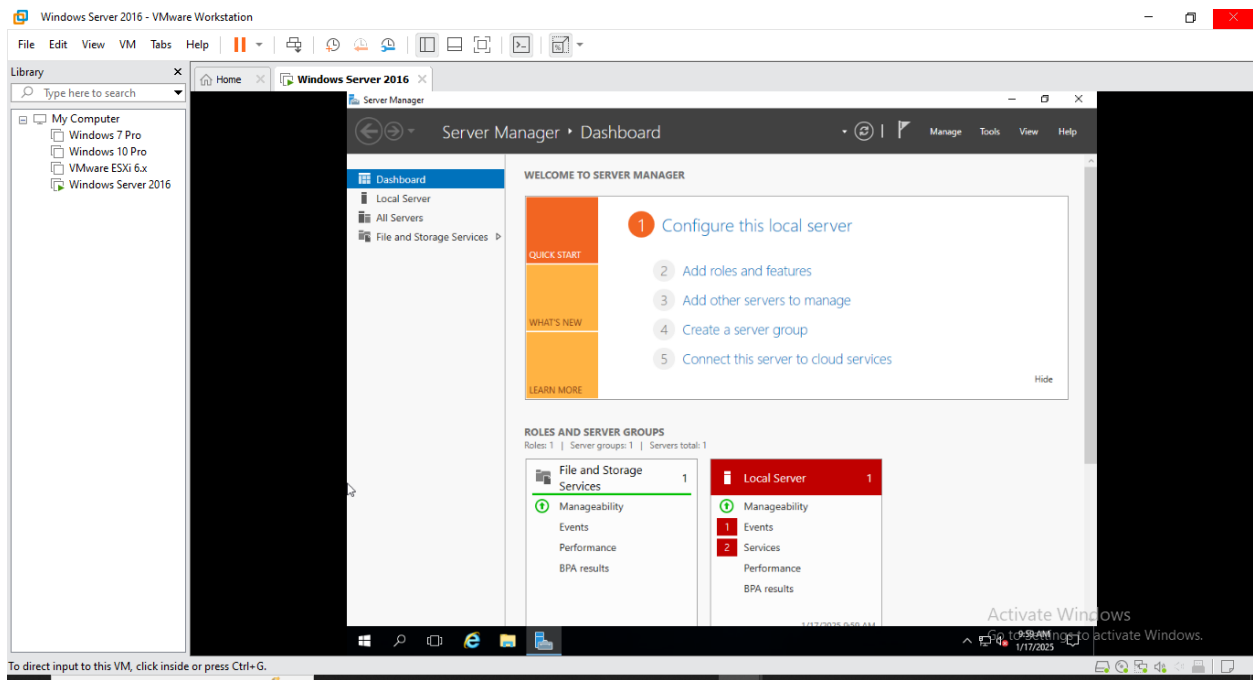


12.8.8. Windows Server 2016 Installed Successfully

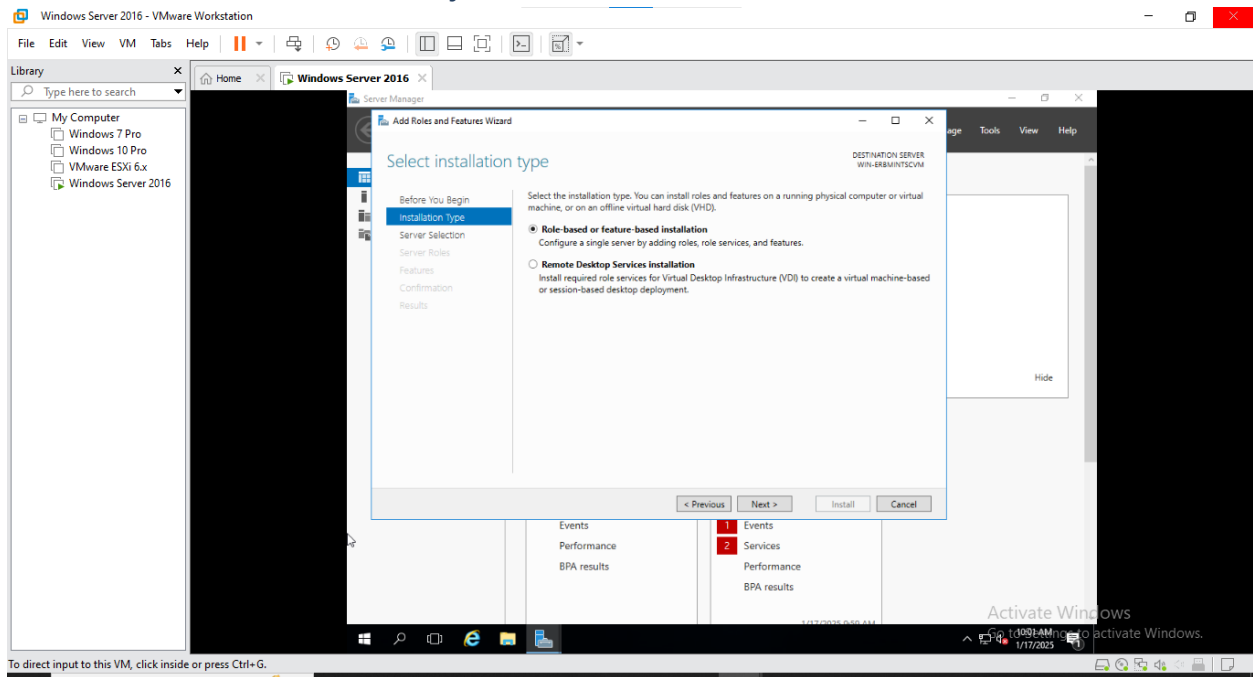


VIRTUALIZED DATA CENTER & DOMAIN SETUP

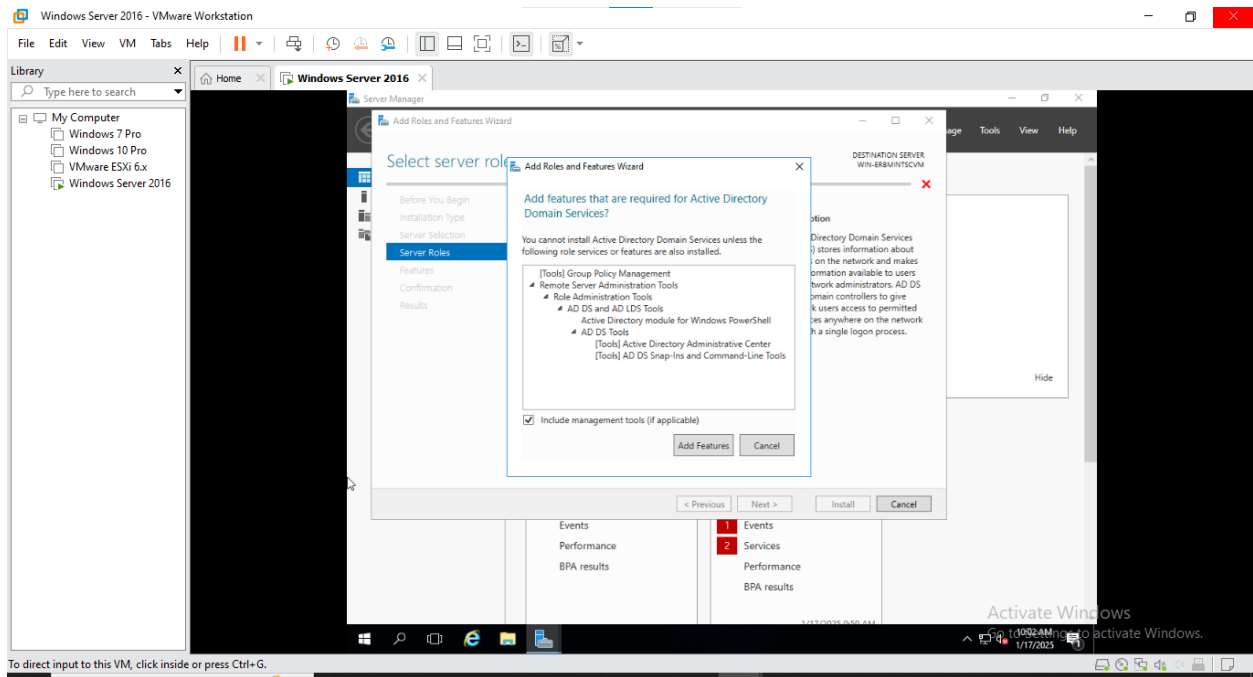
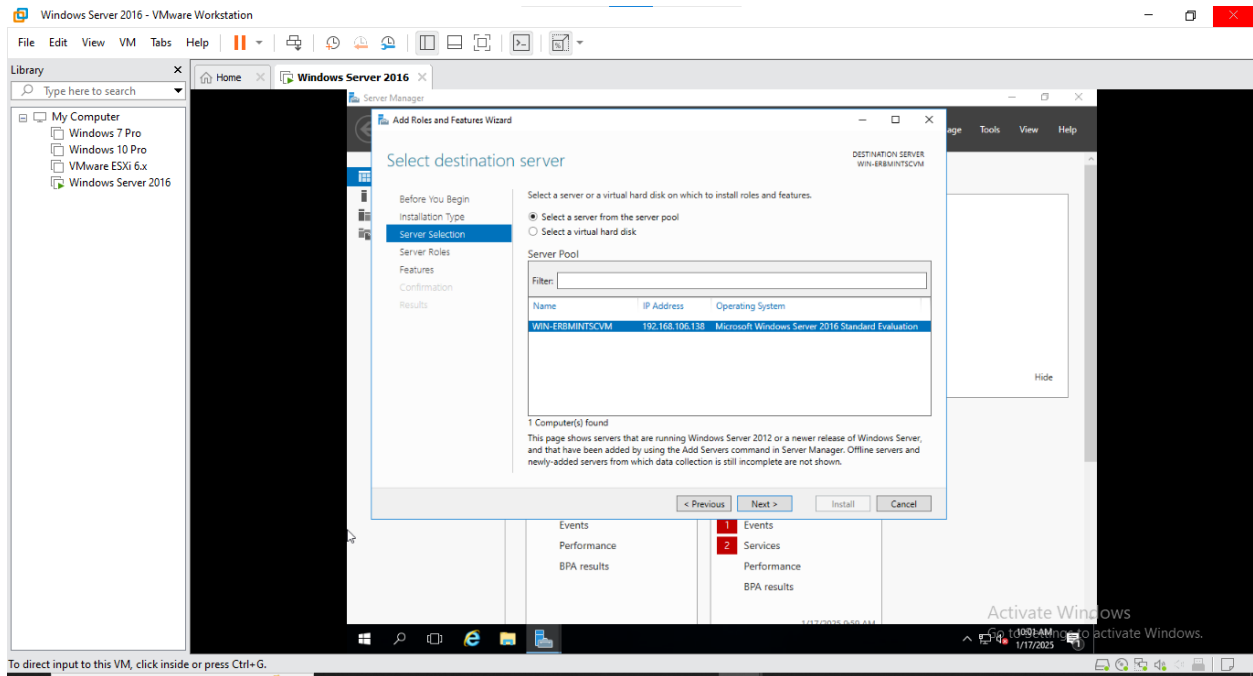
12.8.9. Add Roles and Features



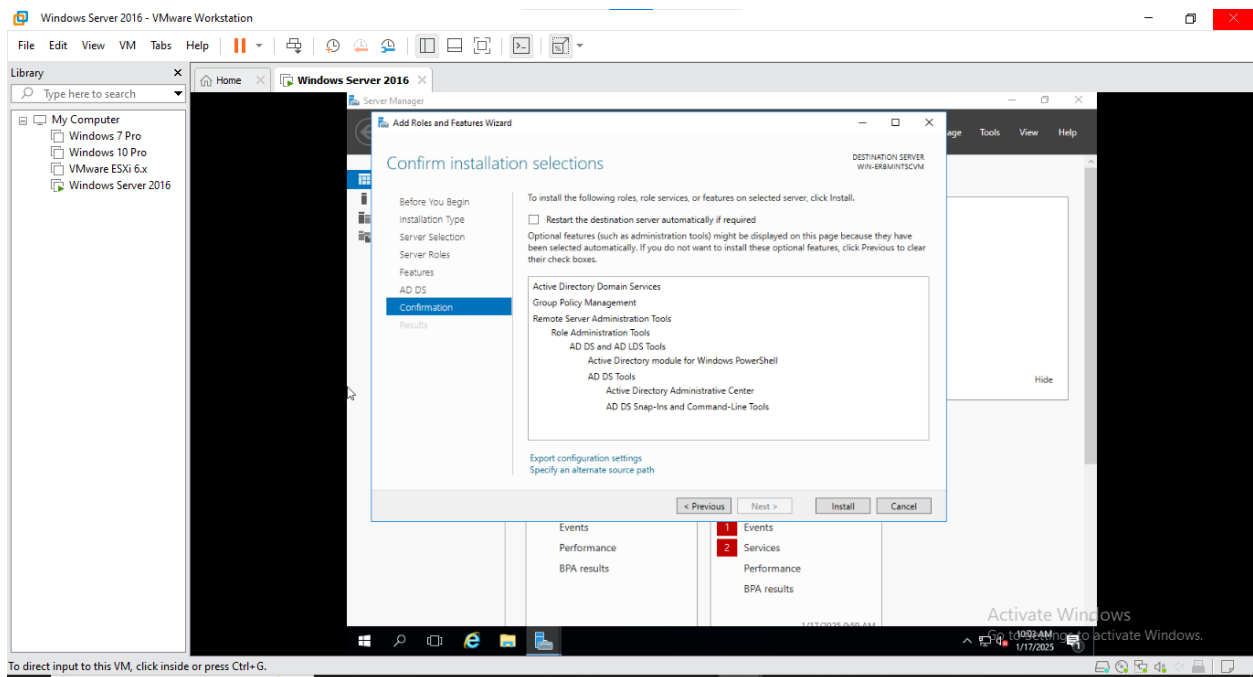
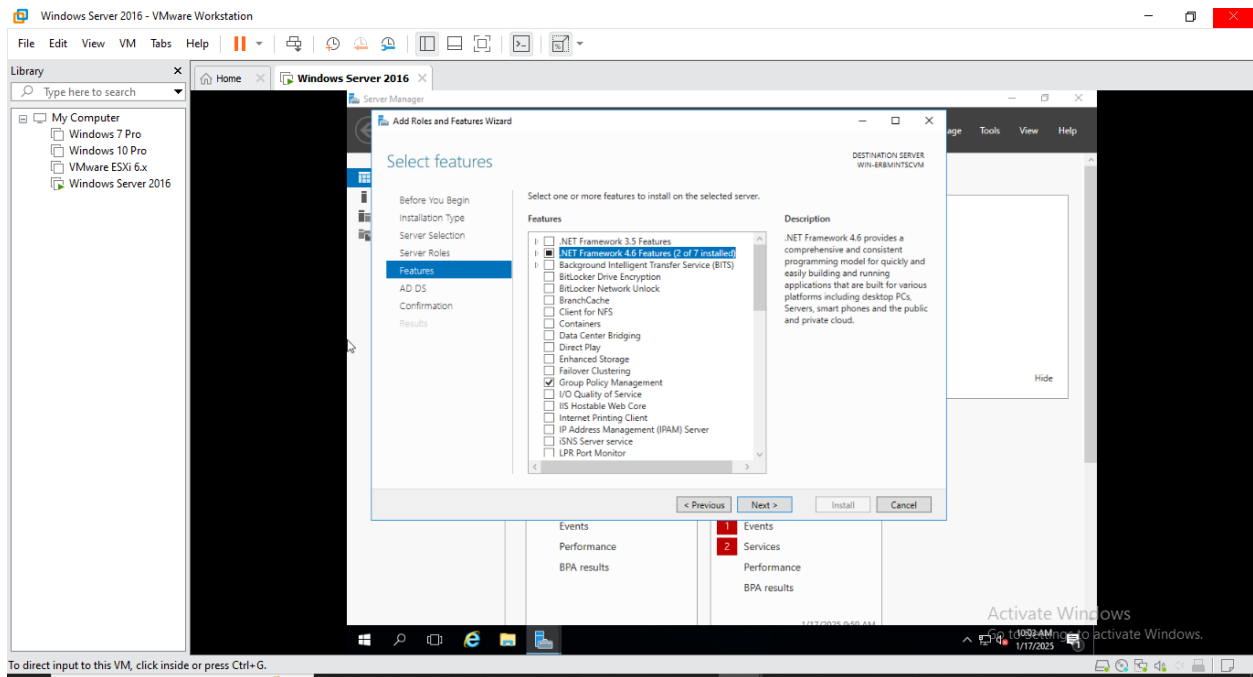
12.8.10. Enable Active Directory



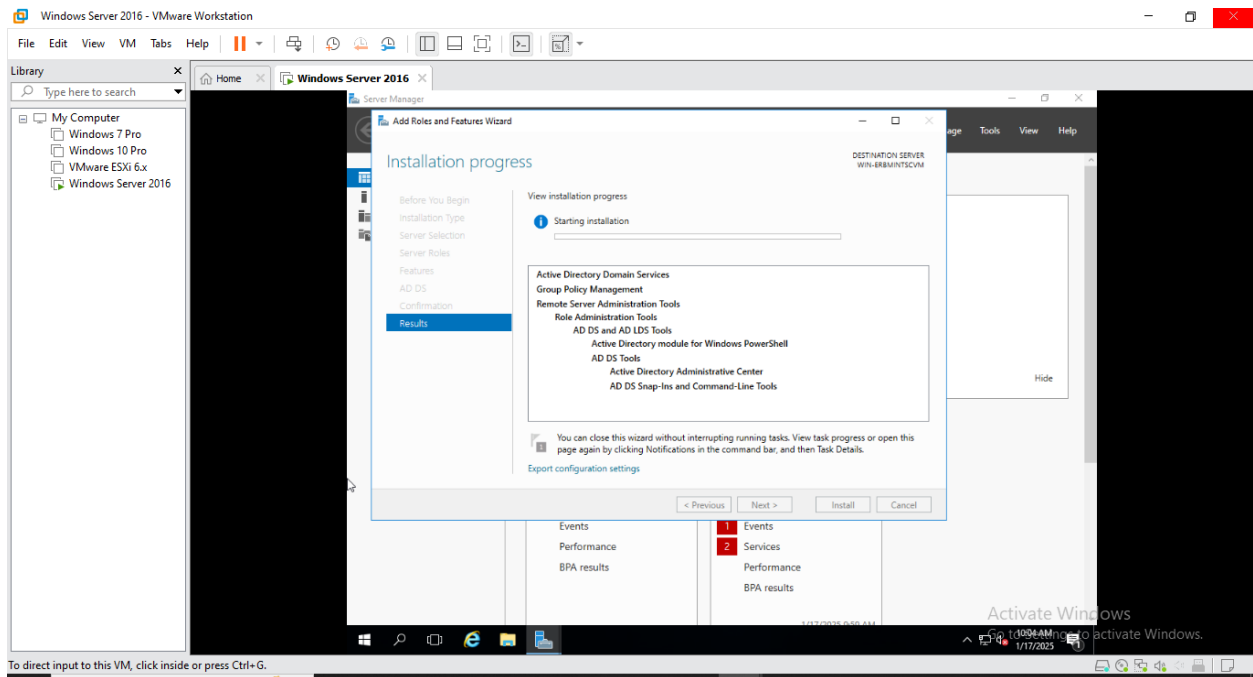
VIRTUALIZED DATA CENTER & DOMAIN SETUP



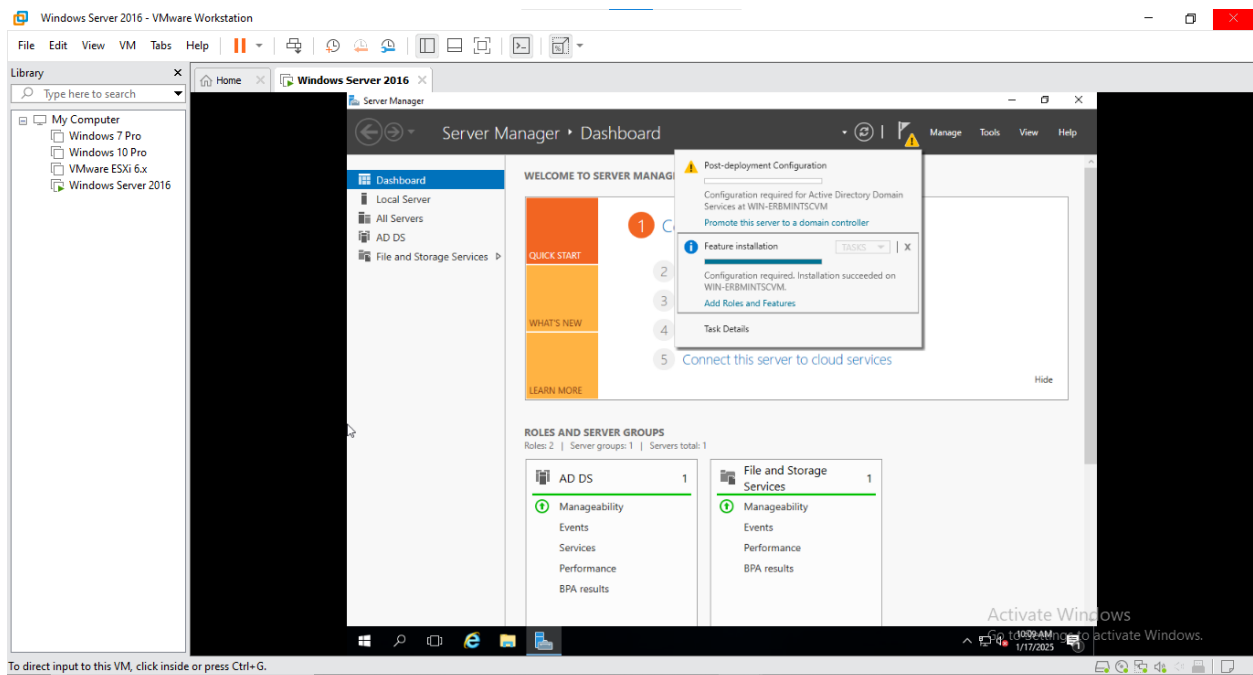
VIRTUALIZED DATA CENTER & DOMAIN SETUP



VIRTUALIZED DATA CENTER & DOMAIN SETUP

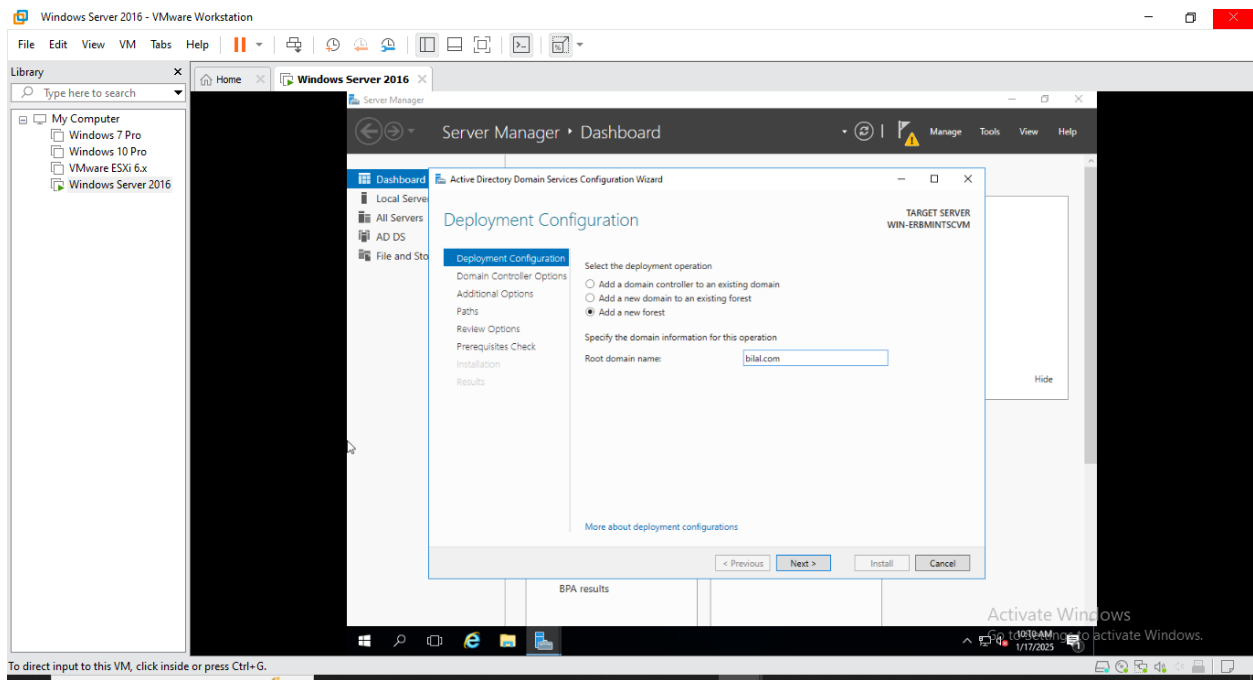


12.8.11. Promote Server to Domain Controller

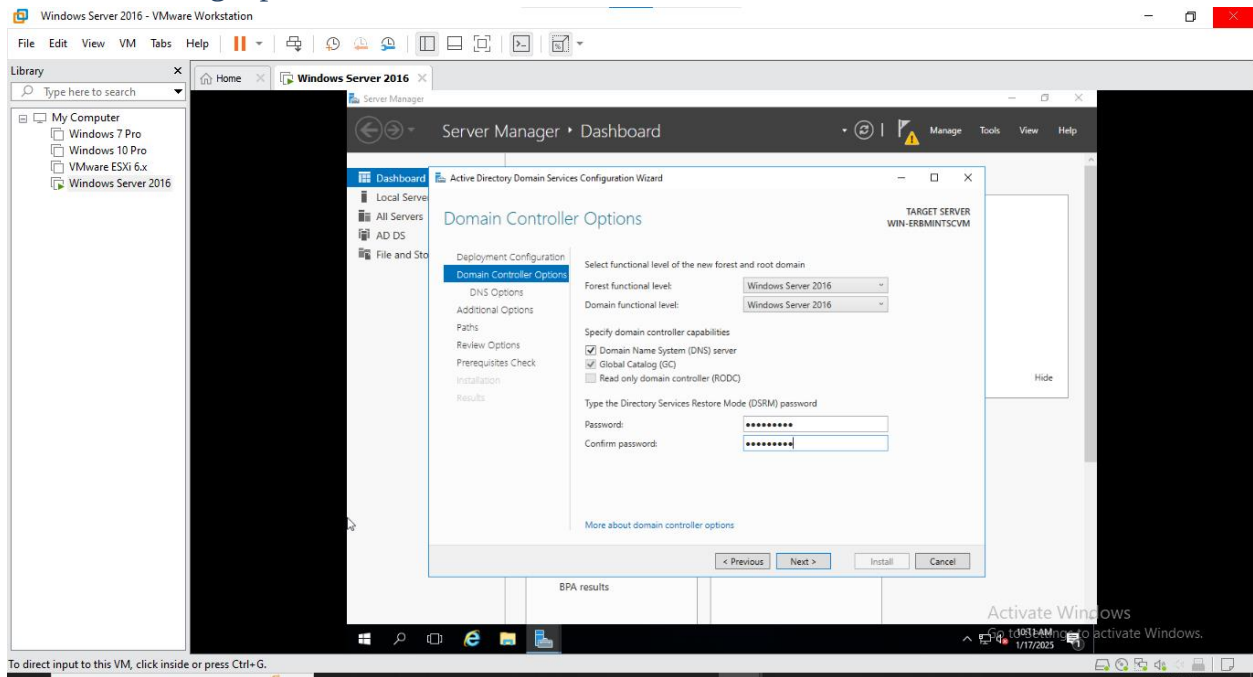


VIRTUALIZED DATA CENTER & DOMAIN SETUP

12.8.12. Add New Forrest and Domain Name

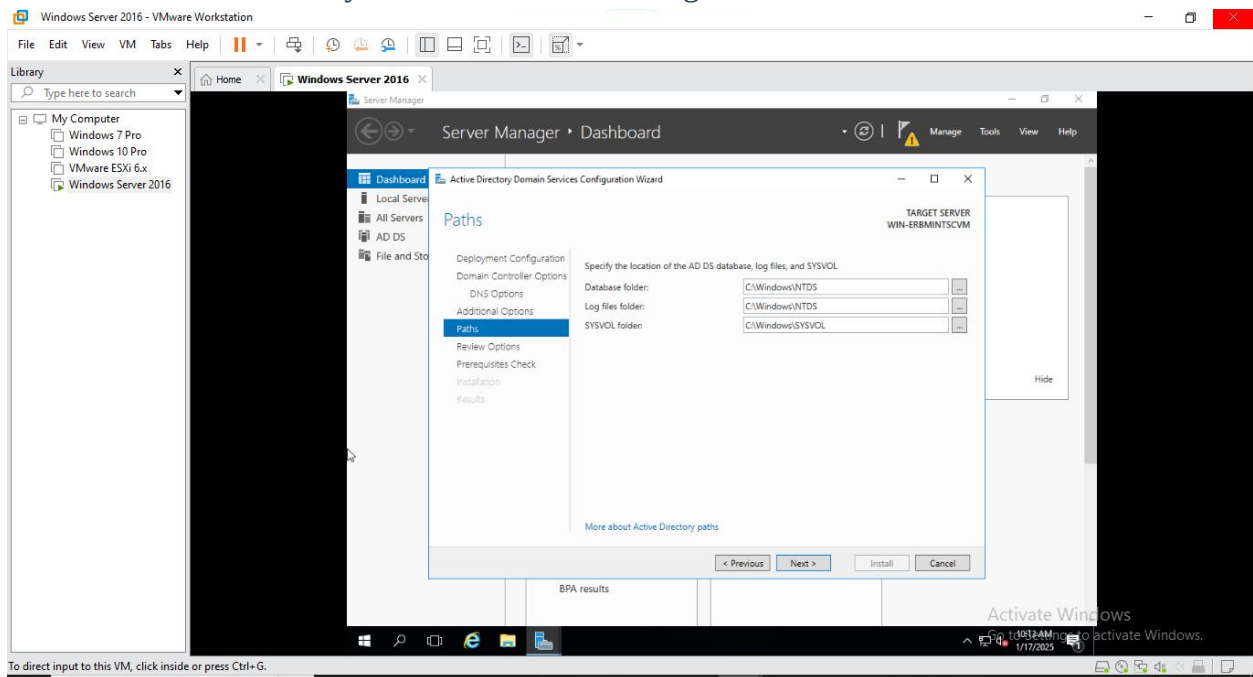


12.6.13. Setting Up Domain Password

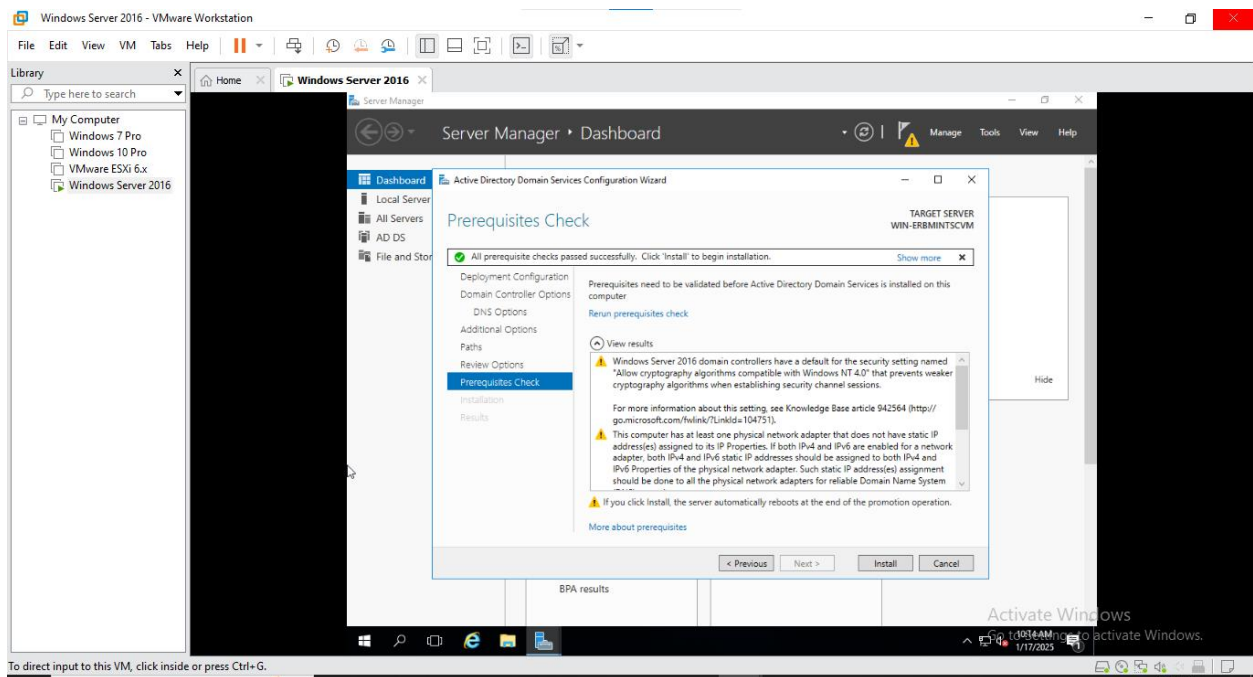


VIRTUALIZED DATA CENTER & DOMAIN SETUP

12.8.14. Active Directory Domain Service Configuration

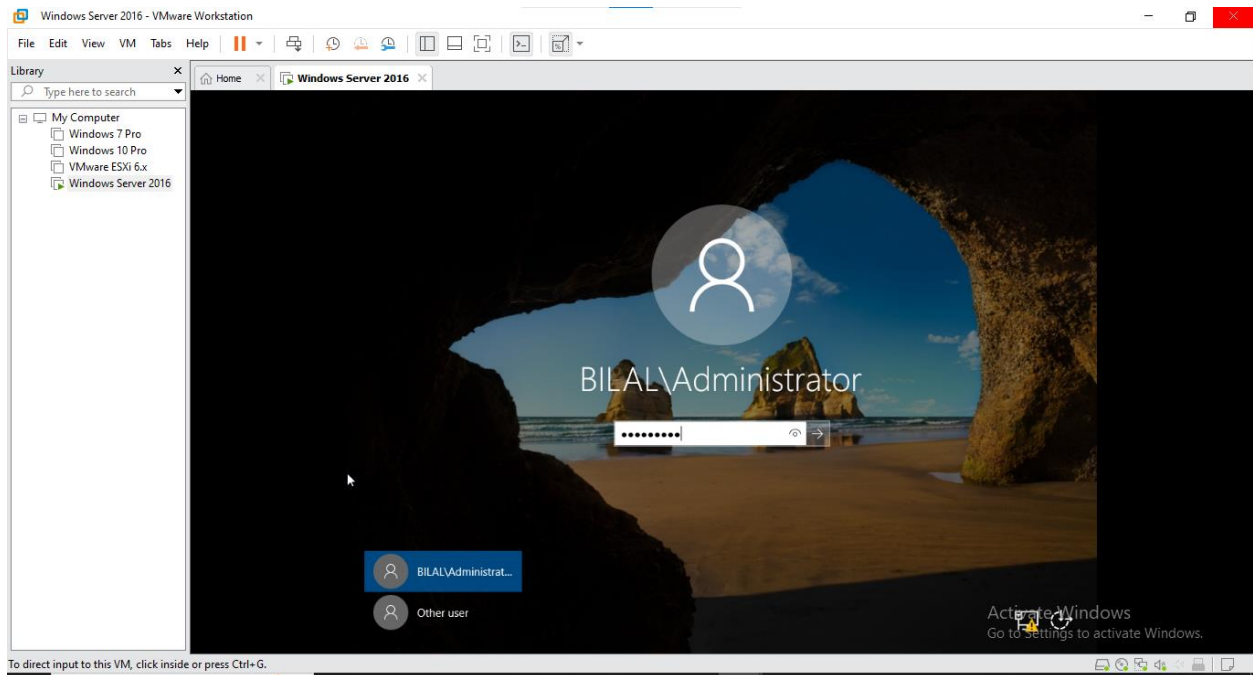
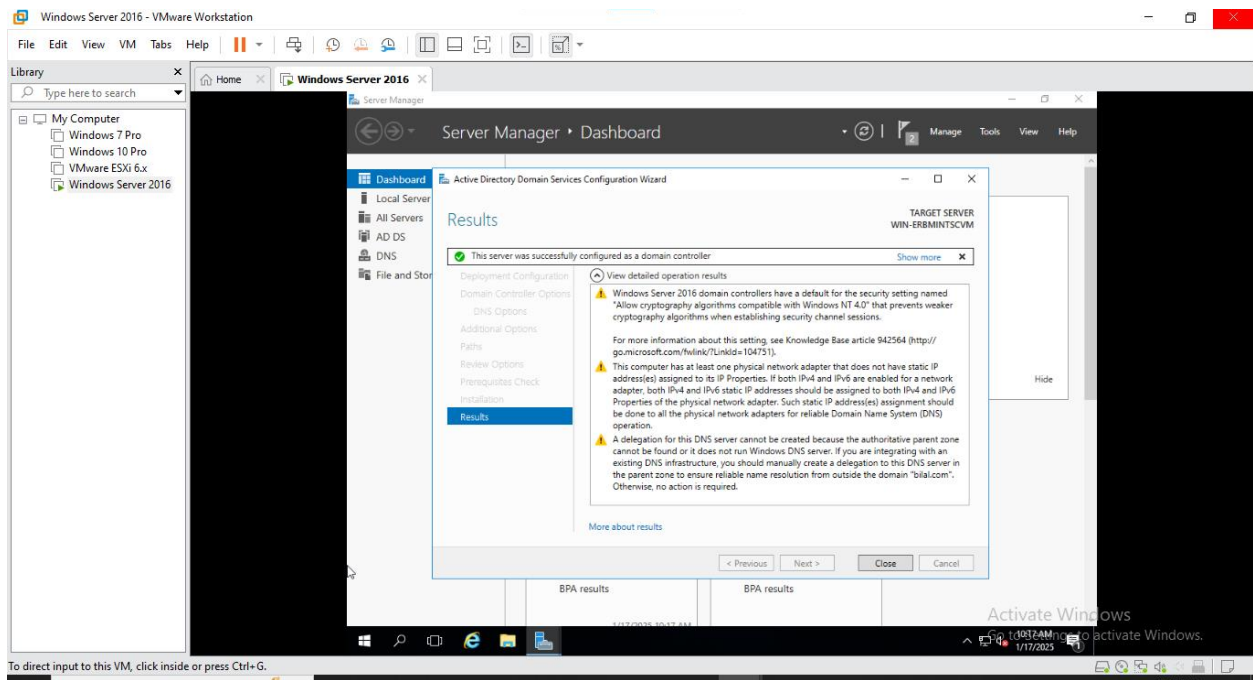


12.8.15. Install Domain Services

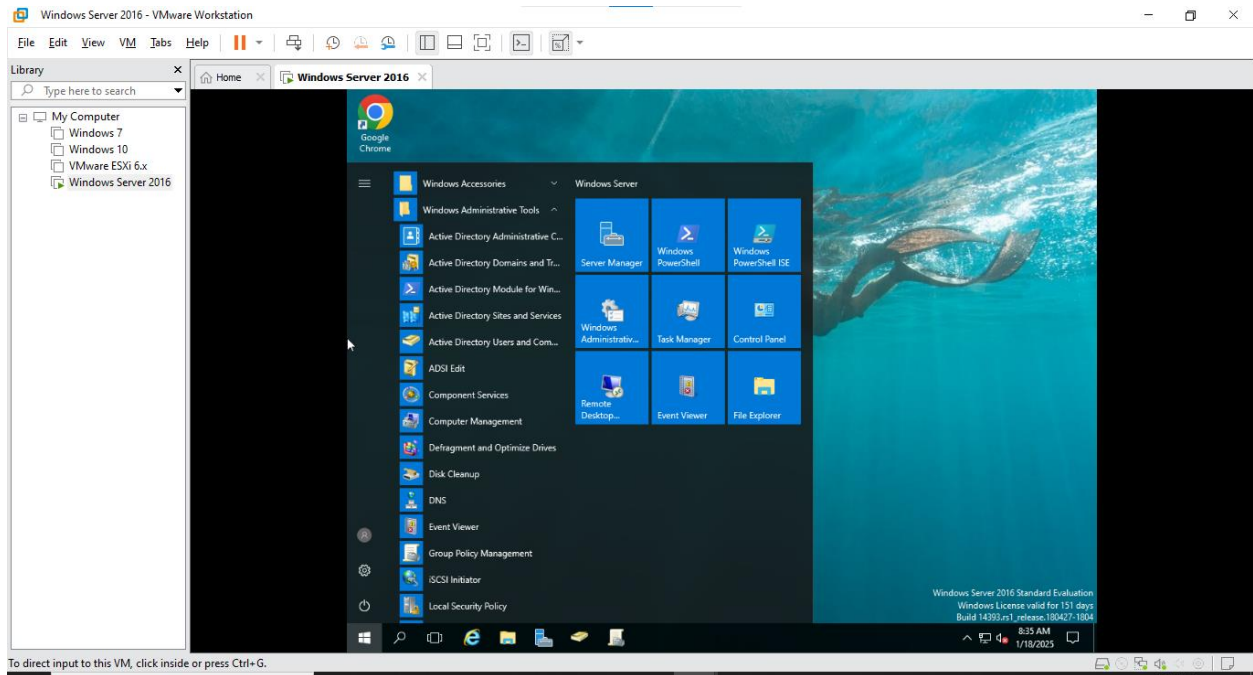


VIRTUALIZED DATA CENTER & DOMAIN SETUP

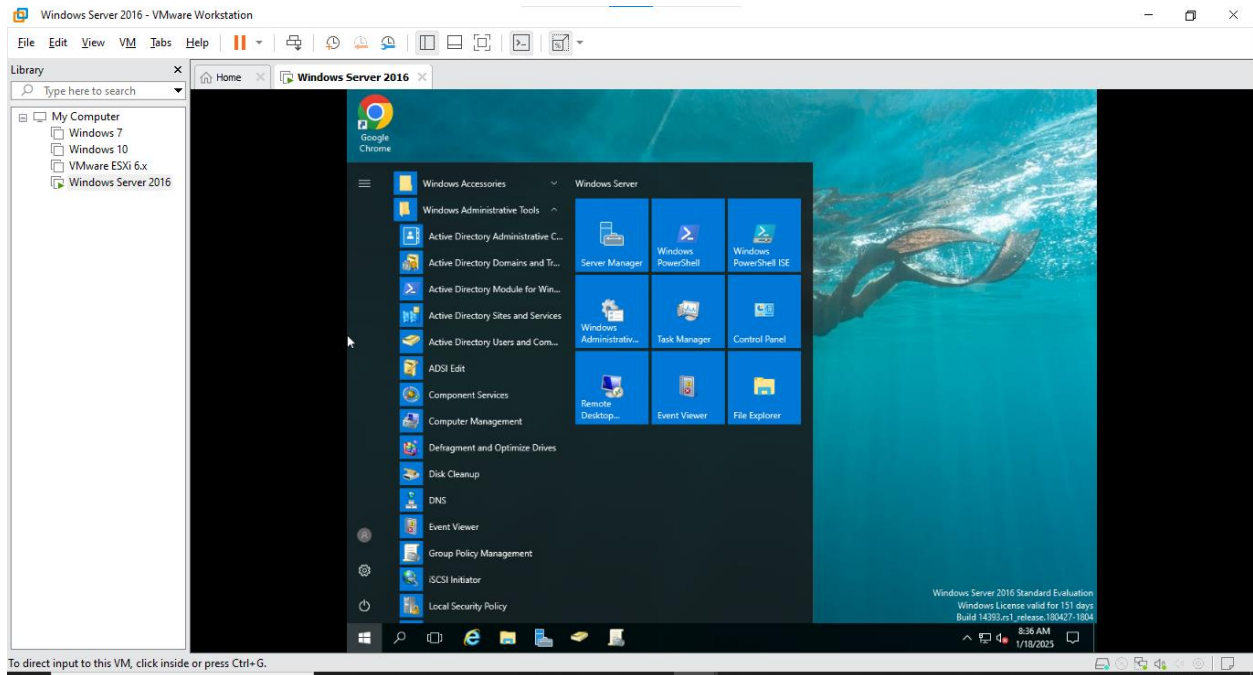
12.8.16. Server Promoted to Domain



13. Creating Users Account in Active Directory

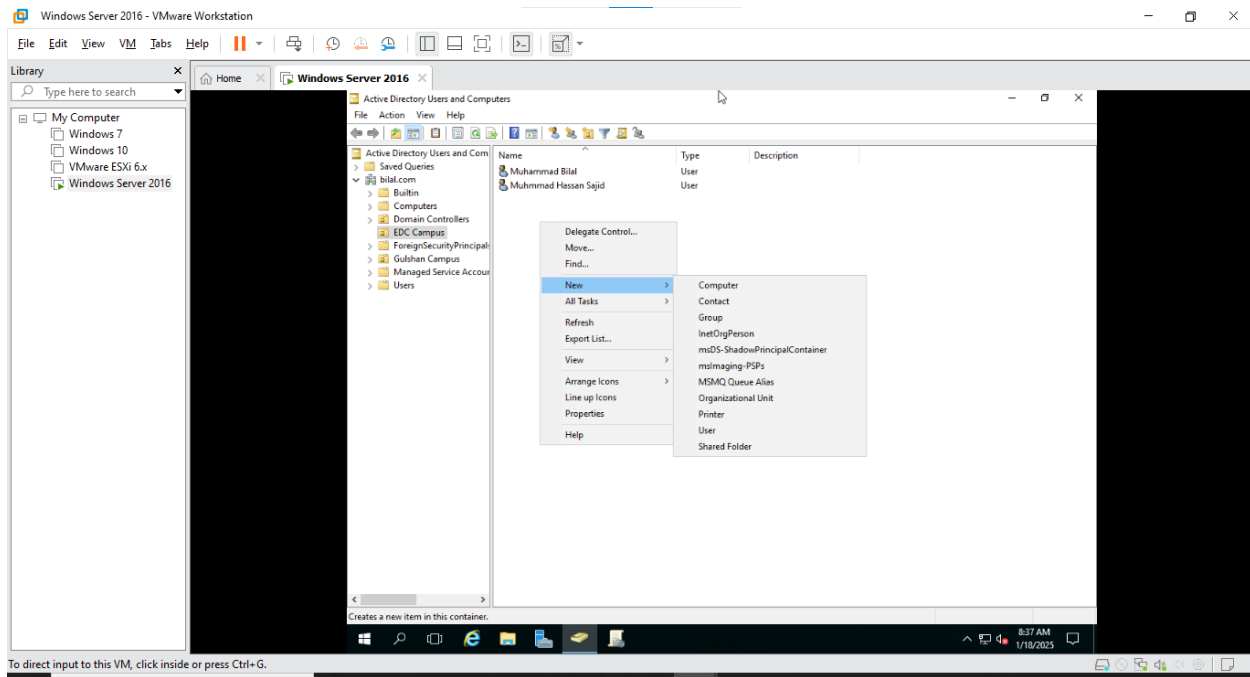


13.1. Open Active Directory Users and Computers

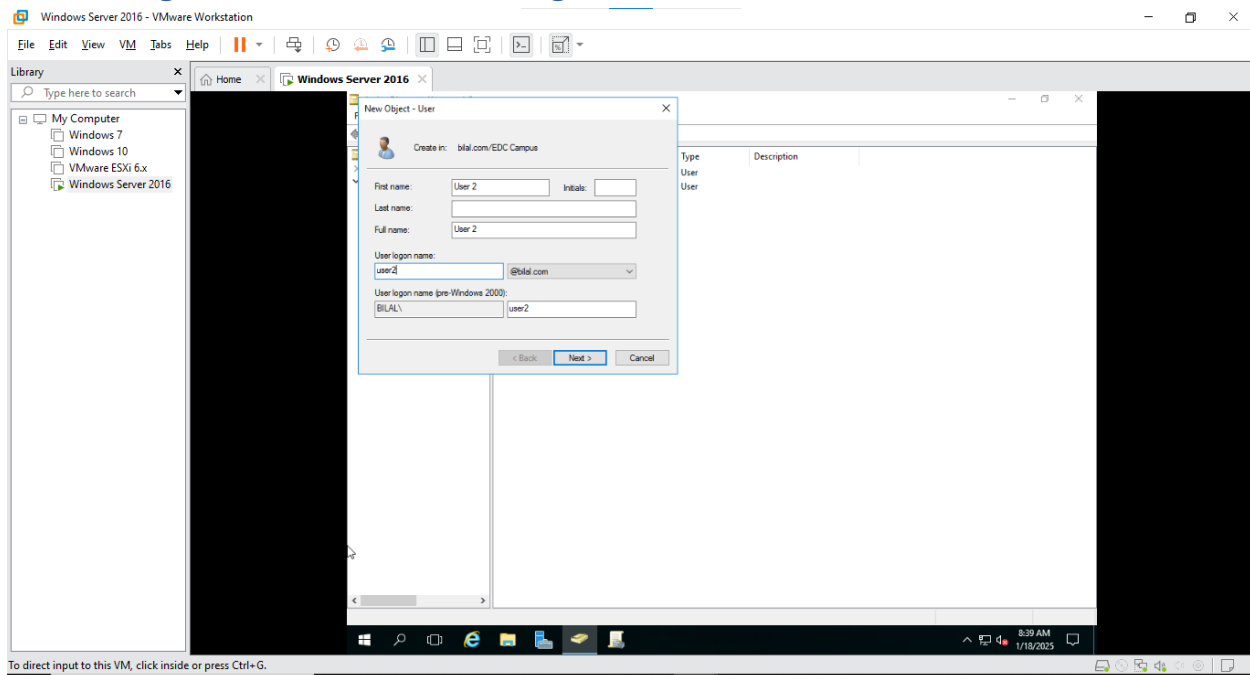


VIRTUALIZED DATA CENTER & DOMAIN SETUP

13.2. Create New User (User 2)

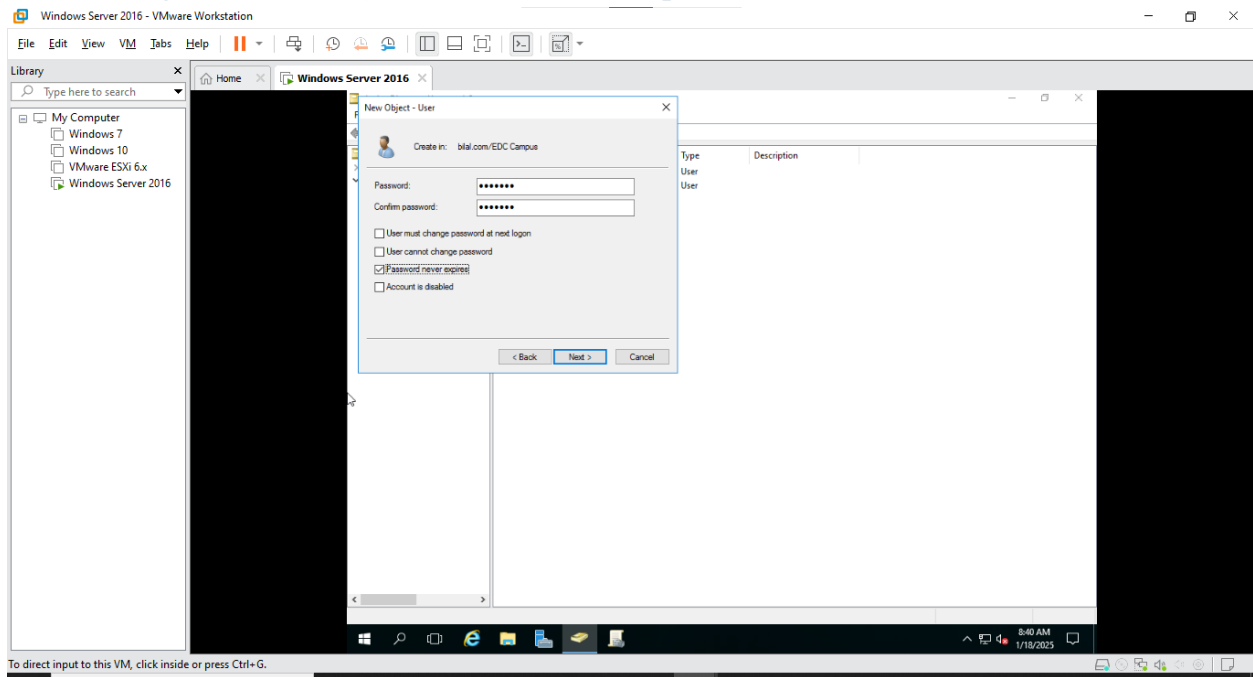


13.3. Assign Username and User Logon Name

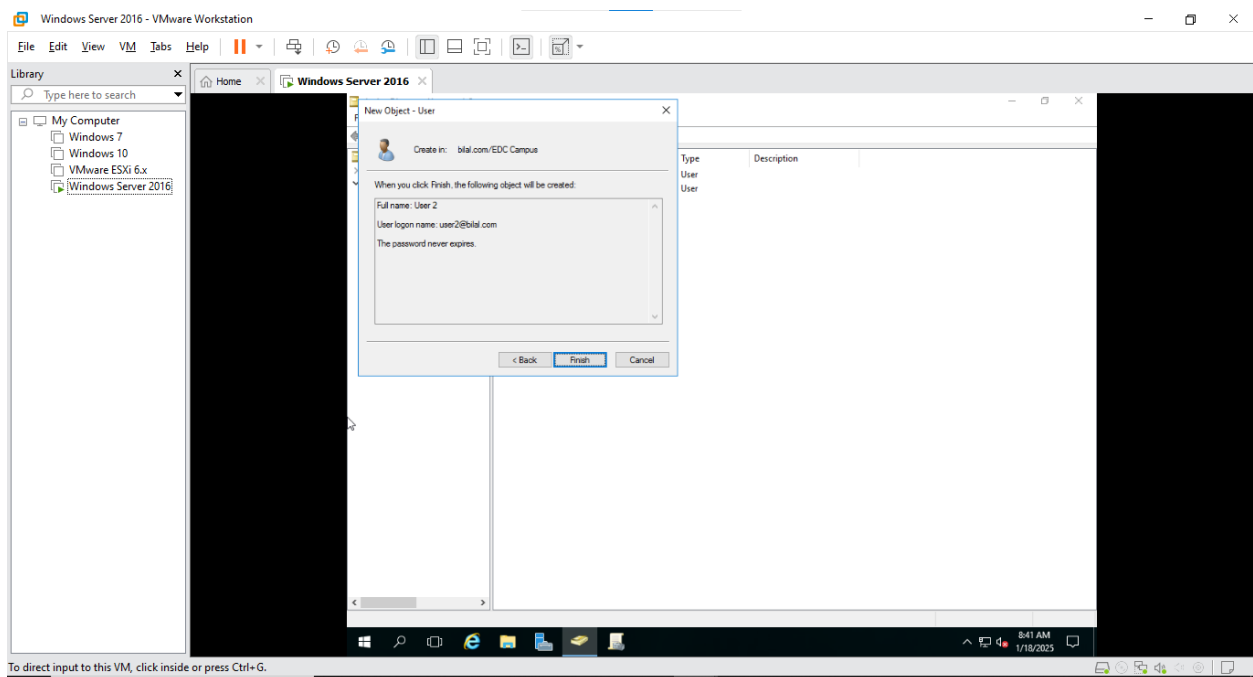


VIRTUALIZED DATA CENTER & DOMAIN SETUP

13.4. Assign Password (Password Never Expire)

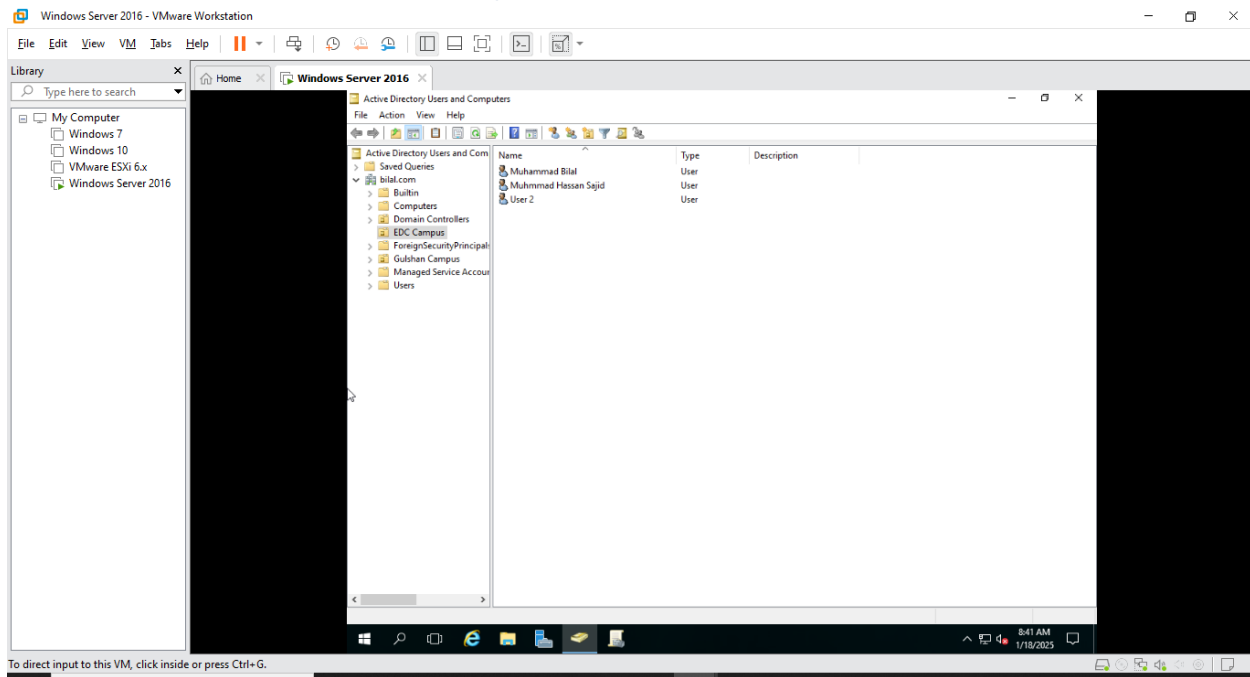


13.5. Finish Process

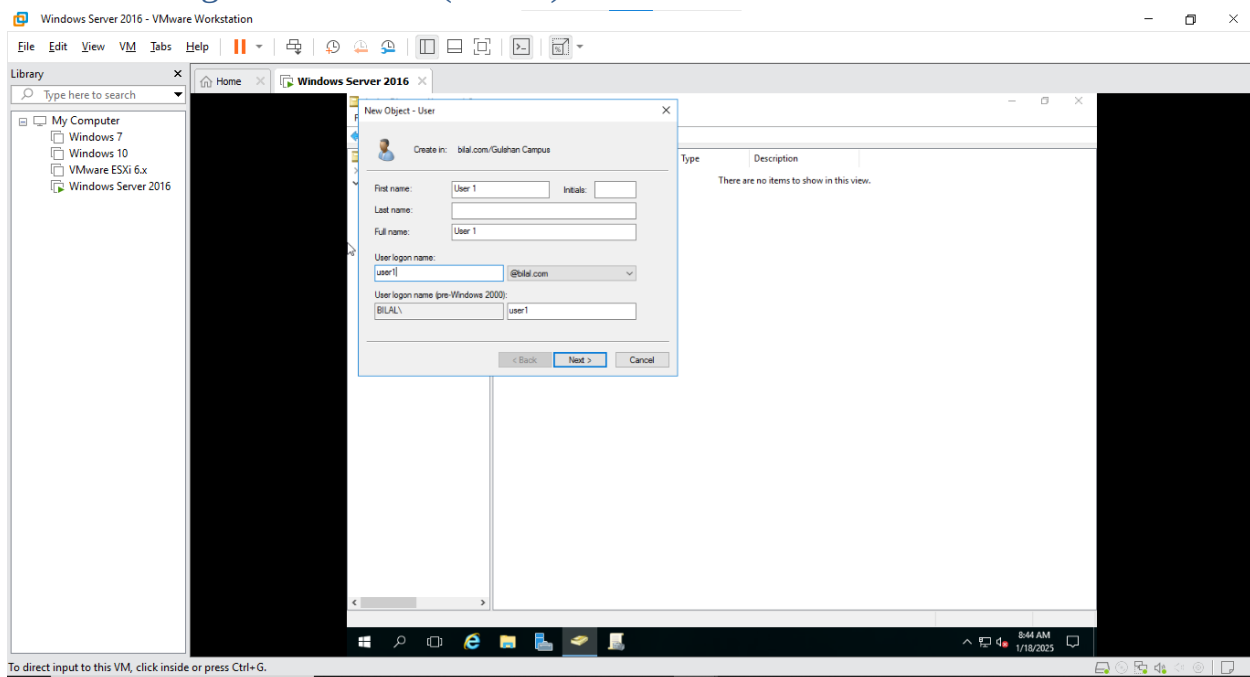


VIRTUALIZED DATA CENTER & DOMAIN SETUP

13.6. User 2 Created Successfully

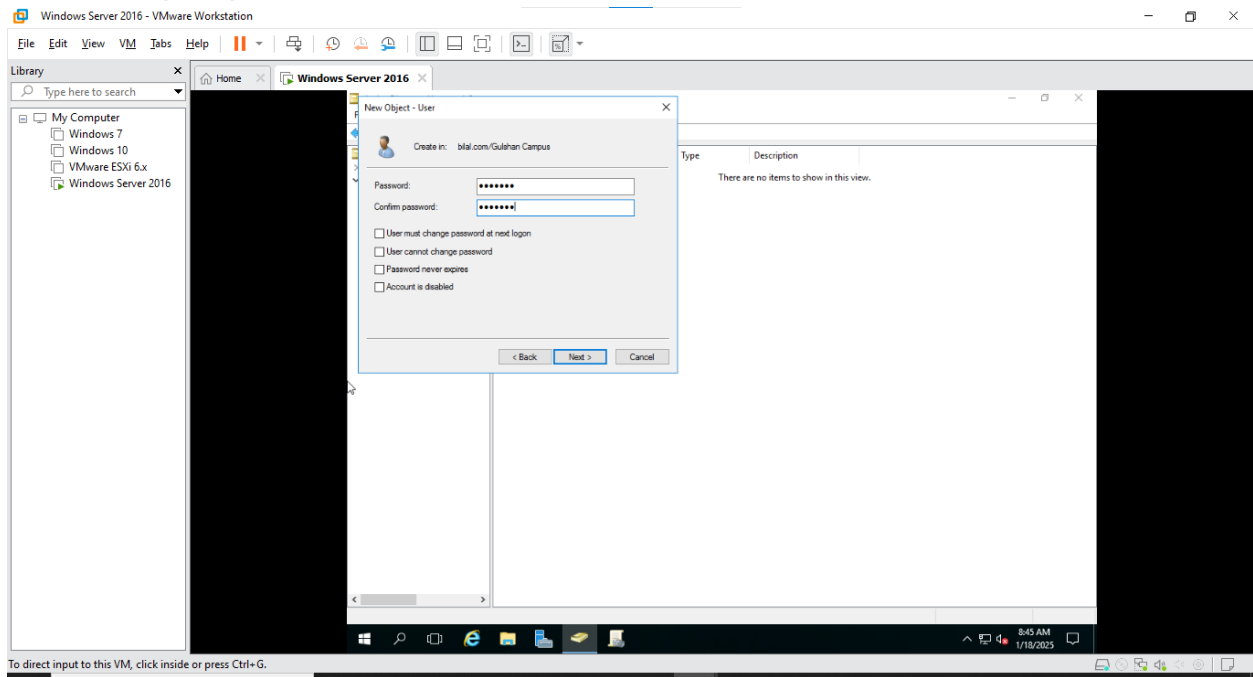


13.7. Creating Another User (User 1)

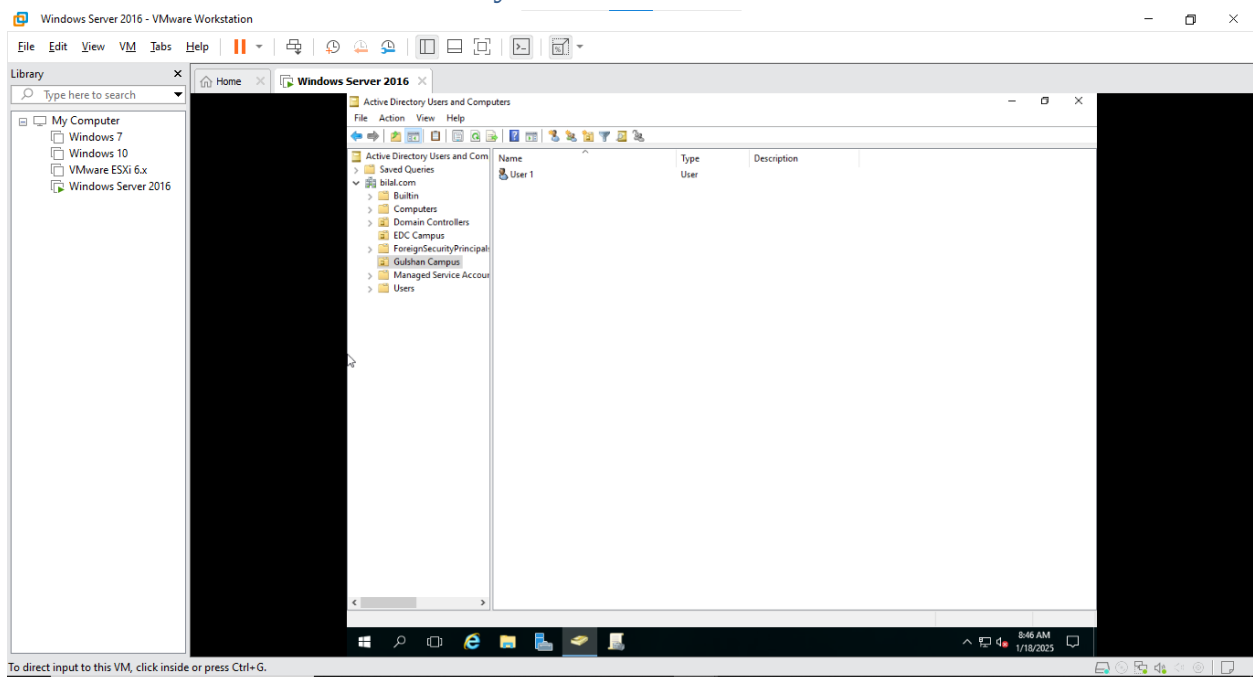


VIRTUALIZED DATA CENTER & DOMAIN SETUP

13.8. Assigning Password

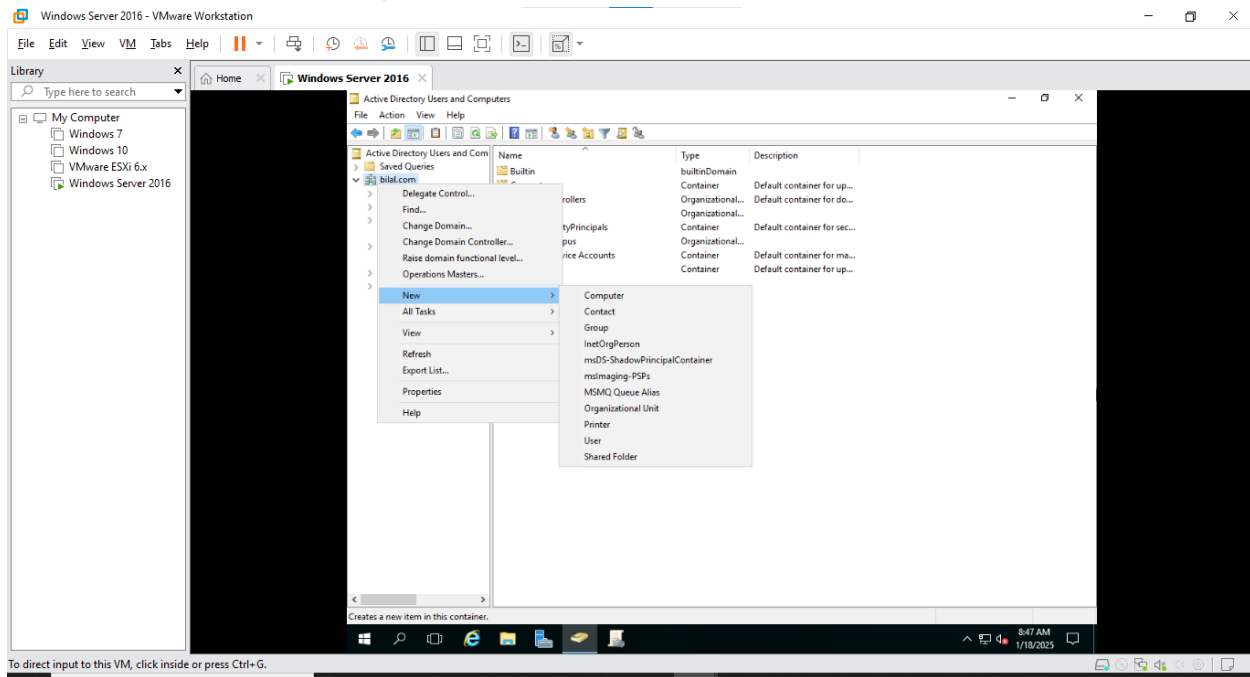


13.9. User 1 Created Successfully

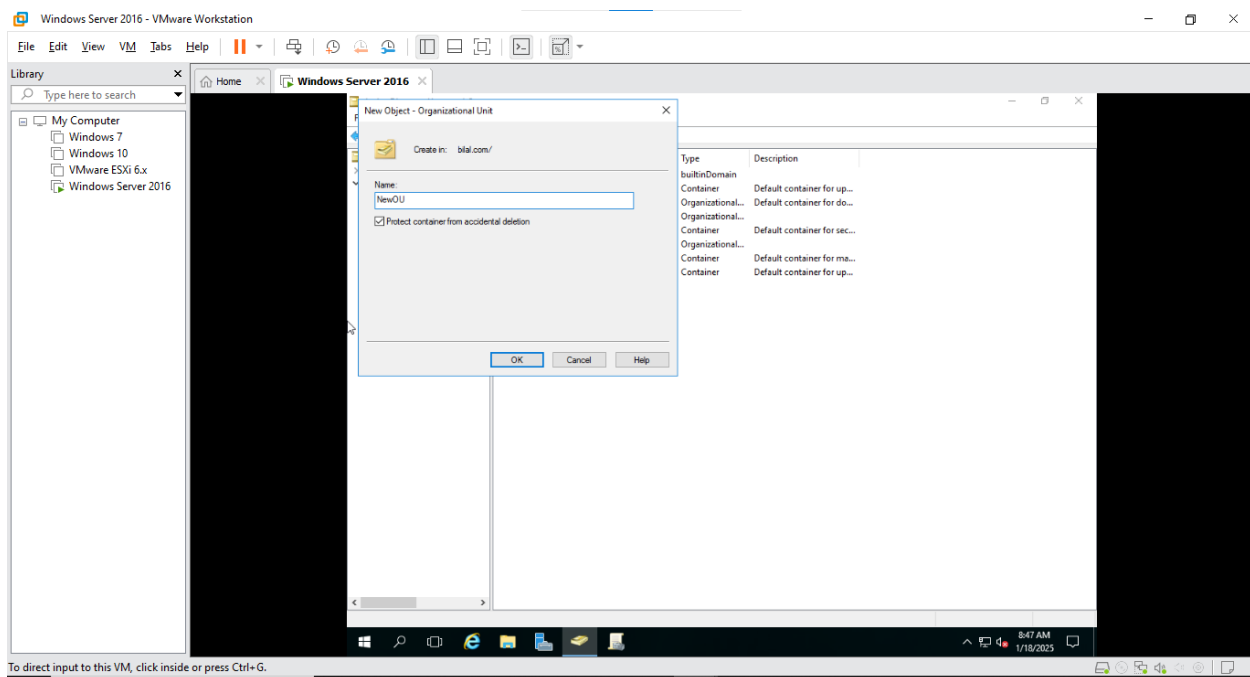


VIRTUALIZED DATA CENTER & DOMAIN SETUP

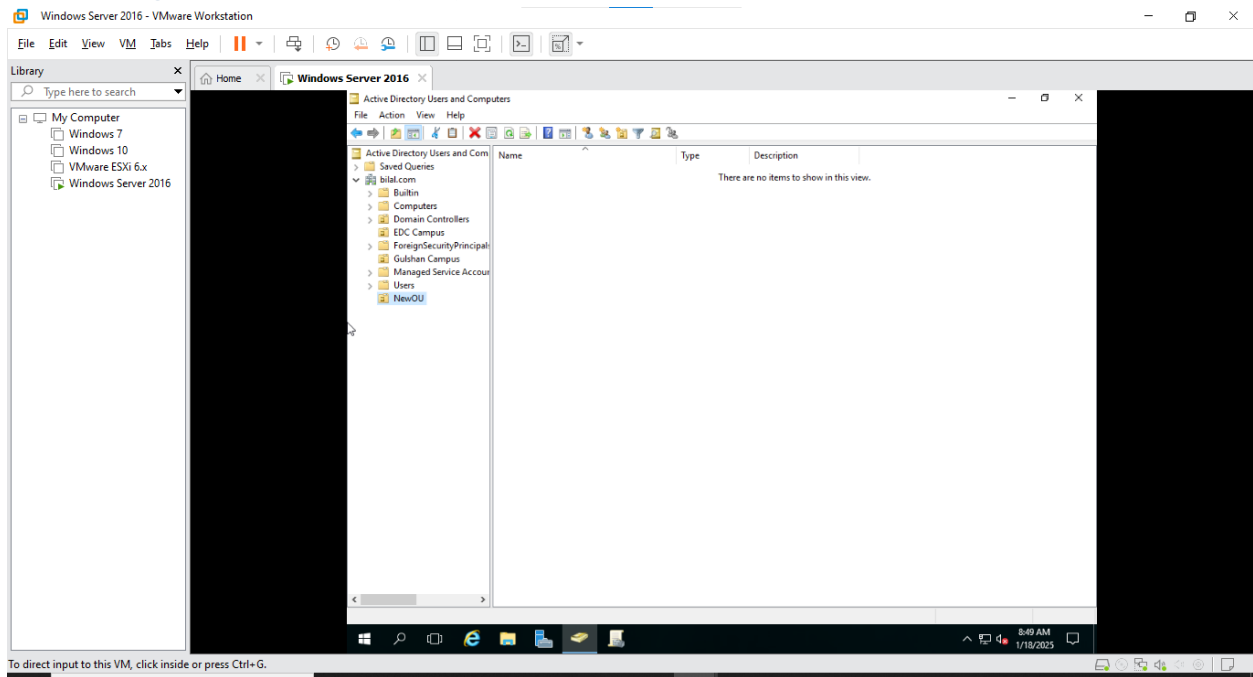
13.10. How to Create Organization Unit



13.11. Name OU and Check Accidental Deletion Box

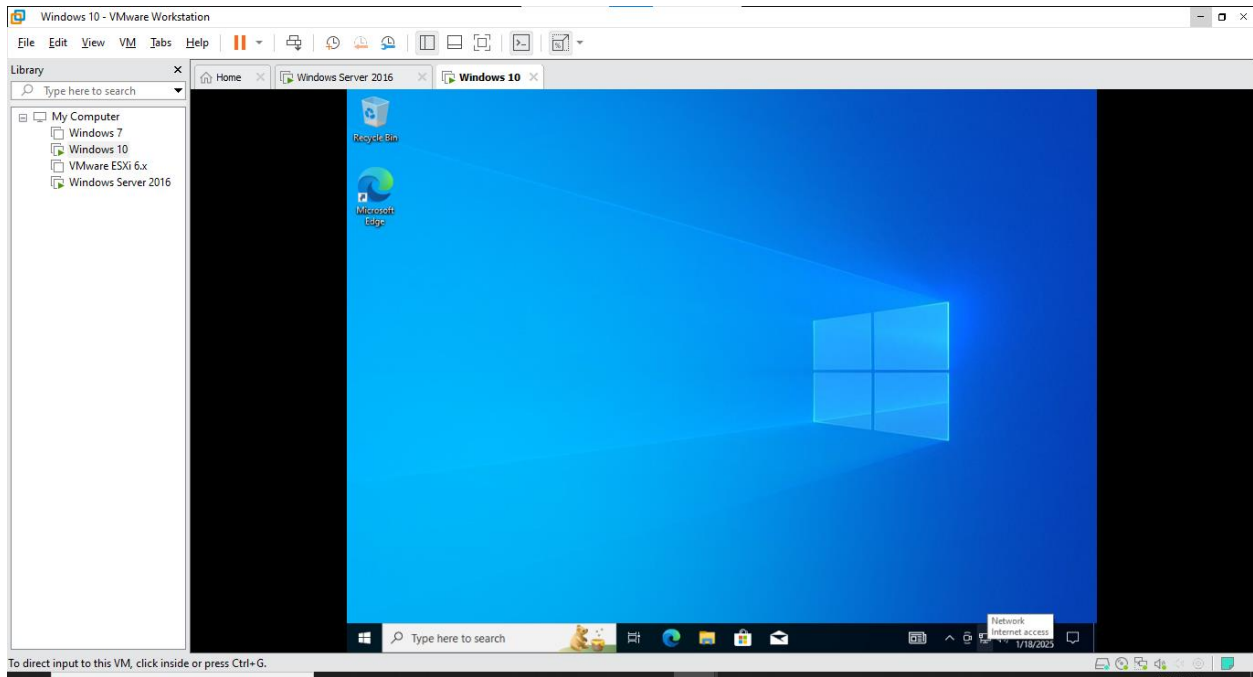


13.12. Organization Unit (OU) is Created



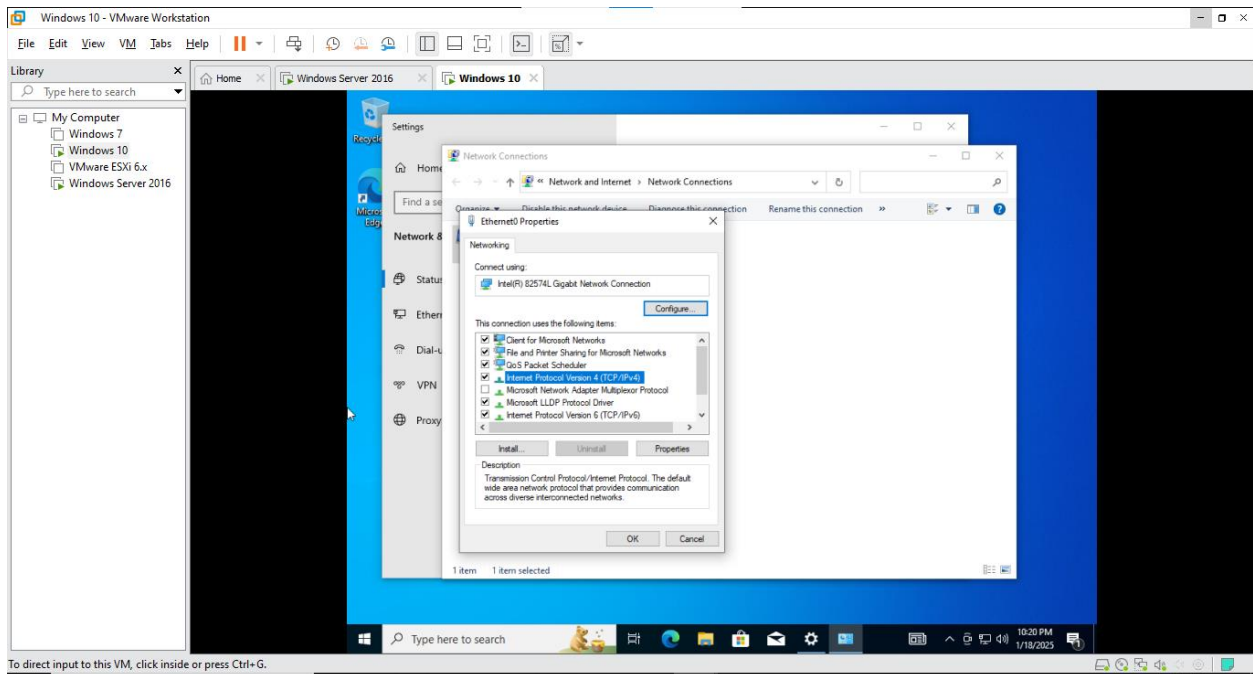
Note: In our case we have already created OU's naming 'EDC Campus' and 'Gulshan Campus' to manage student and apply GPO's accordingly to OU's and Domain.

14. Upgrading Windows 10 Machine to Domain (bilal.com)



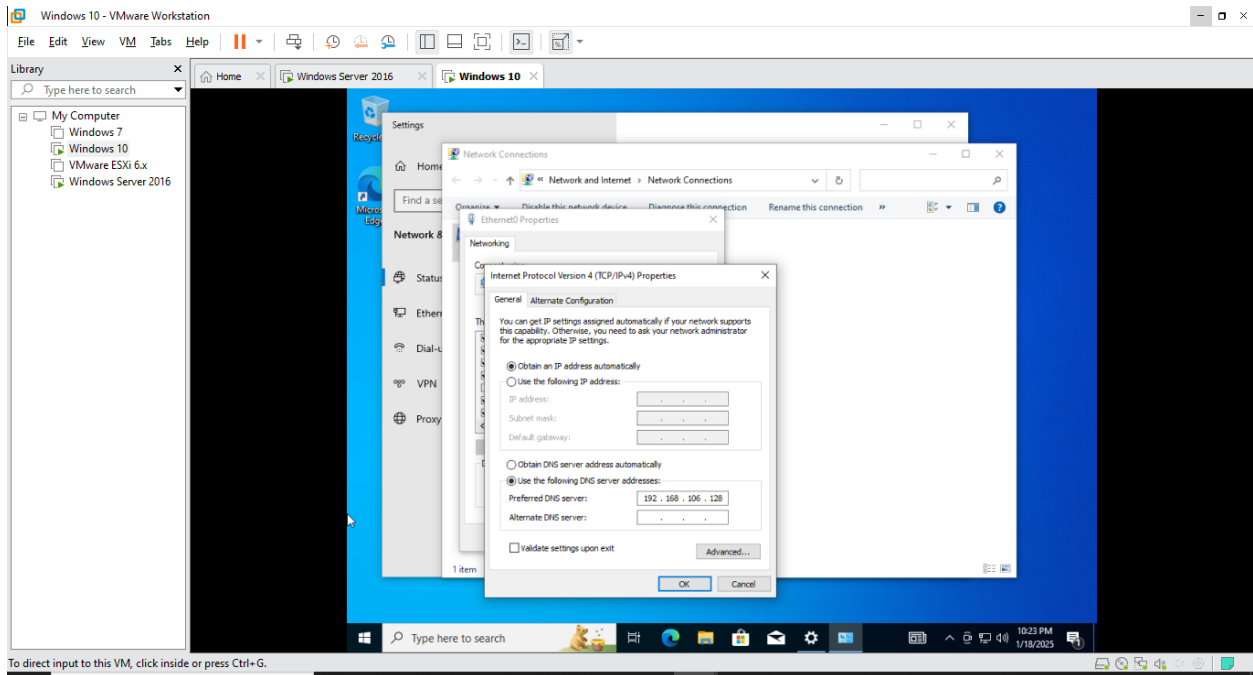
14.1. Update Adapter Setting (Obtain DNS Statically)

Settings → Network & Internet → Change adapter options → Right Click Ethernet → Properties



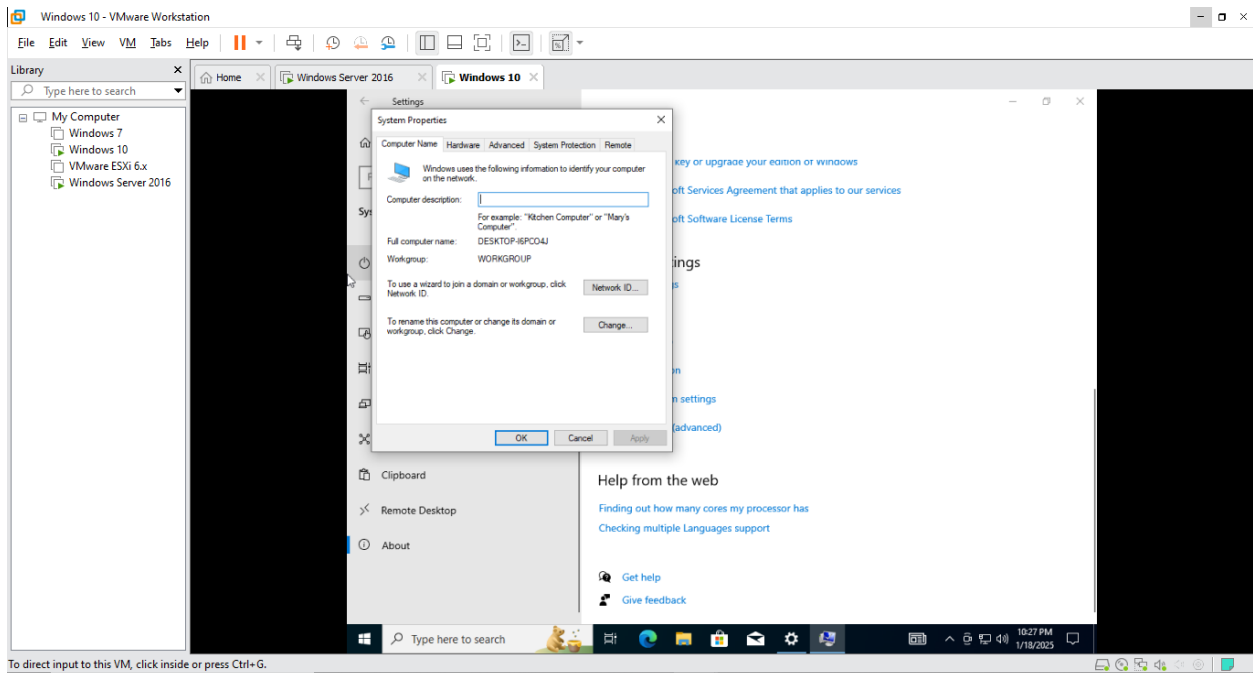
VIRTUALIZED DATA CENTER & DOMAIN SETUP

Click on Internet Protocol Version 4 (TCP/IPv4) → Right Click Properties



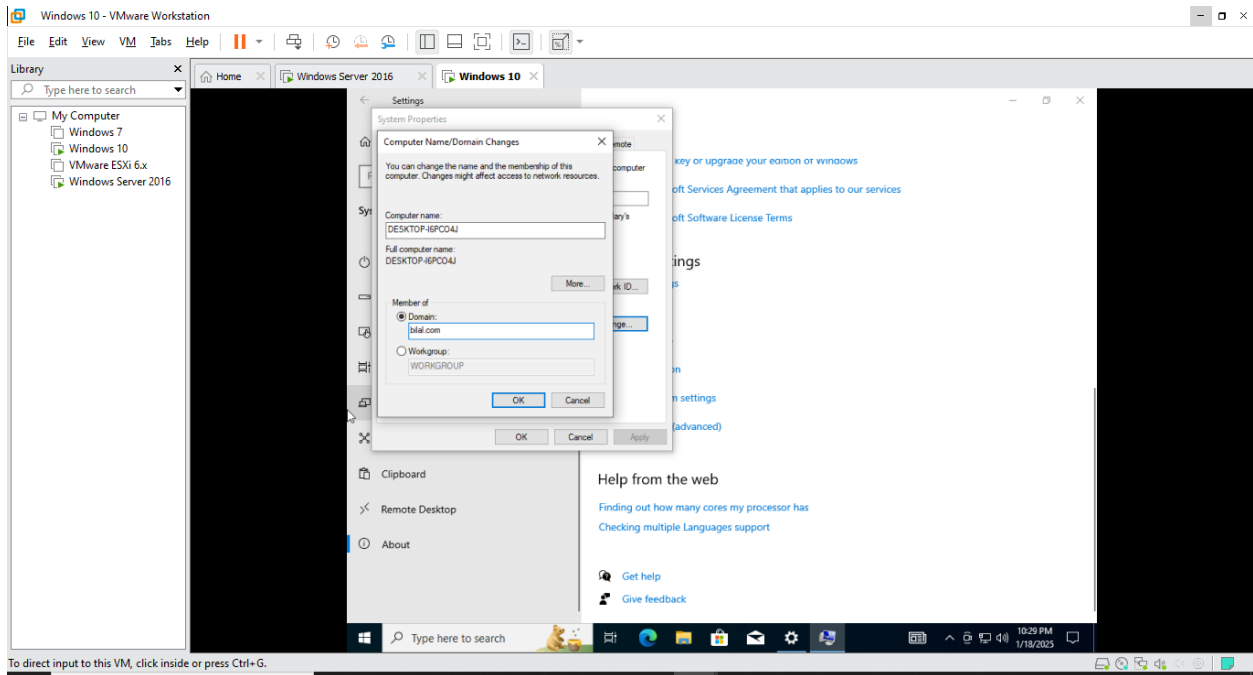
14.2. Upgrade Machine to Domain

Settings → System → About → Advance System Settings → Change Workgroup Environment

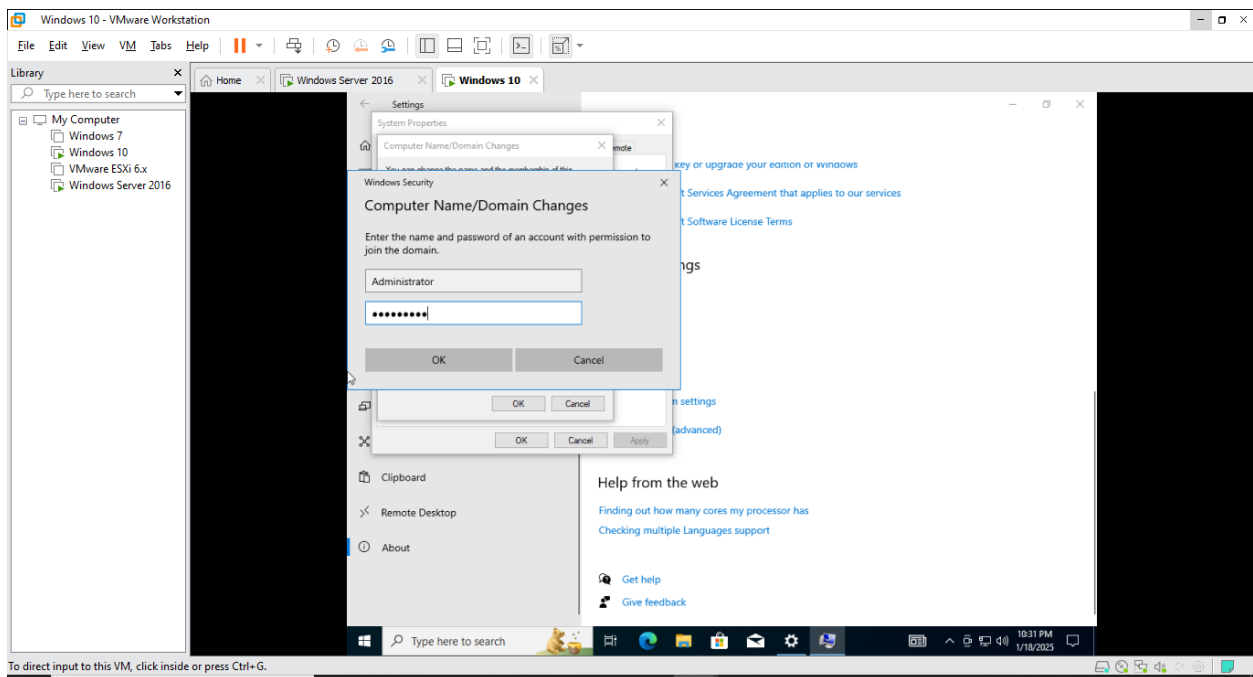


VIRTUALIZED DATA CENTER & DOMAIN SETUP

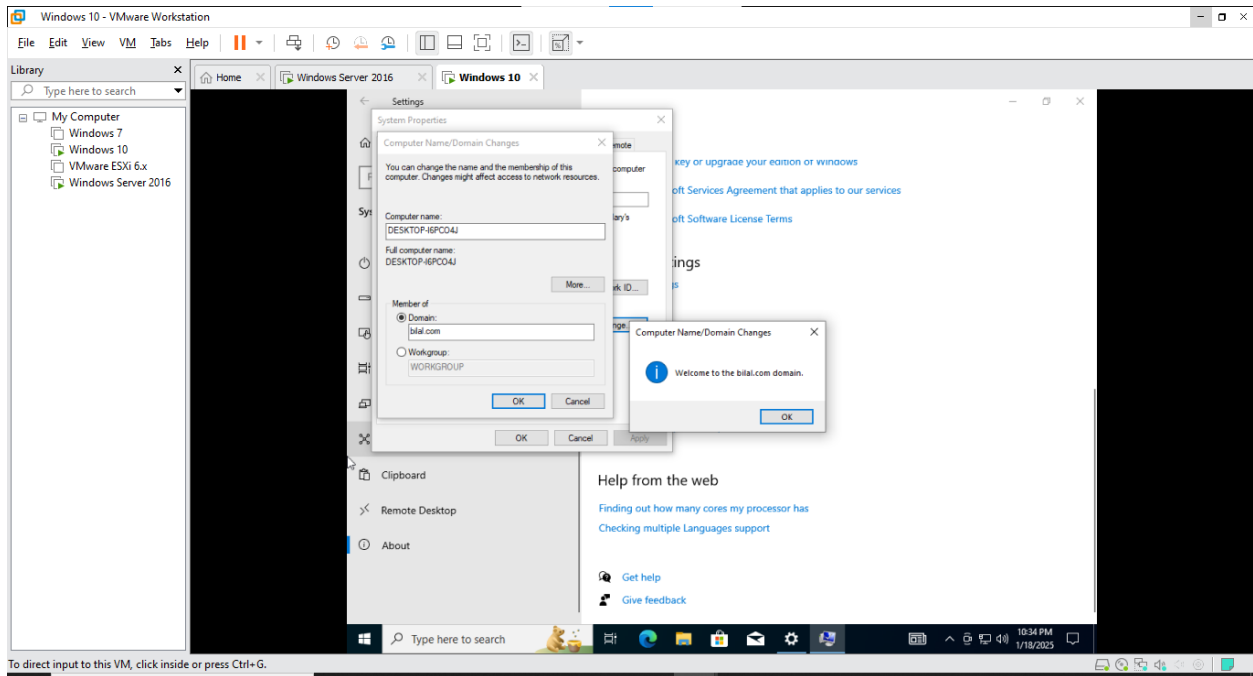
Insert Domain Name → Click Ok



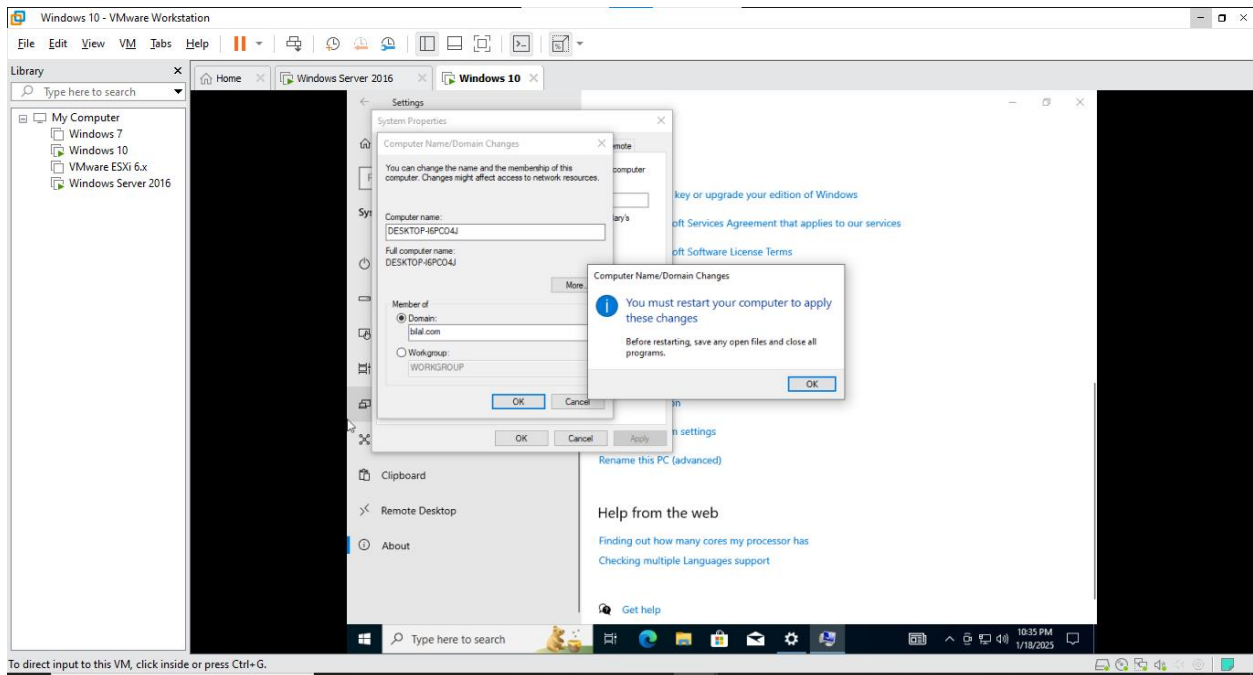
Insert Administrator Username and Password → Click Ok



VIRTUALIZED DATA CENTER & DOMAIN SETUP

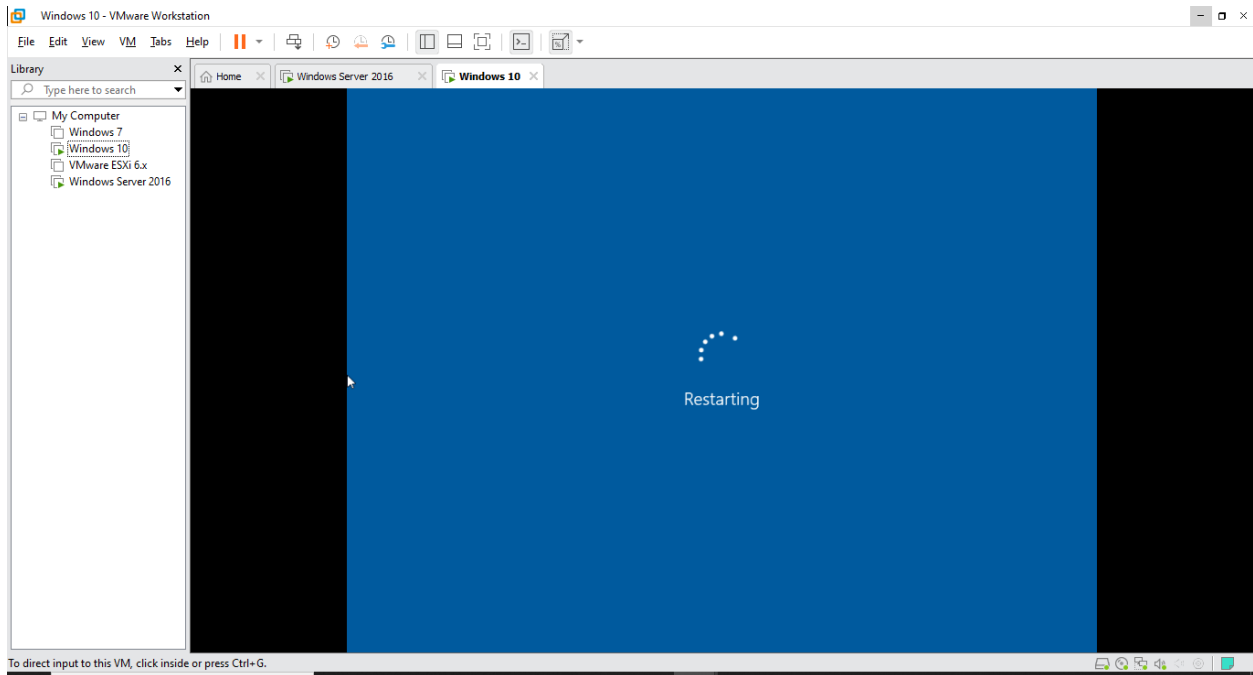


Click Ok → Restart the Computer to Finish

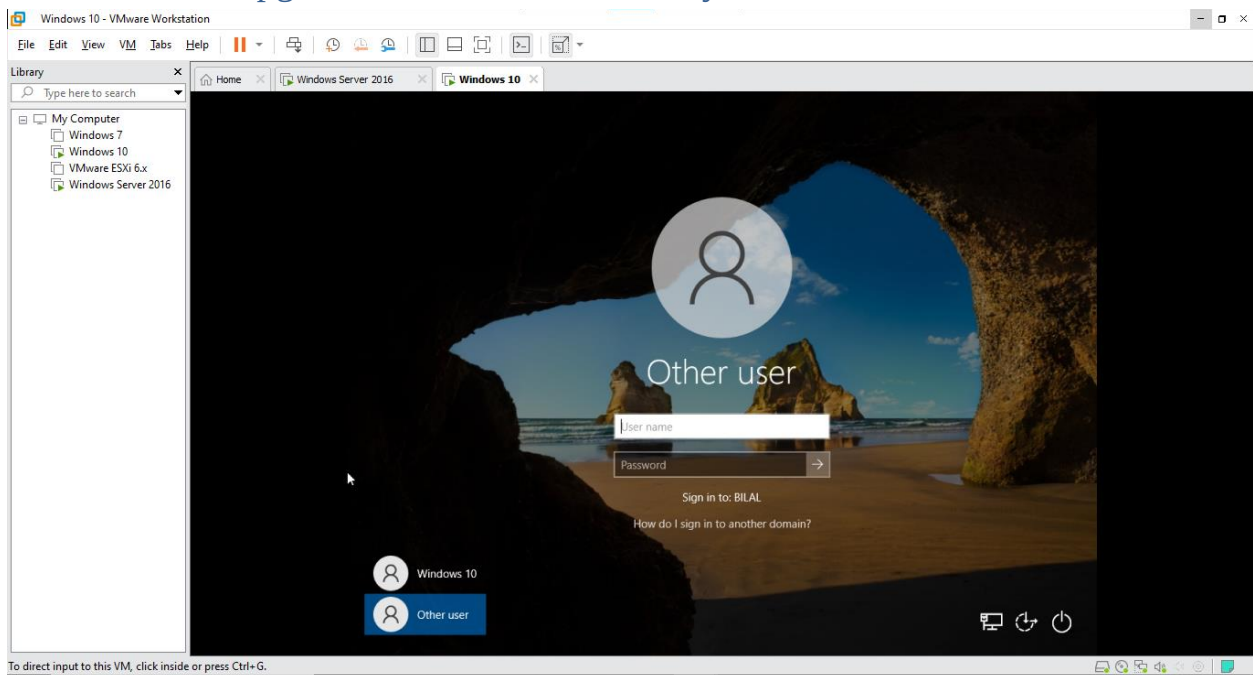


VIRTUALIZED DATA CENTER & DOMAIN SETUP

Computer Restarting

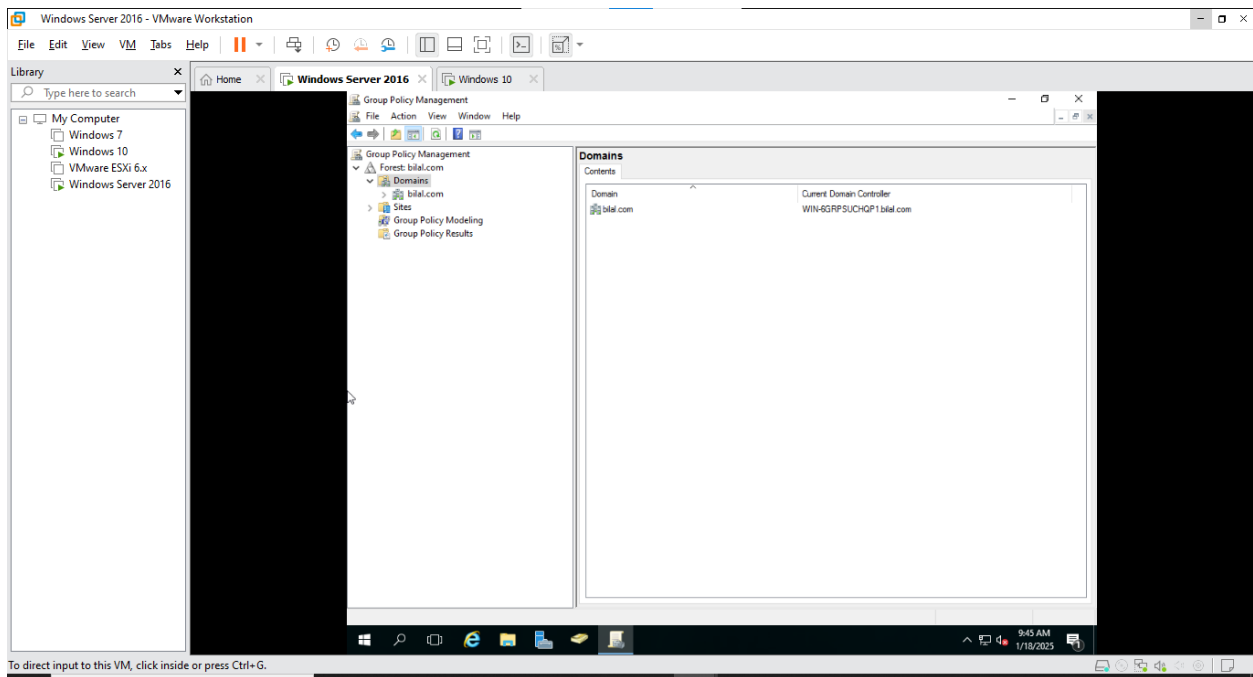


14.3. Machine Upgraded to Server Successfully

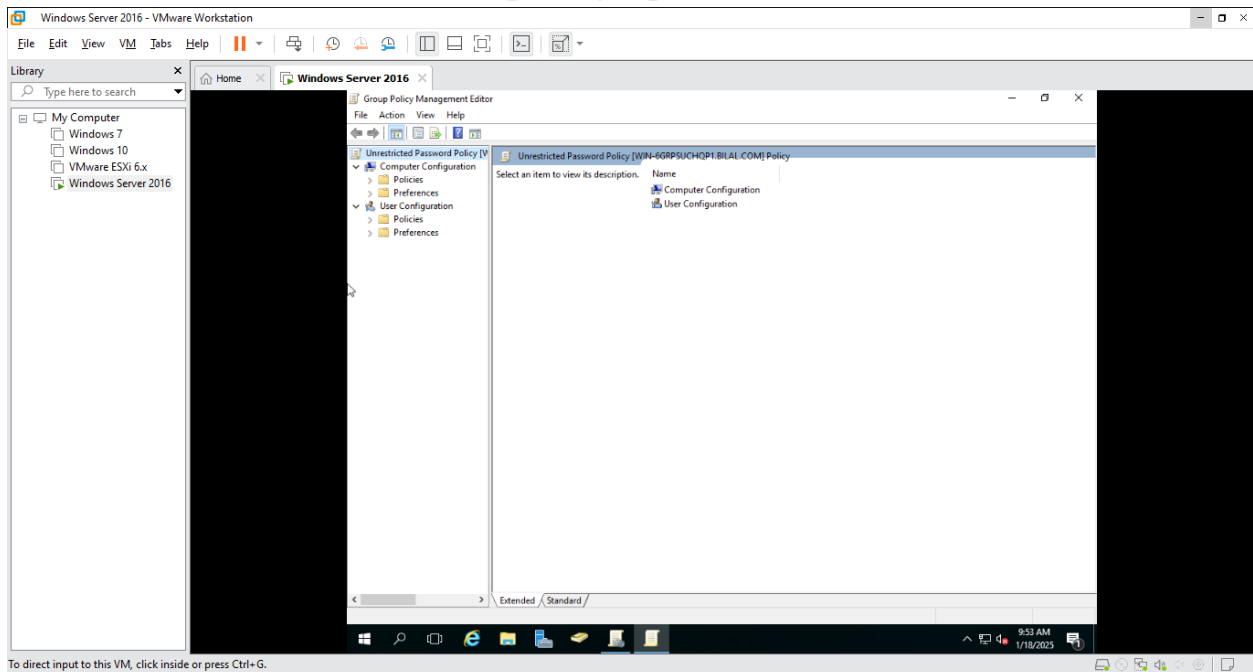


15. Group Policy Management Editor (Windows Server)

Start → Windows Administrative Tools → Group Policy Management

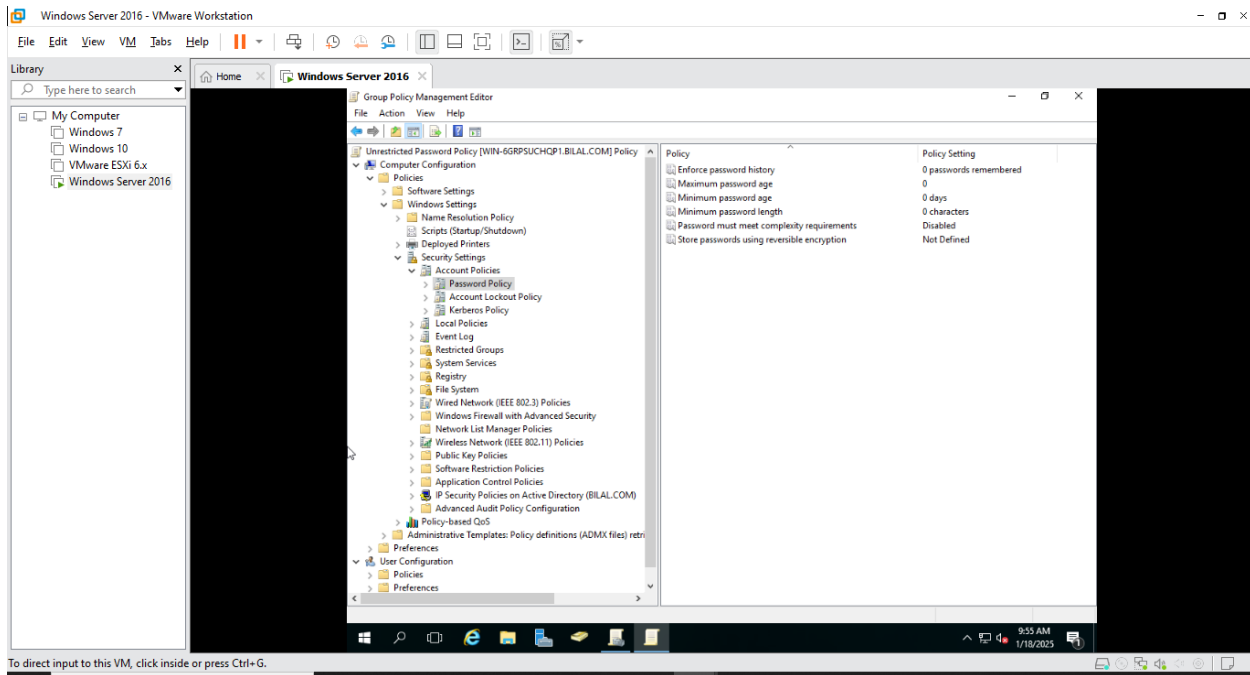


15.1. Users should be able to setup a login password with no restriction.

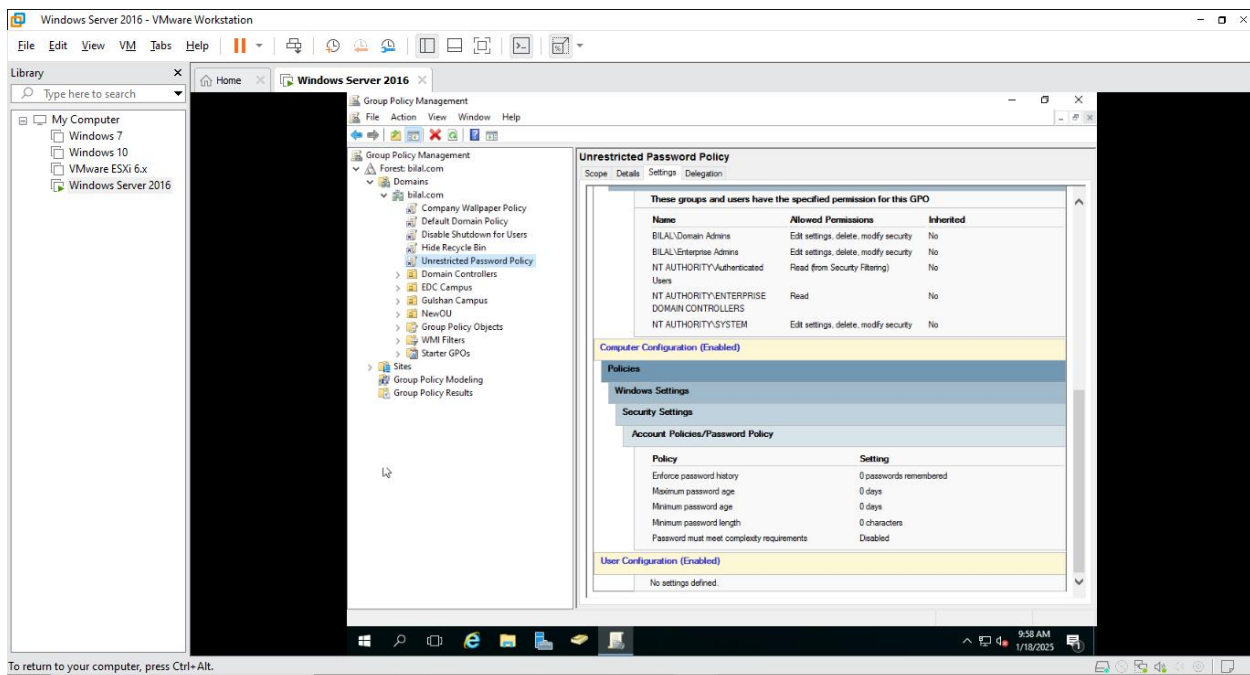


VIRTUALIZED DATA CENTER & DOMAIN SETUP

Policies → Windows Settings → Security Settings → Account Policies/Password Policy

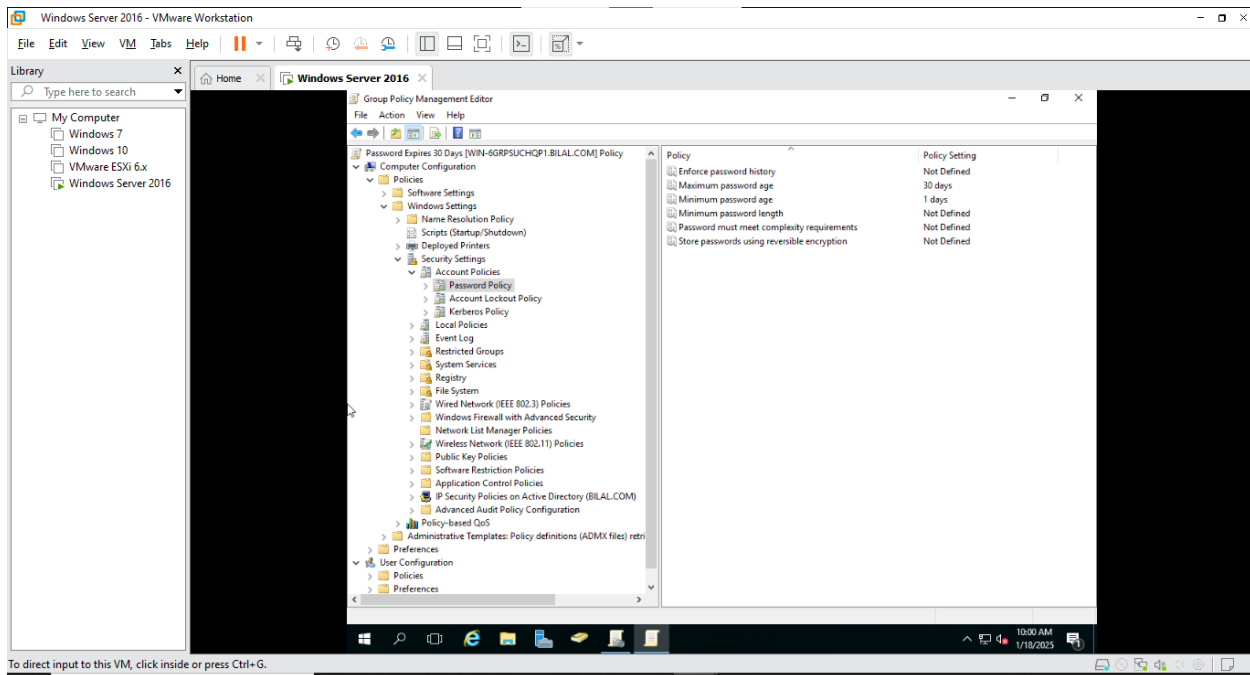


Group Policy Created Successfully

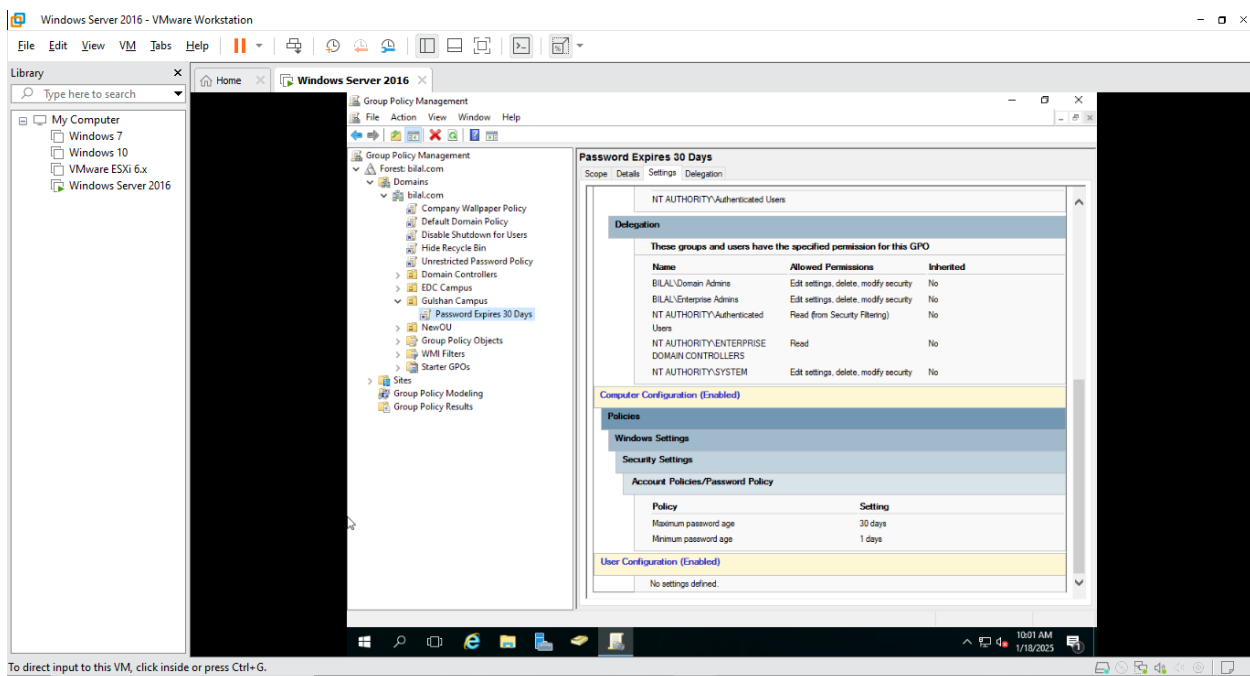


VIRTUALIZED DATA CENTER & DOMAIN SETUP

15.2. User1 password will expire in 30 days, User2 password will never expire.
Policies → Windows Settings → Security Settings → Account Policies/Password Policy



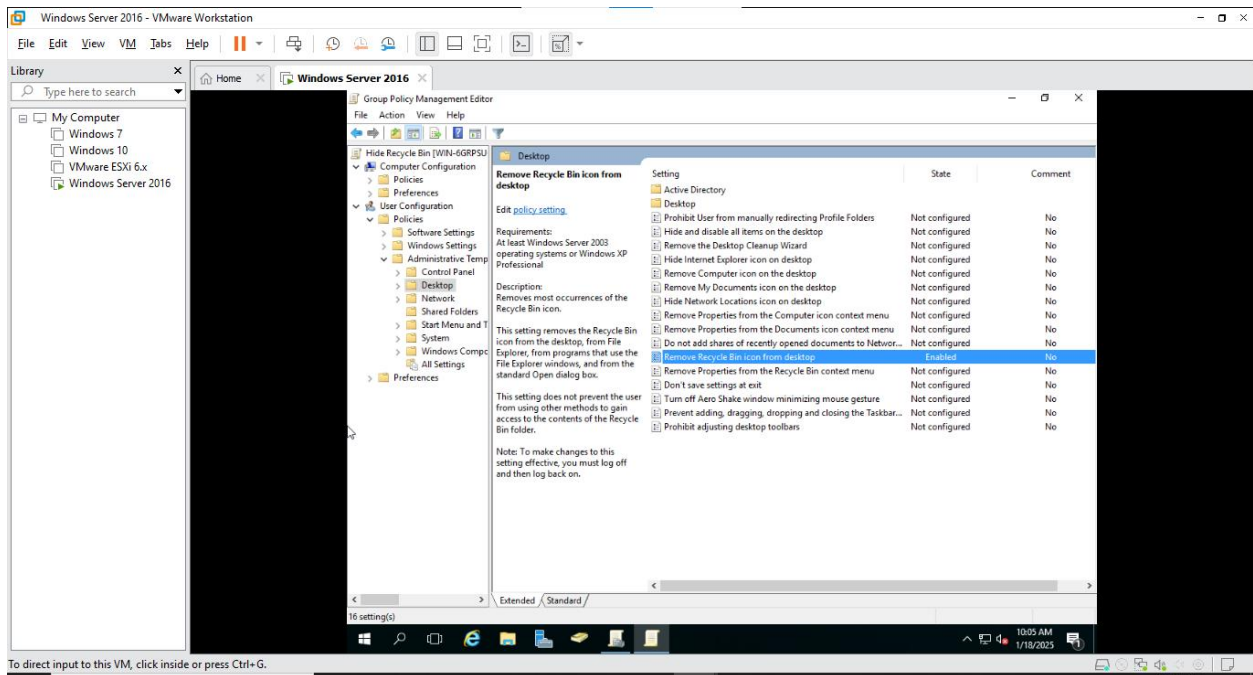
Group Policy Created Successfully



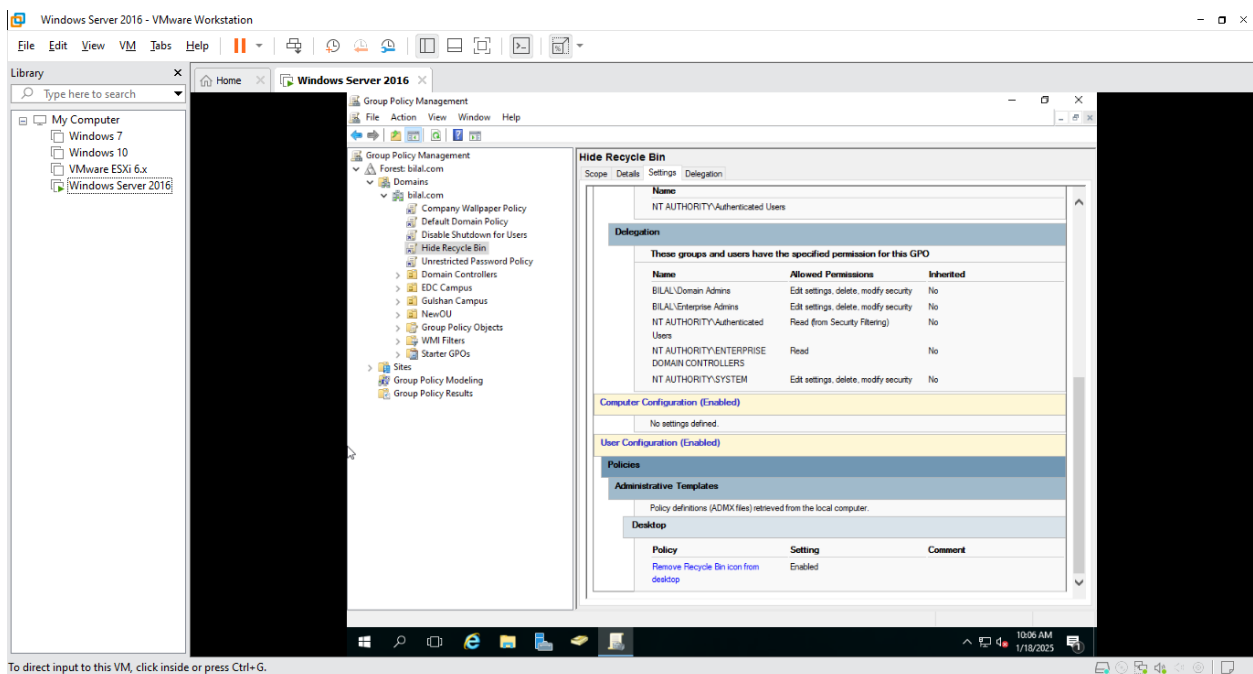
VIRTUALIZED DATA CENTER & DOMAIN SETUP

15.3. Recycle bin should not appear on any user profile.

User Configuration → Policies → Administrative Template → Desktop → Remove Recycle Bin



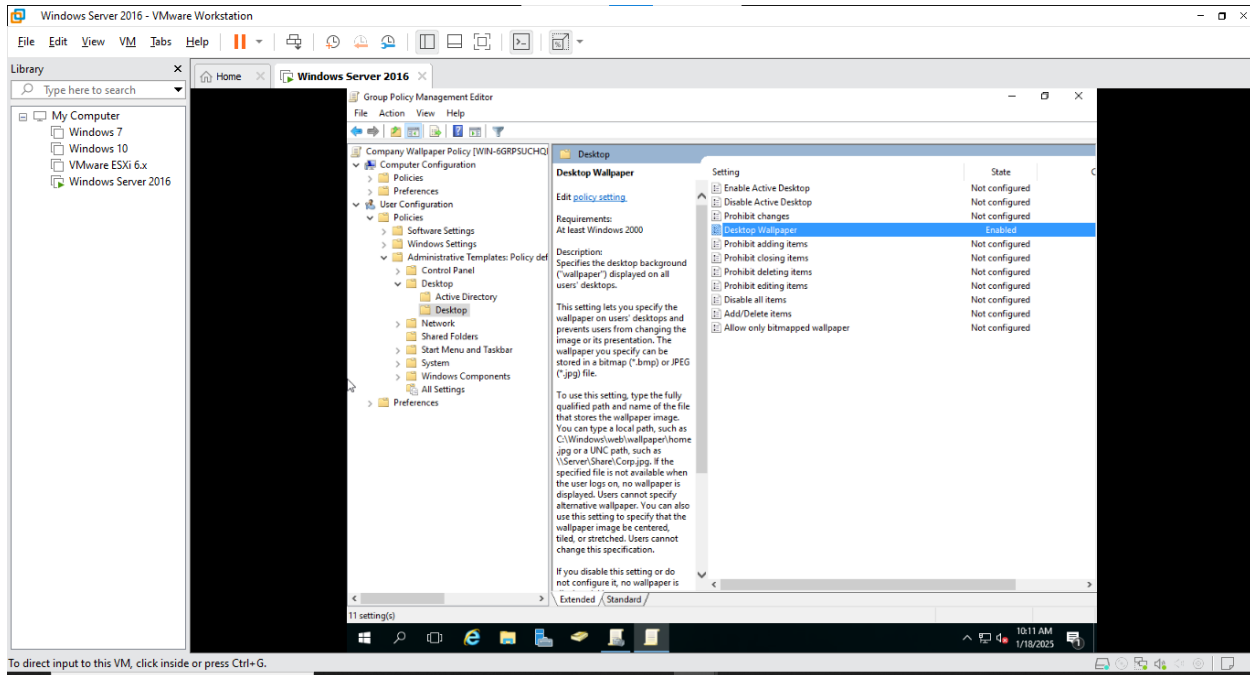
Group Policy Created Successfully



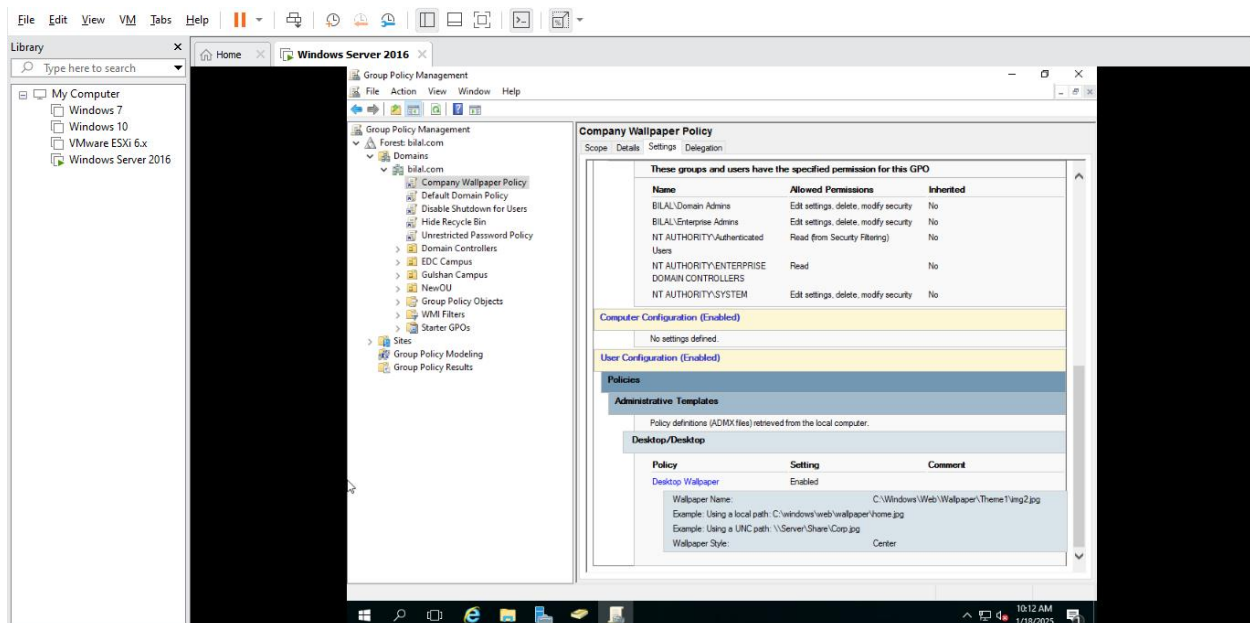
VIRTUALIZED DATA CENTER & DOMAIN SETUP

15.4. Customized Company's Wall Paper should appear on all user's desktop.

User Configuration → Policies → Administrative Template → Desktop/Desktop → Policy → Desktop Wallpaper



Group Policy Created Successfully



16. Group Policy Implementation (according to given scenario)

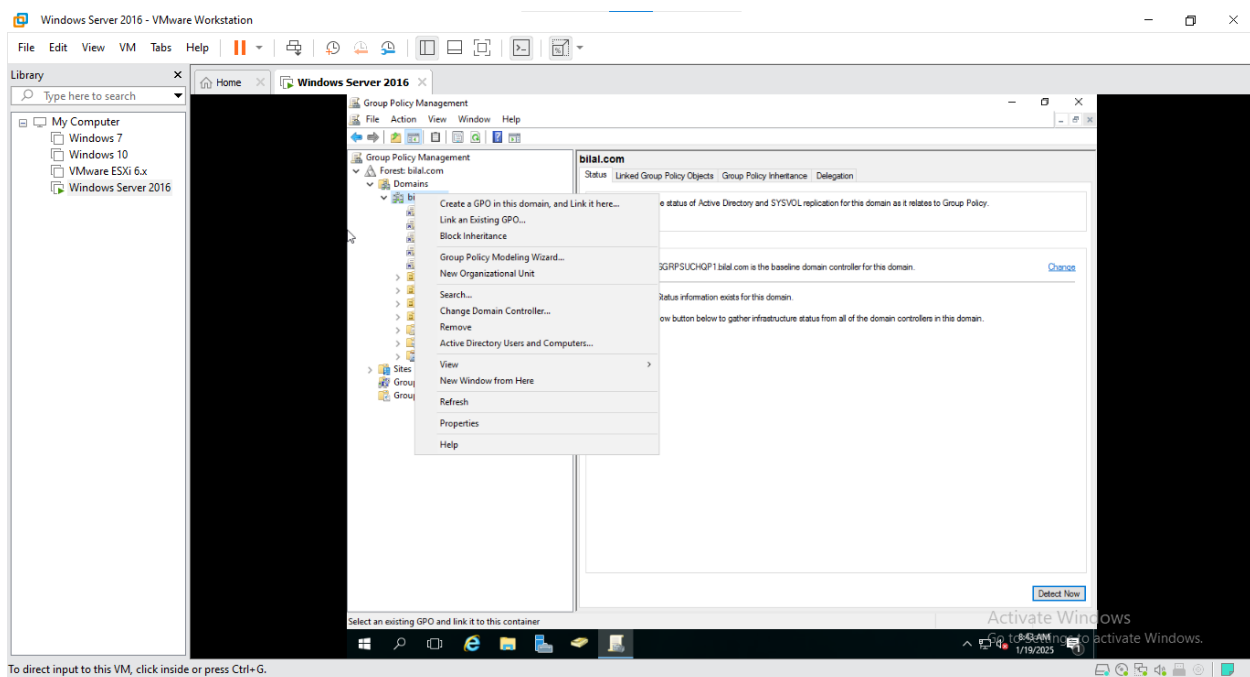
Following is the conceptual implementation of GPO's:

16.1. Users should be able to setup a login password with no restriction.

As users should be able to login password without restriction, this GPO is for all users in domain. So best scenario is to apply/link this GPO on Domain as the Organization Units (OU) will inherit it as block inheritance is disable on OU's.

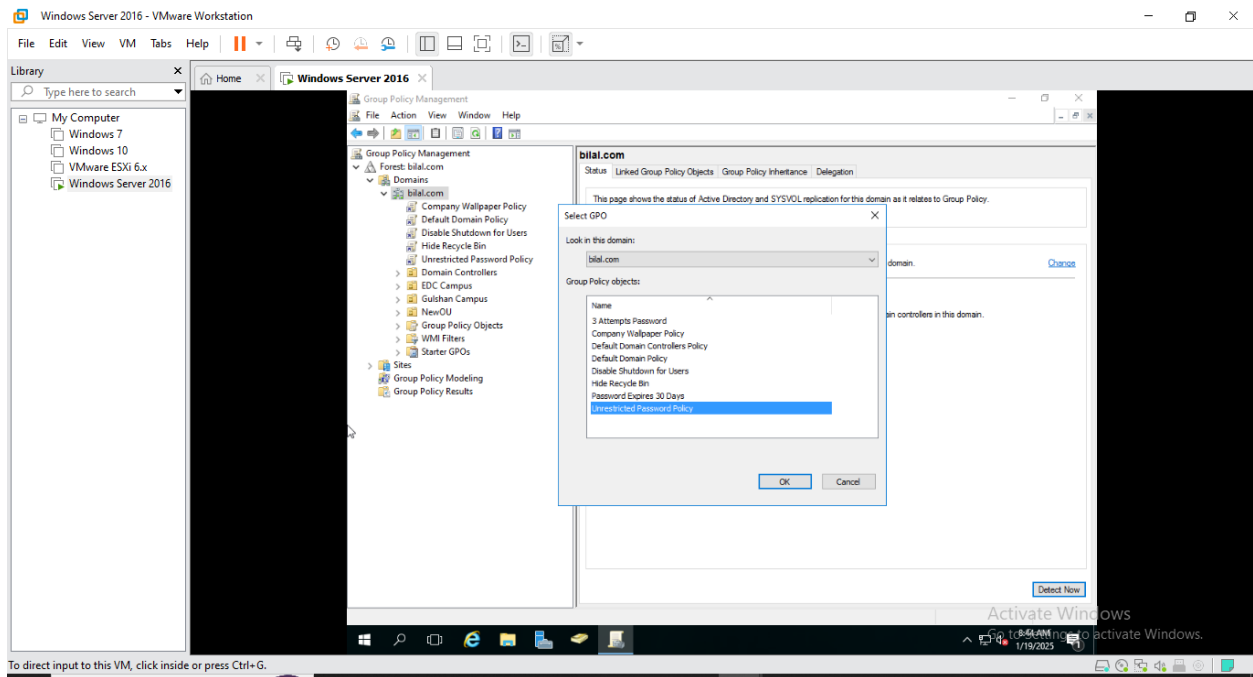
How to link GPO on Domain?

Open Group Policy Management Editor → Right Click on Domain → Link a Existing GPO (if created in group policy objects)

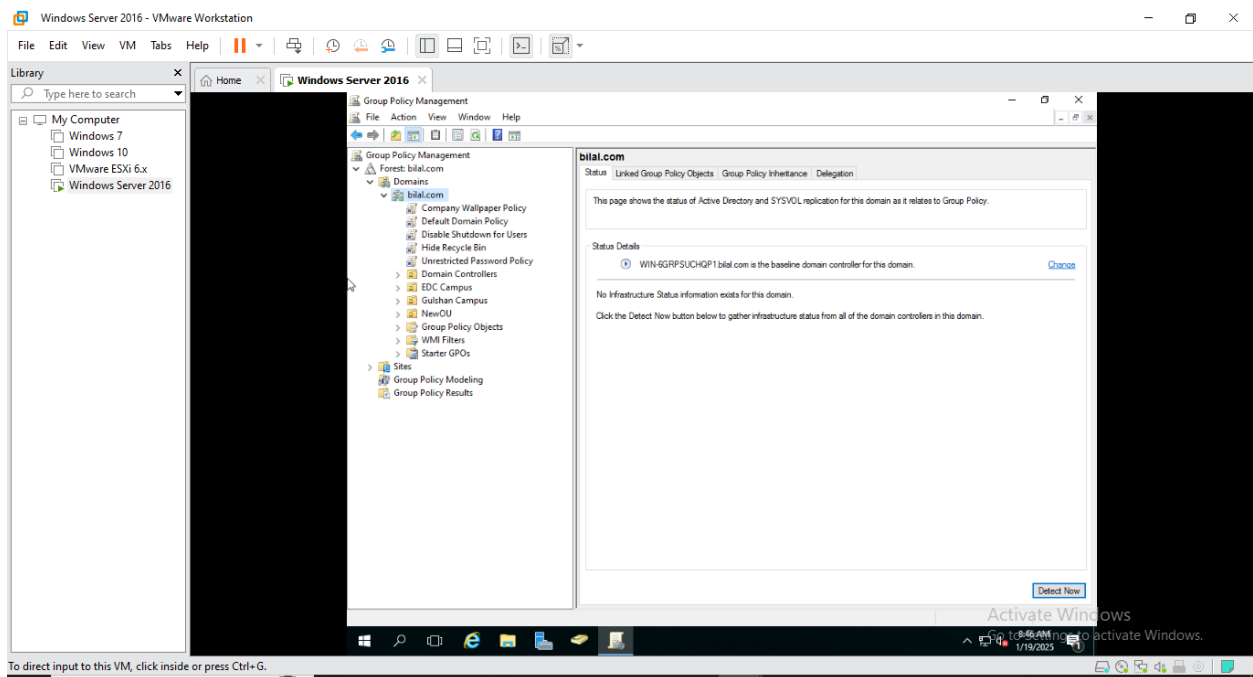


VIRTUALIZED DATA CENTER & DOMAIN SETUP

Select GPO → Click Ok



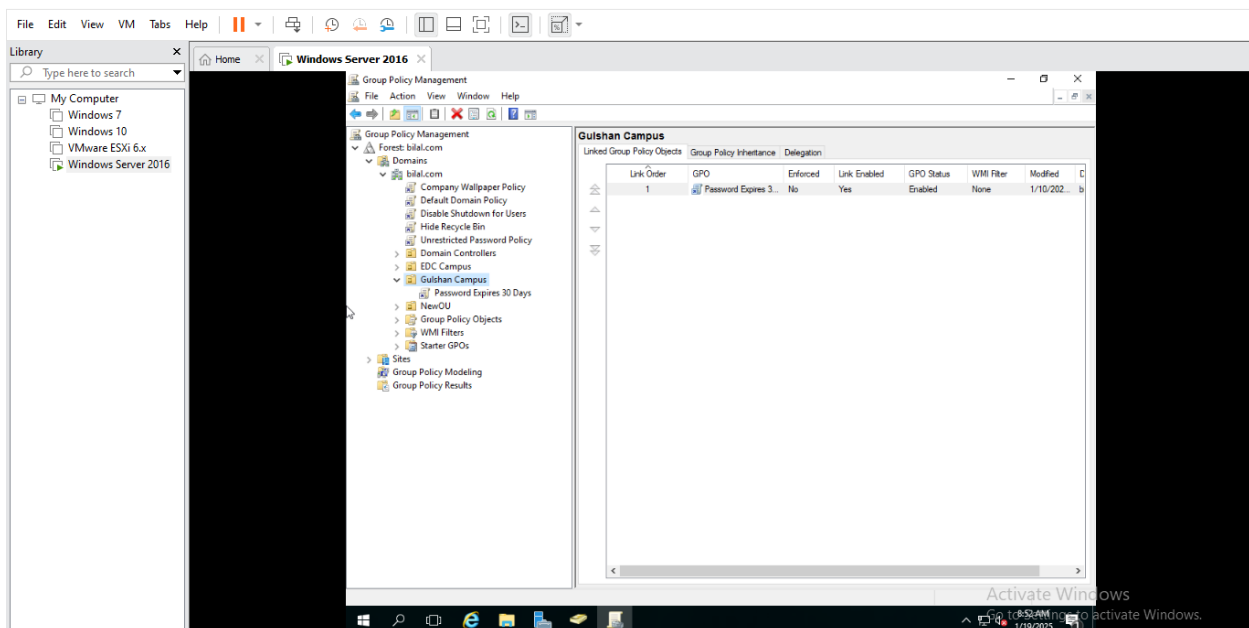
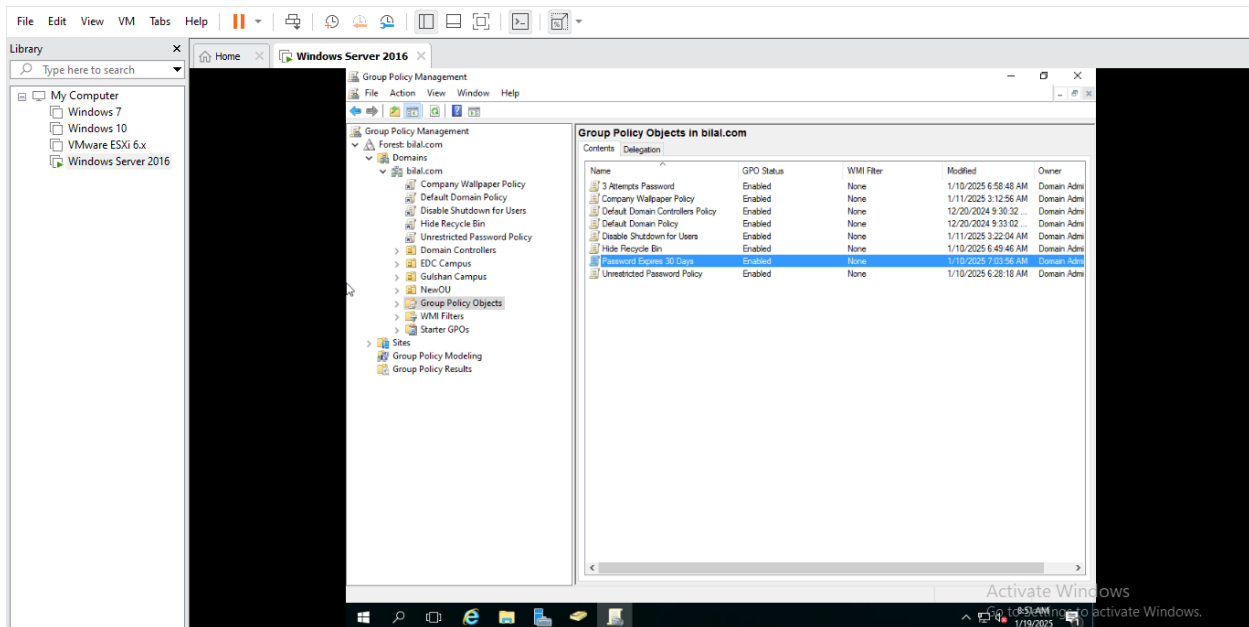
GPO has been linked



VIRTUALIZED DATA CENTER & DOMAIN SETUP

16.2. User1 password will expire in 30 days, User2 password will never expire.

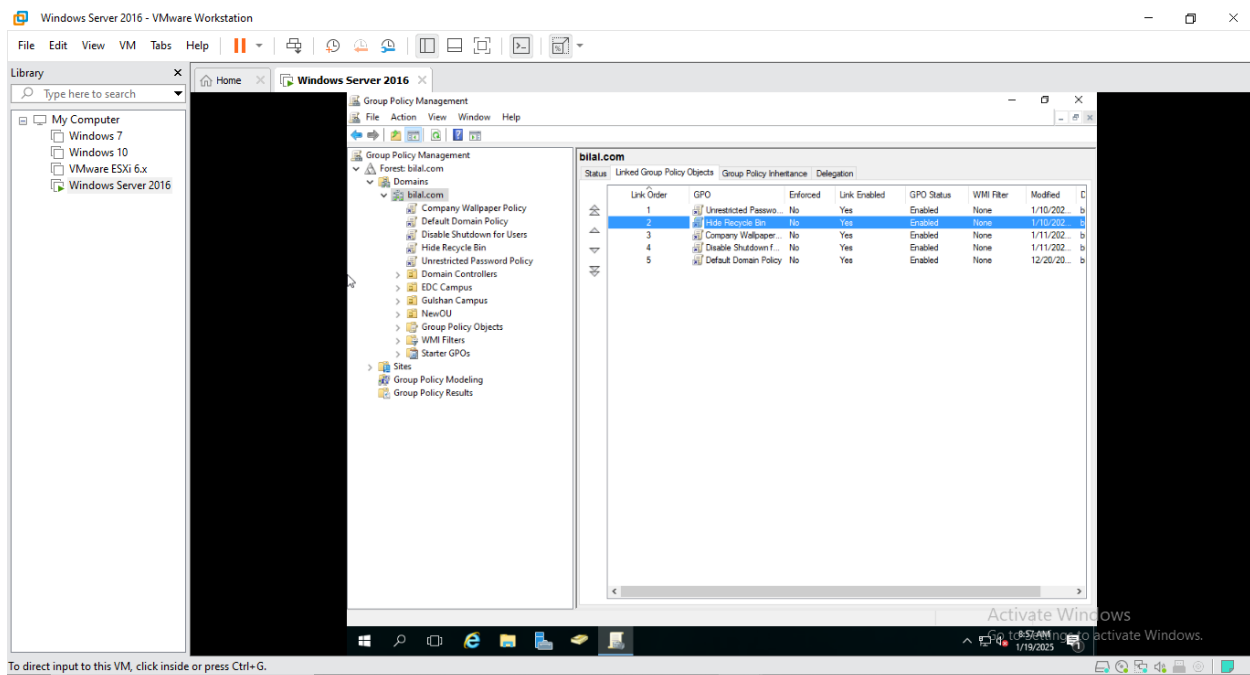
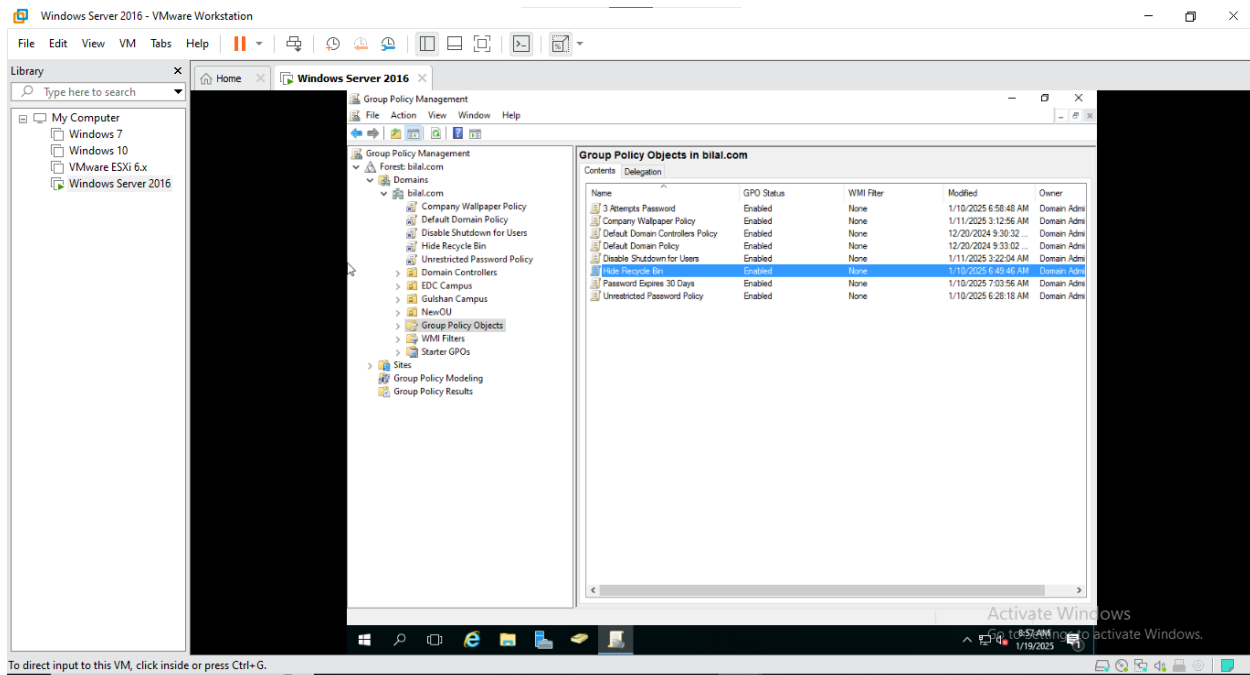
As per our GPO1, user should be able to setup password without no restriction, this GPO contradicts with it. To make this possible we have created an OU name '**Gulshan Campus**' on which such user whose password will be expired will lay. As per priority basis GPO1 applied on Domain will not apply and instead of it GPO2 (Password Expired After 30 Days) will enable. Not effecting global GPO's.



VIRTUALIZED DATA CENTER & DOMAIN SETUP

16.3. Recycle bin should not appear on any user profile.

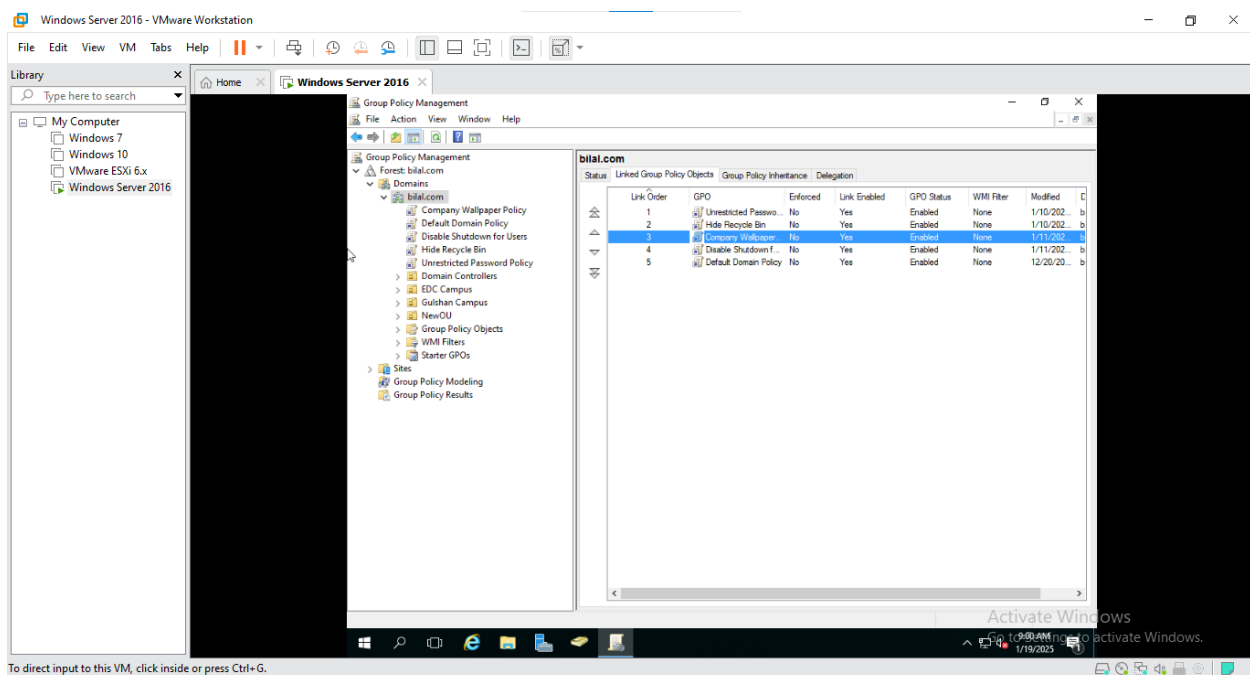
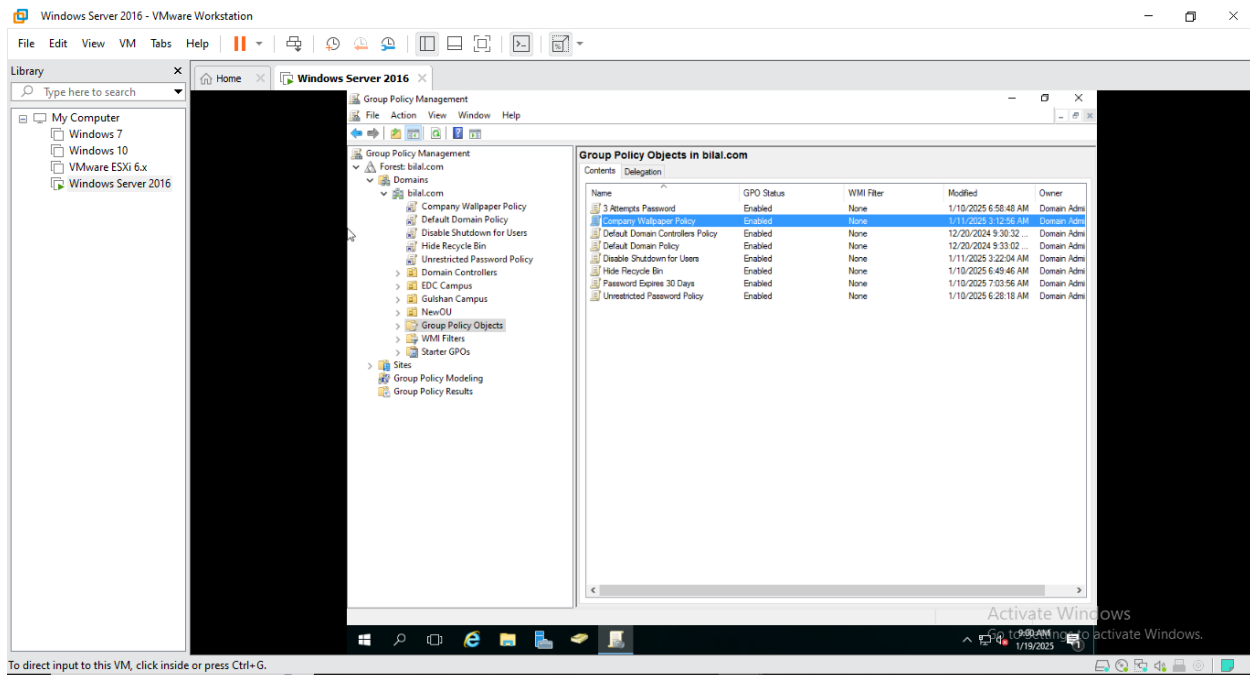
As per description of this GPO, any user should not see Recycle bin Icon of Desktop. To ensure this GPO enable/affective for all users it will be applied on Global Domain.



VIRTUALIZED DATA CENTER & DOMAIN SETUP

16.4. Customized Company's Wall Paper should appear on all user's desktop.

As per description of this GPO, all user should see **Company's Wall Paper** on Desktop. To ensure this GPO enable/affective for all users it will be applied on Global Domain.



17. Conclusion

This project provided a comprehensive understanding of virtualization technologies and enterprise network management. By overcoming challenges and implementing effective solutions, the team gained critical insights into virtual infrastructure deployment and domain administration. The knowledge and skills acquired will serve as a foundation for future complex computing problems in virtualization and cloud computing.